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AOE 4165/AOE 4166 - Space Vehicle Design

Project Hestia

"Payload Design for Lunar Infrastructure Development"

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EXECUTIVE SUMMARY

VT LUNA's Project Hestia presents a specialized mission architecture designed for the C3 Cosmic Lunar Operations Track 2 competition, focusing on the advancement of In-Situ Resource Utilization (ISRU) and scalable lunar infrastructure. To address the complexities of sustained operations at the lunar south pole, the mission leverages a modular rover system engineered to perform high-precision robotic tasks while surviving extreme environmental fluctuations.

Strategic landing site selection was driven by a comprehensive trade study evaluating terrain slope, solar illumination, and Earth visibility. The selected location ensures continuous power generation and reliable direct-to-Earth communication, establishing a stable foundation for the primary mission objective: a fully autonomous sequence for regolith collection and processing.

To validate this conceptual architecture, the team conducted physical hardware prototyping of the autonomous rover that is able to pickup, transport, and deposition simulant. By utilizing advanced kinematics and sensor fusion, the physical system successfully manipulated regolith-simulant without human intervention, demonstrating the feasibility of autonomous lunar mining operations. Furthermore, to ensure mission longevity, the architecture incorporates a robust thermal management system tailored for extreme cryogenic survival. This system actively protects critical avionics and power distribution units, preventing hardware failure during the extended lunar night.

Project Hestia's design is the culmination of a highly coordinated effort across four specialized sub-teams: Microwave/Mold, Rover, Power Thermal and Environmental (PTE), and Filtration Model. By seamlessly integrating these technical disciplines, VT LUNA has produced a high-fidelity proposal that strictly fulfills all requirements derived form the Request for Proposal (RFP). Ultimately, this mission serves as a critical technology demonstration, bridging the gap between conceptual lunar infrastructure and the reality of a sustained human presence on the Moon.

Contents

Acronyms

ACS Attitude Control System.

APXS Alpha Particle X-ray Spectrometer.

CAD Computer-aided design.

CCHP Constant Conductance Heat Pipe.

ECLSS Environmental Control and Life Support Systems.

FEA Finite Element Analysis.

GaN Gallium Nitride.

GNC Guidance, Navigation, and Control.

HLS Human Landing System.

IMU Inertial Measurement Unit.

ISAM In-Space Assembly and Manufacturing.

ISRU In-Situ Resource Utilization.

LiDAR Light Detection and Ranging.

MGA Mass Growth Allowance.

MLI Multi-Layer Insulation.

MW Microwave.

NASA National Aeronautics and Space Administration.

OPAL Optical Precision Autonomous Landing.

PCM Phase Change Material.

PDR Preliminary Design Review.

RF Radio Frequency.

RFP Request for Proposal.

RHU Radioisotope Heater Unit.

ROS Robot Operating System.

SEE Single Event Effects.

SLS Space Launch System.

STiM Stationary Tile Maker.

TRIDENT The Regolith and Ice Drill for Exploring New Terrain.

VRPN Virtual Reality Peripheral Network.

VSD Value System Design.

VT Virginia Polytechnic Institute and State University.

WLAN Wireless Local Area Network.

Nomenclature

δ	Microwave Penetration Depth
ϵ_r	Relative Permittivity
η	Efficiency
λ	Wavelength
μ	Coefficient of Friction
ρ	Density
σ	Electrical Conductivity
σ_y	Yield Strength
τ	Shear Stress
c_p	Specific Heat Capacity
E	Electric Field Magnitude
f	Frequency
g	Gravitational Acceleration
H	Magnetic Field Magnitude
I	Current
k	Thermal Conductivity
m	Mass
P	Power
Q	Thermal Energy
T	Temperature
V	Voltage

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0.1 Preface

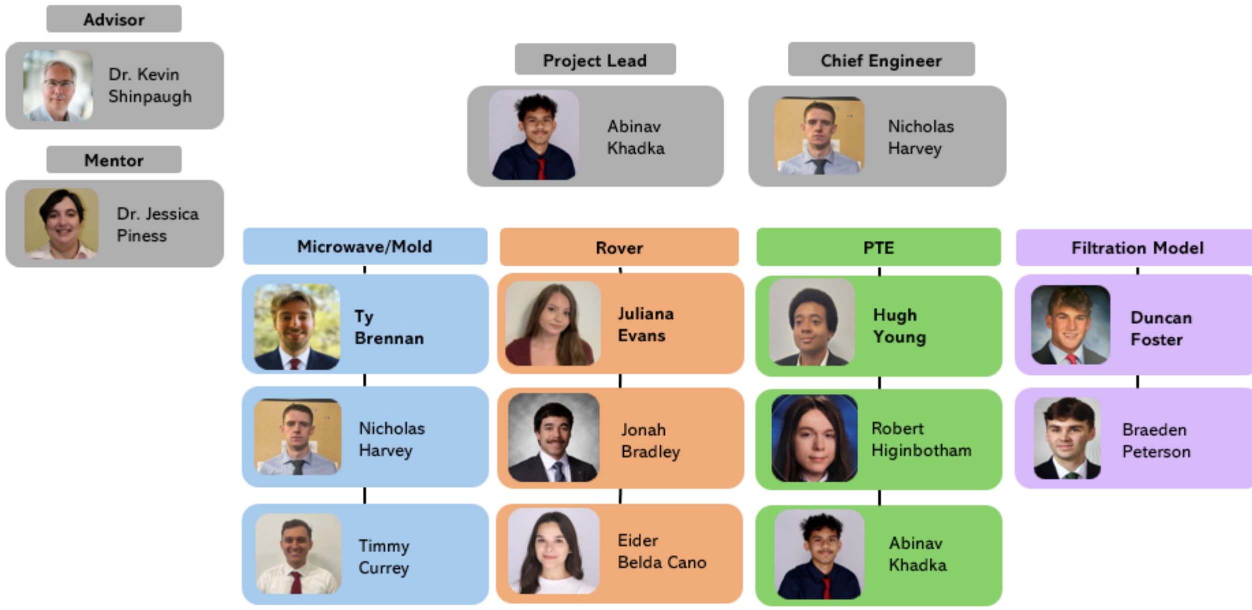


Figure 1: **VT LUNA Team Organizational Chart** Members are organized by system technical discipline. Members in leadership positions denoted with a bold font on their names. The advisor Dr. Shinpaugh and the mentor Dr. Piness for this project are pictured on the top left.

This design report details the architecture and proposal for the C3 COSMIC Lunar Operations competition. Detailed is the establishment of a problem definition for the Track 2 competition theme, as well as comprehensive mission and design considerations. The design process utilizes the core principles of systems engineering to determine the optimal and most favorable lunar mission architecture.

The deliverable below also partially fulfills the course requirements for AOE 4166, a senior design course at Virginia Tech. Therefore, deadlines were dictated by both the competition requirements and the academic course schedule. The analysis presented is the summation of two semesters of work for the project, and efforts will continue following the C3 COSMIC submission and presentation on April 23.

The VT LUNA Team, pictured in Figure 1, focuses on the Track 2 theme of the C3 competition. Throughout the report, the RFP is referenced in relation to this specific competition track. The team members are divided into four technical sub-teams and are led by a Team Lead and a Chief Engineer. The technical divisions of the team are Microwave/Mold, Rover, Power Thermal and Environmental (PTE), and Filtration Model. Each sub-team is managed by a dedicated sub-team lead. The Team Lead, Chief Engineer, Sub-team Leads, and sub-team members are explicitly denoted in Figure 1.

The name for this space mission, Project Hestia, is inspired by Greek Mythology, in which Hestia is the goddess of the hearth, fire, and architecture. This name was selected as it perfectly encapsulates the competition's Track 2 motivation: establishing sustained operations and foundational infrastructure on the lunar surface. Furthermore, the mythological connection to serve as a direct parallel to the project's core technical focus on ISRU.

1 Introduction

1.1 Project Background

NASA's Artemis program represents a fundamental shift in humanity's approach to lunar exploration, transitioning from the short-duration missions of the Apollo program to a sustained and infrastructure-driven presence on the Moon. Rather than simply demonstrating the capability to reach the lunar surface, Artemis is designed to enable long-term habitation, scientific discovery, and technology maturation in preparation for future human missions to Mars and beyond [1].

The program began with Artemis I in November 2022, which successfully demonstrated the integrated performance of the Space Launch System (SLS) and the Orion spacecraft in a cislunar environment. This uncrewed mission validated critical systems, including deep-space navigation, propulsion, avionics, and thermal protection during high-energy reentry [1].

Building on this success, Artemis II was successfully completed in 2026, marking the first crewed mission beyond low Earth orbit since 1972. The mission carried four astronauts on a 10-day free-return trajectory around the Moon, demonstrating life-support systems, crew operations, and mission architectures required for sustained human exploration. The flight validated Orion's Environmental Control and Life Support Systems (ECLSS), communication systems, and crew interfaces under deep-space conditions, significantly reducing risk for subsequent surface missions [1].

The progression to surface operations begins with Artemis III, which aims to return astronauts to the lunar surface for the first time in over five decades. This mission targets the lunar south polar region, an area of high scientific and strategic interest due to the presence of permanently shadowed regions believed to contain water ice and other volatiles. Artemis III integrates Orion with a commercial Human Landing System (HLS), currently being developed by SpaceX using a lunar-optimized Starship variant. In addition to advancing understanding of lunar geology and resource distribution, the mission will test next-generation spacesuits, surface mobility systems, and operational concepts required for extended stays [1].

Subsequent missions, particularly Artemis IV, expand the scope of lunar operations through the introduction of the Lunar Gateway. Positioned in a near-rectilinear halo orbit, Gateway will serve as a staging node for surface missions, a platform for scientific research, and a testbed for deep-space habitation technologies. Artemis IV is expected to deliver key modules, including the International Habitation module (I-Hab), in partnership with international space agencies, enabling a sustained and flexible lunar architecture.

Despite these advancements, establishing a sustained human presence on the Moon requires the development of robust surface infrastructure. The lunar environment presents extreme challenges, including abrasive regolith, wide thermal gradients, and the absence of an atmosphere. One of the most critical hazards is plume-induced regolith ejection during landing and ascent, which can generate high-velocity particles capable of damaging equipment and endangering crew. Mitigating these risks necessitates the construction of durable landing and launch surfaces, making infrastructure development a mission-critical requirement.

Recognizing the importance of in-space construction capabilities, the White House released the 2022 National Strategy for advancing In-Space Assembly and Manufacturing (ISAM). This strategy outlines a road map for enabling construction, manufacturing, and servicing capabilities directly in space, reducing reliance on Earth-based logistics. In response, the Consortium for Space Mobility and ISAM Capabilities (COSMIC) was established as a collaborative network of over 250 organizations spanning government, academia, and industry to accelerate ISAM development.

Through its C3 Capstone Challenge, COSMIC invites student teams to address key infrastructure gaps in space exploration. Within Track 2: Lunar Operations (C3-Lunar), teams are tasked with designing payloads capable of enabling a permanent lunar presence, with the constraints of 200 kg and within 3/4 of the volume of the Griffin Lander. The Virginia Polytechnic Institute and State University (VT) Luna team has elected to focus on autonomous construction technologies as a foundational capability [2].

Specifically, VT Luna proposes the development of an autonomous payload deployable via the Griffin Lunar Lander, capable of fabricating durable landing pad tiles using ISRU techniques. By leveraging lunar regolith as a primary construction material, the system minimizes mass launched from Earth while enabling scalable infrastructure development. The payload will integrate excavation, processing, and fabrication subsystems designed for autonomous operation in the lunar environment [2].

The proposed mission aligns with the Artemis campaign timeline, targeting deployment in the late 2020s to early 2030s to directly support sustained surface operations initiated by Artemis III and expanded through Artemis IV and beyond. By addressing the critical need for landing surface stabilization, this effort contributes to the foundational infrastructure required for a long-term human presence on the Moon and the continued expansion of human activity into deep space.

1.2 Problem Definition

1.2.1 Scope

Currently, there is no established lunar base or any accompanying infrastructure, and there are many such examples of infrastructure to choose from this blank slate. With a problem statement as general as 'create infrastructure for a permanent lunar outpost,' one might wonder how VT Luna arrived at the conclusion of constructing lunar landing pads. The first step in creating a lunar outpost is the consistent delivery of Earth technologies and payloads. To deliver logistical equipment, build surface assets, and rotate crew members, numerous landings and launches are required on the lunar surface. These launches and landings must be safe, reliable, and within a relative range of the established outpost.

Lunar regolith, or Moon dust, poses a great risk to the launch/landing requirements, as well as the safety of other previously delivered Earth technologies. Lunar regolith is extremely fine, abrasive, and electrostatically active, with a study done by National Aeronautics and Space Administration (NASA) concluding that the regolith is a "significant and pervasive risk [3]." When a Lander fires near the surface of the Moon, it erodes the loose regolith beneath it and accelerates the particles to velocities up to 2 km/s, a speed high enough to sandblast and damage hardware even kilometers away [4]. This uncontrolled erosion or plume ejecta can also crater and destabilize the landing zone for vehicles and even threaten the Lander's own structures and sensors.

This is where lunar landing pads become essential first-order infrastructure. Lunar landing pads replace loose, erodible regolith with a consolidated surface capable of withstanding the force of a rocket engine. With the addition of lunar landing pads, risk to all other infrastructure is mitigated. They remove erosion and cratering under the vehicle, minimizing the threat to the landing system. They also prevent the high-velocity plume ejecta that poses a risk to other existing infrastructure, such as power systems, habitats, rovers, or other science payloads. In addition to mitigating regolith-related mission risks, the landing pads also provide a central delivery zone, enabling repeated, closely spaced landings. This allows the aforementioned delivery process to be streamlined and local, eliminating the need to venture in a rover to a distant landing point for earth shipments whenever a landing occurs. This aspect significantly decreases the overhead costs of missions and creates a more efficient environment for the astronauts to live and work in. VT Luna is not alone in its conclusion on the importance of lunar landing pads as a necessary first step. NASA's Lunar Surface Innovation Initiative has identified landing and launch pads as a key infrastructure alongside ISRU and power [5].

Lunar dust and surface stability present significant hazards for repeated lunar landings. To support sustained lunar presence, landing pads are required to mitigate plume effects and provide stable landing zones. The system must operate autonomously to collect regolith, process it, and manufacture dense, durable tiles using microwave sintering.

The Project Hestia mission directly addresses several shortfalls identified in the NASA 2026 Civil Space Shortfalls document, released on 12 January 2026 [6]. This document identifies the most critical capability gaps that must be closed to enable sustained human presence on the lunar surface, and if the payload is positioned to contribute meaningful progress toward future Artemis Missions .

Table 1: **NASA shortfalls.** NASA 2026 Civil Space Shortfalls addressed by the Project Hestia mission [6].

Need ID	NASA Need Statement	Project Hestia Relevance
4.01	Perform site preparation and bulk regolith manipulation for infrastructure construction, including landing pads, berms, and regolith overburden for radiation protection.	This shortfall explicitly calls for the construction of landing pads using bulk regolith manipulation, which is the primary objective of the Stationary Tile Maker (STiM) payload.
4.04	Construct structures on the lunar surface through advanced manufacturing techniques.	Microwave sintering of lunar regolith is a leading example of an advanced in-situ manufacturing technique, and Project Hestia demonstrates this capability at mission scale.
3.01	Predict plume surface interaction surface erosion, crater width and depth, and ejecta energy flux during landing.	While the STiM payload does not predict plume surface interaction directly, the entire mission architecture is designed to eliminate the surface erosion and ejecta energy flux hazards this shortfall characterizes.
2.09	Mitigate and survive high velocity impacts of small particles on lunar surface assets.	By constructing consolidated landing pad tiles, Project Hestia removes the loose regolith that is accelerated into high-velocity ejecta, directly mitigating the particle impact hazard to all surrounding surface assets.
5.02	Excavate and transport lunar regolith at a scale relevant for a demonstration mission.	The STiM collection rover is designed to excavate and transport regolith at demonstration scale, feeding the sintering mold for each production cycle.

1.2.2 Stakeholders

NASA's Artemis program represents the United States' next major initiative in human space exploration, with the goal of establishing a sustainable human presence on the lunar surface while laying the groundwork for future crewed missions to Mars. Building upon the technological legacy of the Apollo program, Space Shuttle program, and the International Space Station, Artemis integrates next-generation systems such as the Space Launch System, and the Orion spacecraft. In addition to advancing transportation capabilities, Artemis emphasizes the development of surface infrastructure, including power systems, habitats, and construction technologies that rely on in-situ resource utilization (ISRU). These efforts are supported through a combination of government leadership and commercial partnerships, positioning Artemis not only as a scientific endeavor but also as a driver of technological innovation and economic activity in cislunar space.

Within this framework, stakeholders extend beyond NASA to include international space agencies, private aerospace companies, academic institutions, and research initiatives such as the VT Luna C3 project. A critical area of shared interest among these stakeholders is material acquisition and processing and conducting high-value scientific investigations of the Moon's south polar region particularly the ability to collect and utilize lunar regolith for construction purposes. Technologies for autonomous excavation, transport, and deposition of material are essential for enabling large-scale infrastructure development, such as landing pads, radiation shielding, and structural components manufactured through additive processes. Should support from Artemis diminish, systems like the rover proposed in this work could be transitioned to alternative stakeholders, including international programs led by organizations such as the European Space Agency or emerging commercial lunar initiatives. At a smaller scale, terrestrial applications in automated construction and material handling also present viable pathways for adaptation, reinforcing the broader relevance and flexibility of autonomous material collection technologies.

1.2.3 Timeframe

VT Luna has outlined a 6-month period to complete its mission beginning in September of 2029, and ending in March of 2030. This timeline was selected for a multitude of reasons but primarily it is to be in accordance with NASA's lunar architecture timeline. The Artemis program is scheduled to send preliminary missions to the lunar surface in the late 2020s and will be followed by many more missions in the 2030s [7]. Landing pads are a critical piece of initial infrastructure, as they help ensure safer and less damaging landings on the surface, and thus will likely be constructed in the early 2030s. As the mission is a proof of concept, it will predate this construction period and therefore occur in the late 2020s. In this time frame, it is expected that the mission will have access to very little lunar architecture, as it is among the first pieces of infrastructure to be constructed. A key piece of infrastructure in landing pad construction could be water extraction architecture, as water can act as a binder; however, this technology is unlikely to be operational before the early 2030s to mid-2030s and thus cannot be assumed for the mission. Some assumptions that can be made include a 5 kW solar power supply from the Griffin Lunar Lander, as well as a local on-surface astronaut for tile relocation and testing. These assumptions are provided by COSMIC, the overseeing consortium of this project. Another primary driver behind this mission window is the illumination. In order for whatever payload VT Luna designed to have access to the full 5 kw power supply from the Griffin Lander, the site must be illuminated. Based on the site selected, which will be outlined later in the report, this specific 6 month window will provide almost constant illumination, consisting of 26 lunar days and 3 lunar nights. This mission window will maximize the accessible power for the payload, as well as reduce the time spent in thermal life support mode. With this multitude of considerations, the specific mission timeframe was selected.

1.2.4 Assumptions

To develop a mission addressing the COSMIC challenge, several assumptions were made. The first five were provided directly by COSMIC: the Griffin Lander will deliver the payload to the lunar surface, supply the payload with 5 kW of continuous power, provide an accessible rover charging port on top of the Lander, and provide deployment capabilities for the system. COSMIC also provided that a lunar surface astronaut will be accessible for the mission. These assumptions establish a reliable power source, a charging method for the rover, a deployment method for the system, and an astronaut available for troubleshooting and returning manufactured tiles to Earth.

Additional assumptions include near constant illumination based on the selected landing site and mission timeframe, and no access to water. Given the selection of a highly elevated landing site during a high illumination period, lunar nights are not expected to exceed four days. Water was also assumed to be unavailable, as the mission takes place in 2029 and 2030, prior to any anticipated lunar water harvesting capabilities. While water would be a valuable resource as a potential binder material, it will simply not be accessible during this timeframe. Working within these assumptions, the concept for constructing lunar landing pads was developed.

1.3 Astrobotic's Griffin Lunar Lander

Astrobotic Technology's Griffin Lunar Lander is a next-generation, medium-class lunar Lander designed to support missions requiring substantial payload capacity, high-precision landing, and extended operations in challenging lunar environments, particularly within the Moon's south polar region. Unlike smaller commercial Landers, Griffin is specifically engineered to deliver large, complex payloads—including rovers, scientific instruments, and infrastructure systems—directly to the lunar surface. Its adaptable configuration allows it to be arranged and augmented to meet a variety of mission profiles and landing site conditions, providing flexibility for payload integration and deployment. This versatility enables Griffin to support a wide range of mission objectives, including scientific research, technology demonstration, resource prospecting, commercial payload delivery, and symbolic or commemorative missions [8].

The Griffin Lander builds upon Astrobotic's proven spacecraft bus architecture, emphasizing modularity, subsystem reliability, and scalability across missions. Its design incorporates enhanced propulsion systems for controlled descent and precision landing, advanced thermal management to withstand extreme temperature variations in permanently shadowed or high-latitude regions, and robust structural components capable of supporting heavier payload classes [8]. Griffin also features upgraded power systems, including high-efficiency solar arrays and energy storage solutions, to sustain longer-duration surface operations. Guidance, navigation, and control systems are optimized for autonomous landing in complex terrain, reducing reliance on real-time human intervention. Collectively, these capabilities position Griffin as a key platform for supporting sustained lunar exploration efforts where reliable delivery of materials, instruments, and infrastructure components to the lunar surface is essential for enabling long-term human presence.

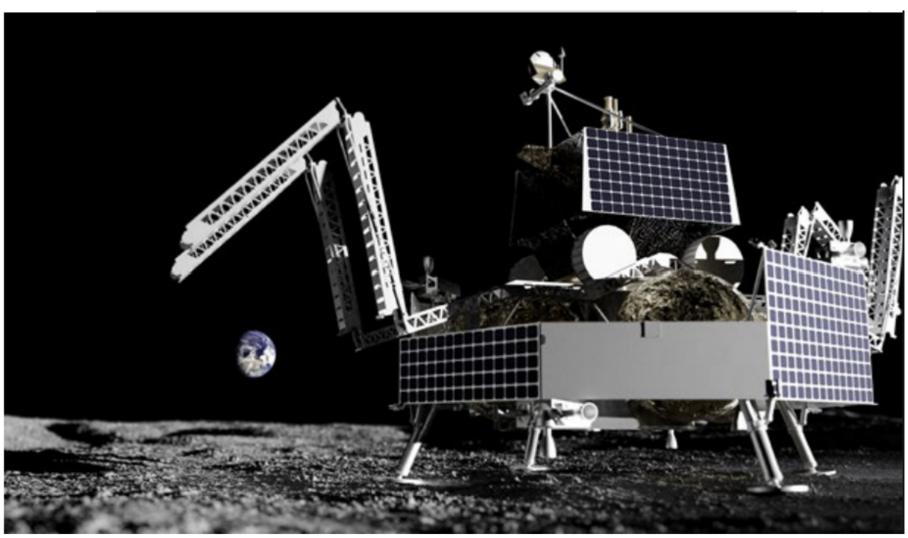


Figure 2: **Astrobotic's Griffin lunar Lander.** Delivery platform selected for Project Hestia as outlined by the C3 COSMIC RFP. This defines the main payload constraints for the mission, including available mass, power, volume, and lunar surface deployment considerations.[9]

1.3.1 Structure

The Griffin Lunar Lander features a structurally robust yet lightweight design that is optimized to withstand the mechanical and environmental stresses associated with launch, transit, and lunar landing. The primary structure is composed of high-strength aluminum alloys, selected for their favorable strength-to-weight ratio, corrosion resistance, and proven performance in aerospace applications. This structural framework is divided into three main components: the launch vehicle adapter cone, the central deck panels, and the radiator panels. The launch vehicle adapter cone serves as the primary interface between the Lander and the launch vehicle, transmitting loads experienced during ascent while maintaining structural integrity. The deck panels form the core platform of the Lander, providing mounting points for payloads, avionics, and critical subsystems, while also distributing mechanical loads throughout the structure. Radiator panels are integrated into the design to support both payload accommodation and thermal regulation, enabling efficient heat rejection in the extreme thermal environment of the lunar surface [8].

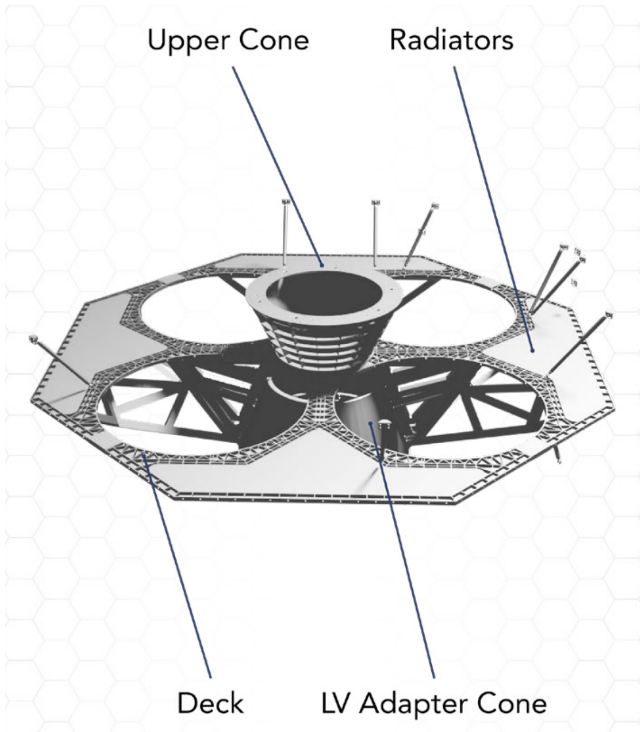


Figure 3: **Diagram of Griffin's adapter cone and deck.** Main layout of the adapter cone and deck. This adapter informs the arrangement of the missions payload during transport to the lunar surface [9]

To ensure stability and survivability during touchdown, the Lander is equipped with four landing legs that are engineered to absorb impact forces and adapt to uneven terrain. These legs typically incorporate damping elements that dissipate kinetic energy during landing, reducing stress on the main structure and sensitive onboard systems. The wide stance of the landing gear enhances stability, particularly in the relatively uneven and regolith-covered surfaces commonly found in polar regions. Structurally, the Lander is designed to maintain a low center of gravity, further improving landing stability and reducing the risk of tipping-over during surface operations.

In addition to its primary structural elements, the Griffin Lander integrates six major subsystems: propulsion, Guidance, Navigation, and Control (GNC), power, avionics, communications, and thermal control—within its structural framework [8]. The structure not only supports these subsystems mechanically, but also facilitates their functional integration by providing routing paths for wiring, propellant lines, and thermal interfaces. This tight integration ensures that the Lander can operate as a cohesive system while maintaining modularity for payload accommodation and mission-specific adaptations. In general, the structural design of Griffin reflects a balance between strength, mass efficiency, and adaptability, enabling it to support a wide range of lunar missions while maintaining reliability in one of the most demanding operational environments.

1.3.2 Propulsion System

The Griffin Lunar Lander propulsion system is designed to provide reliable thrust for trajectory correction, descent, and precision landing in the challenging lunar environment. It is equipped with five 700 lbf-class pulsed main engines that generate 3100 N of thrust as the primary thrust required for deorbiting and controlled descent, as well as twelve 110 N Attitude Control System (ACS) thrusters [8]. The system uses a pressure-fed propellant architecture, in which propellants are delivered to the engines using pressurized gas rather than turbopumps, simplifying the design and improving reliability. The propellant combination is hypergolic, meaning the fuel and oxidizer ignite spontaneously upon contact, eliminating the need for a separate ignition system and reducing potential failure points.

The pulsed nature of the main engines allows thrust to be modulated through rapid, repeated combustion cycles, providing fine control over descent velocity and enabling soft landing capabilities. Meanwhile, the ACS thrusters play a critical role in controlling the Lander's orientation by adjusting pitch, roll, and yaw. This precise attitude control is pivotal for tasks such as trajectory corrections during cruise, maintaining proper orientation for communications and power generation, and executing hazard-avoidance maneuvers during landing. Together, the main propulsion and ACS systems help the Lander to perform highly controlled and autonomous operations, including navigating uneven or hazardous terrain on the lunar surface, where precise positioning and stability are critical for mission success [9].

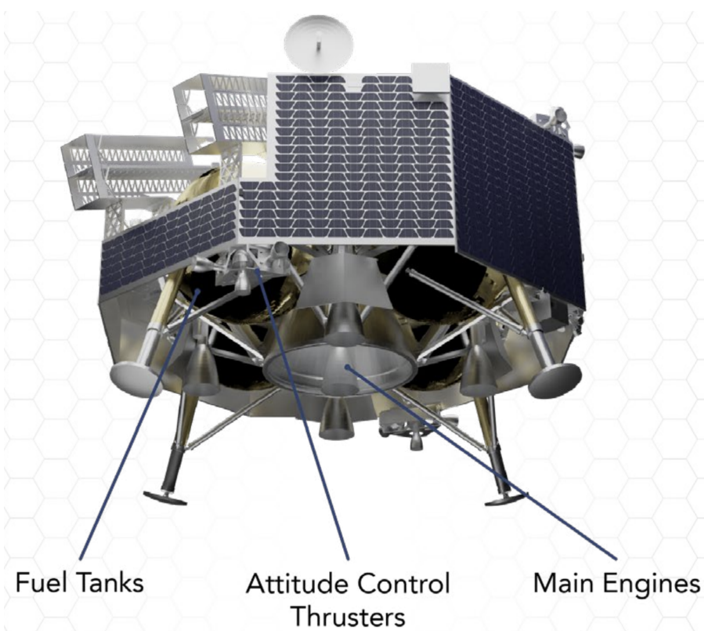


Figure 4: **Diagram of Griffin's propulsion system.** Main propulsion layout of Astrobotic's Griffin Lunar Lander. The propulsion system is relevant to Project Hestia because it affects Lander operations, payload accommodation, and the overall delivery environment for the lunar surface payload.[9]

1.3.3 Guidance, Navigation, Control System

The Griffin Lunar Lander GNC system is responsible for determining the Lander's state and executing precise maneuvers throughout all phases of the mission, from cruise to touchdown. It processes inputs from a suite of onboard sensors, filtering and correcting for noise, bias, and other measurement uncertainties to maintain an accurate estimate of the Lander's position, velocity, and attitude. This state estimation is continuously updated and used to generate control commands that guide the vehicle along its intended trajectory. During cruise and orbital phases, Earth-based ranging data contributes to refining position and velocity estimates, enabling accurate trajectory correction maneuvers. The system also utilizes star trackers and sun sensors to determine orientation in space, allowing the Lander to maintain proper attitude for communication and to keep its solar arrays pointed toward the Sun for optimal power generation [9].

As the Lander transitions into its descent and landing phase, the GNC system shifts to a sensor-driven mode to ensure a safe and precise touchdown. A Doppler Light Detection and Ranging (LiDAR) instrument provides real-time measurements of altitude (range) and descent velocity (range rate), enabling the system to make fine adjustments to thrust and orientation during the final approach. Complementing this capability are the integrated Optical Precision Autonomous Landing (OPAL) sensors, an experimental vision-based navigation system consisting of a camera and onboard processing unit [9]. These sensors compare real-time surface imagery with preloaded lunar terrain maps to estimate the Lander's position relative to the surface and predict landing conditions. This terrain-relative navigation capability enhances hazard detection and avoidance, allowing the Lander to adjust its landing site in response to obstacles such as slopes, rocks, or uneven terrain. Together, these systems enable a high degree of autonomy, reducing reliance on real-time ground control and significantly improving the reliability and accuracy of lunar landing operations.

1.3.4 Power System

The Griffin Lunar Lander power system is designed to be power-positive for all mission phases except descent, meaning it generates more energy than it consumes during cruise, orbital operations, and surface operations. The system is responsible for power generation, storage, distribution, and overall power management across all subsystems and payloads. Primary power generation is provided by a panel of GaInP/GaAs/Ge triple-junction solar cells with extensive flight heritage in both orbital and deep space missions. These high-efficiency solar arrays are continuously oriented toward the Sun, in coordination with the GNC system, to maximize energy collection throughout the mission [9].

Energy storage is provided by a space-grade lithium-ion battery system, which ensures continuous power availability during periods of eclipse, peak load demand, and the descent phase when solar generation is limited. The stored and generated power is regulated and distributed through a 28 V power bus that supplies all onboard avionics, subsystems, and payload interfaces. After landing, the power system continues to operate in conjunction with solar array pointing strategies to maintain consistent energy production on the lunar surface, enabling sustained payload operations and long-duration mission support in the harsh thermal and illumination conditions of the Moon [9].

1.3.5 Avionics System

The avionics on board perform all command and data handling, managing the multiple different inputs and outputs of the Lander's subsystems. On board, the Griffin Lunar Lander has an Integrated Avionics Unit (IAU) that houses ten boards, each with distinct functions that seamlessly help aspects of the avionics system and flight computer. The Griffin's flight computer consists of a 32-bit high-performance dual-core LEAON 3 FT microprocessor. To stay protected, the computer is equipped with radiation-hardened circuits, fault-tolerant and single-event upset (SEU) -proof characteristics. Also encompassed within the IAU is the payload computer.

This computer monitors payload power consumption and directly communicates with the payloads [9]. Data and commands from the payload ground software are sent to the payload via the payload computer, and payload telemetry is packaged for downlink to Earth.

1.3.6 Communications System

Griffin's communications system is responsible for providing reliable command, telemetry and data relay between the Lander, Earth, and hosted payloads throughout all phases of the mission. The system enables two-way communication, allowing mission operators to send commands to the Lander while simultaneously receiving health status, navigation data, and scientific payload information. A high-powered transponder onboard the Lander serves as the primary interface for deep-space communication, supporting long-distance data transmission back to Earth. Communications with Earth are conducted using multiple frequencies within the X-band range, which is commonly used for deep space missions due to its balance of bandwidth efficiency and resistance to signal degradation over long distances [9].

To ensure continuous connectivity during different mission stages, the Lander is equipped with multiple antennas optimized for varying operational conditions. Low-gain antennas are used during cruise and lunar orbital operations, providing broad coverage and maintaining stable communication links. Once the Lander successfully touches down on the lunar surface, it transitions to using its medium-gain or high-gain antennas to increase communication bandwidth and improve data transmission rates. In addition to Earth communication systems, Griffin also supports local surface operations through a 2.4 GHz IEEE 802.11n compliant wireless local area network (Wireless Local Area Network (WLAN)) modem [9]. This system enables direct wireless communication between the Lander and deployed payloads on the lunar surface, facilitating efficient data exchange, coordination, and autonomous operation of instruments and robotic systems without requiring constant Earth-based intervention. More information and its relevance about the communications system is provided in subsection 5.3.1.

1.3.7 Thermal Control System

Finally, the Griffin Lunar Lander uses a thermal control system that relies mainly on passive methods to regulate temperature in the extreme lunar environment. Since the Moon lacks an atmosphere, the Lander must manage large temperature swings between sunlit and shadowed regions, especially in polar areas where lighting conditions can be highly variable. To address this, Griffin utilizes radiator panels that dissipate excess heat by radiating it directly into space, preventing overheating of onboard electronics and payload systems [9]. These radiators are strategically placed to maximize thermal efficiency while minimizing interference with other subsystems and payload operations.

In addition to radiators, the Lander employs heat pipes to redistribute thermal energy throughout the structure [9]. These heat pipes transfer heat from warmer components to cooler regions of the Lander, helping to maintain a uniform temperature distribution and prevent hotspots. This is important for sensitive avionics and scientific instruments, which must operate within strict temperature limits to operate reliably. The use of heat pipes also improves overall system efficiency by allowing excess heat to be reused where necessary rather than simply discarded.

To further protect the Lander from extreme thermal conditions, Multi-Layer Insulation (MLI) is applied to critical components and structural surfaces [9]. MLI consists of multiple thin layers of reflective material that reduce heat transfer through radiation, effectively insulating the Lander from both intense solar heating and the cold of deep space. This combination of radiators, heat pipes, and insulation creates a balanced thermal management system that minimizes the need for active heating or cooling, reducing power consumption and system complexity.

For the specific mission outlined by C3 Lunar Track 2, the Griffin Lunar Lander offers a great amount of flexibility and customization.

1.4 Key Requirements

The system requirements were derived from the physical constraints of the Griffin Lander payload envelope, the constraints provided by COSMIC's RFP, and the operational necessities of the lunar surface environment. Each of these requirements defines the capabilities that the subsystem must fulfill, serving as the baseline for the mass, power, and autonomy considerations. Table 2 lists each of the system requirements for this mission. The first eight requirements represent the necessary objectives that the system must be able to achieve.

Table 2: **System requirements for Project Hestia.** An overview of the requirements that the mission must adhere to, derived from the RFPs and the mission definition. The first two requirements are from C3 Cosmic's RFP. The key requirements are from SYS1.0 - 3.0 which highlights the core functions of the system

System Requirements	
RFP1.0	The system shall have a total mass below 200 kg.
RFP2.0	The system shall occupy no more than 75% of the available Griffin payload volume (1.5 m × 1.5 m × 2.5 m).
SYS1.0	The system shall autonomously manufacture structurally sound tiles using in-situ resource utilization (ISRU).
SYS2.0	The system shall accept and process lunar regolith as the primary material for tile production.
SYS3.0	The system shall produce tiles that meet defined structural and thermal performance standards.
SYS4.0	The system shall operate autonomously for the duration of the mission, including health monitoring and fault recovery.
SYS5.0	The system shall survive and operate within the lunar surface environment, including thermal and radiation conditions.
SYS6.0	The system shall perform all operations within the allocated peak power and energy-per-tile budget.
SYS7.0	The system should autonomously produce one structurally sound tile within two weeks.
SYS8.0	The system should detect and respond to failures associated with the regolith collection, conditioning, and sintering subsystems.
SYS9.0	The system should store sufficient regolith to produce two to three tiles without additional delivery.

As seen in the list of system requirements, the design is heavily constrained by mass and density (RFP1.0, RFP2.0) and the necessity for high-level autonomy (SYS4.0). The requirement to process abrasive regolith (SYS2.0) and survive the thermal extremes of the lunar surface (SYS5.0) drives the selection of robust mechanical and thermal subsystems.

2 Value System Design

The most effective approach to identifying an optimal mission design is through the application of systems engineering principles. The Value System Design (VSD) helps outline which factors make one design more favorable than another and how these factors are weighted. The factors are weighted against each other using either qualitative or quantitative data. After analyzing the values of each factor, all potential mission architectures are evaluated. In this case, the results help determine the optimal configuration for the payload to construct lunar landing pads. The requirements that define mission success are used to establish the specific figures of merit for the study. Each of the proposed concepts will be systematically assessed against these weighted factors. The goal of this analysis is to produce a clear, justifiable recommendation on the most viable architecture to pursue for the next phase of design and development.

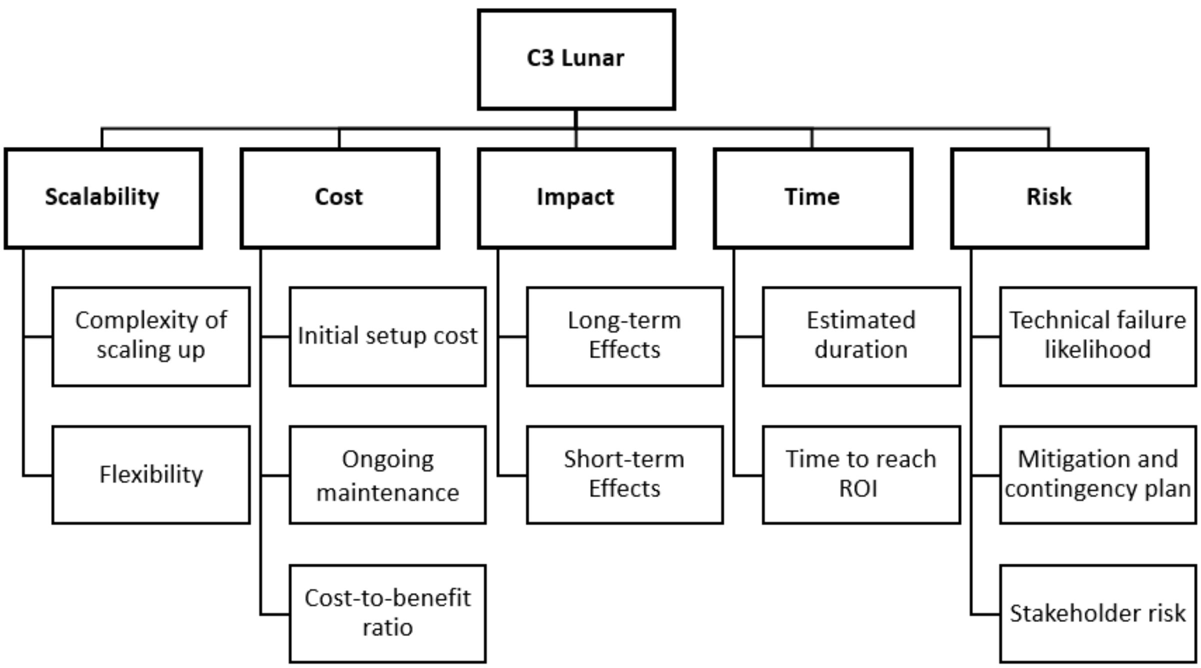


Figure 5: **Project Hestia value system design diagram.** The diagram summarizes the main criteria used to evaluate and compare payload design options for lunar infrastructure development. This shows consideration for elements affecting the mission and provides a framework for decisions made throughout mission development.

2.1 Scalability

This factor measures the potential for the design to be replicated, adapted, and expanded for full-scale infrastructure creation. The payload operates under strict weight and volume constraints. It cannot build a complete lunar base in a single mission; rather, it must demonstrate a viable method that can be scaled up. Therefore, the architecture must prove that the sintering and construction process is not limited to a small sample size but can be expanded to create large-scale landing zones. If the design cannot be scaled, the payload fails to demonstrate long-term utility.

2.2 Impact

Secondly, impact quantifies the importance and effectiveness of the mission beyond its initial success. This factor ensures that the purpose and significance of the mission justify the resource investment. High-impact missions secure support from potential stakeholders (such as NASA and commercial partners) by advancing the end goal of permanent lunar settlement. A mission with low impact implies that even if technical success is achieved, it does not meaningfully advance the state of the art or solve the critical problem of lunar dust mitigation.

2.3 Time

Next, time is a critical criterion driven by funding cycles and planning logistics. This metric evaluates the mission timeline from development to operational capability. Missions with extended timelines, or those requiring upwards of a decade to achieve a minor goal, face significant risks regarding funding continuity. Furthermore, the timeline is constrained by the interdependence of other lunar assets; for example, if the mission relies on equipment that has not yet arrived on the Moon, the timeline is compromised. Therefore, minimizing the time to mission completion is favorable.

2.4 Risk

Risk determines the probability of mission failure and the effectiveness of mitigation strategies. Since the C3 Lunar mission focuses on robotic infrastructure construction rather than human-manned missions, a higher tolerance of risk is accepted compared to crewed standards. However, risk remains a factor regarding the safety of the Lander, the prevention of collisions with orbital debris, and the reliability of the payload after the shocks of landing. The design must ensure that the equipment operates correctly in the harsh lunar environment (vacuum and thermal extremes) to prevent total mission failure.

2.5 Cost

Cost directly affects the feasibility of the mission, though it is the least influential factor in this specific value system. While financial constraints exist, the primary objective of this mission is to demonstrate the feasibility of building infrastructure at a large scale. The mission is not designed to minimize spending at the expense of performance; rather, it focuses on establishing a viable method for construction. An initial investment of higher value is considered acceptable if it ensures the payload works effectively and is scalable.

2.6 Mission Performance Ranking

The following Table 3 describes the value system established for the C3 Lunar mission. The opinions of each team member were taken into account, and a value system matrix was generated to prioritize design decisions.

Table 3: **Value System Matrix.** Weighted ranking of value system evaluation criteria.

Criterion	Weight	Rank
Scalability	0.444	1
Impact	0.292	2
Time	0.129	3
Risk	0.081	4
Cost	0.052	5

As can be seen, it was decided that **Scalability** was the most important criterion. This is due to the weight constraints imposed, meaning the payload would have to demonstrate how it could build an infrastructure instead of building it completely. So, scalability has to be important; otherwise, the payload would not be beneficial. The hope is that the payload could be replicated or scaled up to create the complete infrastructure.

Impact was ranked second in importance. The impact of a mission ensures that the purpose and significance of the mission are worth the amount of money and effort being put into it. If a mission is of low impact, it is not only harder to get support from potential stakeholders, but it also doesn't advance an end goal in any meaningful way.

Time was chosen as the third most important criterion. Time is important due to funding and planning. Depending on the planned time for a mission, it could be difficult to achieve funding. For example, if a mission plans to use water gathered from the Moon, but is reliant on the equipment to do so already being there, it limits the timeline of the mission. It is also important to mention that if a mission is planning to take upwards of ten years, but is achieving a small goal, it reduces the chance of funding.

Risk was chosen as the fourth criterion due to this mission not involving a crew. This mission was created assuming no people will be near the Lander or payload upon landing and during operation, which greatly reduces the risk. There is still some risk in the Lander getting off course and potentially crashing into a satellite or orbital debris and causing damage, but that is low. There is also a risk in the equipment not operating correctly once it reaches its landing site, but that is always a risk in any mission.

The final criterion chosen was **Cost**. This mission is to demonstrate the feasibility of building an infrastructure at a large scale and to demonstrate value. It was not created to try to minimize spending. It focuses on having a viable method for building infrastructure instead of a lower-cost mission that may not work or be scalable.

3 Mission Overview

3.1 Key Mission Elements

The main objectives of this mission are to design a payload for the Griffin Lunar Lander to help build the infrastructure needed for a permanent lunar outpost. The mission serves as a technology demonstration for In-Situ Resource Utilization (ISRU), specifically proving the feasibility of autonomously manufacturing structural tiles from lunar regolith. These objectives are formulated to address the critical hazard of plume ejecta by creating consolidated landing zones, thereby supporting the sustainability of future Artemis operations.

The request for a lunar infrastructure payload allows for significant flexibility in architecture. Unlike a standard rover mission, this design contains both a rover for collection and heavy industrial processing, which is handled by the stationary system. The choice of a different design process would dramatically affect the concept of operations in terms of the different stages of the operation.

The rover architecture system defines the division of labor. To satisfy the mass constraint (RFP1.0), the system was divided into two distinct units: a lightweight mobile rover for collection and a stationary heavy-duty factory for processing. This split allows the heavy sintering hardware to remain on the Lander, reducing the energy cost of locomotion.

The stationary system focuses on the technique for tile creation. The decision was made to utilize microwave sintering (SYS1.0), as it requires no binder material from Earth, fully satisfying the ISRU requirement. This method directly influences the power subsystem design due to the high energy bursts required for heating.

3.2 Concept of Operations

The Concept of Operations for Project Hestia is shown in Figure 6. The mission employs the Griffin Lander to deliver the payload to the lunar surface. Following landing, the system transitions to autonomous surface operations and executes a cyclical operational process designed to collect lunar regolith, process it on-site, and fabricate solid tiles for surface infrastructure applications. The concept of operations proceeds as follows:

1. **Payload Launch**: The payload, mounted to the Griffin Lunar Lander, is launched on board what will most likely be a SpaceX Falcon Heavy. Mission launch occurs in September 2029

2. **Deployment and Initialization:** After landing, the Lander performs system health checks and initializes surface operations. Solar arrays deploy to provide power, and the stationary tile-fabrication system is deployed from the lower deck of the Griffin Lander to the lunar surface.

3. **Regolith Collection:**

    (a) Rover deploys from the Griffin Lander and traverses to the harvest site guided by onboard LiDAR cameras

    (b) Rover uses arm and scoop to collect lunar regolith at a rate of 1 kg/min until bucket volume of 5 L (mass of 16kg) is reached

(c) Rover navigates back to Stationary Tile Maker(STiM) and ascends the ramp

(d) Rover deposits regolith into reception hopper

(e) Rover either returns to harvest site or mounts to wireless charger

4. **Regolith Processing:** Inside the fabrication unit, the regolith is sifted using a vibratory screen to remove oversized particles. The processed material is then weighed and distributed into a mold to ensure consistent tile geometry.

    (a) The reception hopper receives the regolith sample from the rover. The reception hopper acts as the entry point to the Stationary Tile Maker (STiM) as well as a storage unit for excess regolith when processing and sintering are commenced.

    (b) The regolith will then pass from the reception hopper to the vibratory screener. The main purpose of this subsystem is to remove oversized regolith particles from the STiM and deposit them outside of the STiM via a chute. The rover can then remove and dispose of the non-optimal granularity regolith.

    (c) After passing through the vibratory screener, the optimal granularity regolith makes its way into the weigh bin. The weigh bin will be mounted to load compression cells that will measure the mass of the incoming regolith. Once the target mass for a single tile is reached, the vibration cycle will cease and the slide gates between the hopper and vibratory sifter as well as the slide gate between the sifter and weigh bin will seal off any further regolith flow.

    (d) When target mass is reached, a rotary gate in the bottom of the weigh bin will deposit the requisite regolith volume for a single sintering layer into the mold.

    (e) After the regolith is deposited into the mold, the mold will commence a vibration sequence followed by a compaction sequence from the microwave press to ensure even distribution as well as requisite bulk density is achieved.

5. **Sintering:** A microwave laser system melts and sinters the regolith into a solid layer. At which point the mold will return to the deposit configuration and the system will reset to step 4d. This cycle will commence a total of 24 before one full tile is completed.

6. **Cooling & Release:** The fabricated tile undergoes controlled cooling to mitigate thermal stresses and prevent cracking. Once cooled, the tile is released onto the lunar surface.

7. **Rover Collection & Testing**: The rover will collect the ejected tile and take the tile to the testing site to undergo two tests. The first being a mock assembly, in which tiles will be assembled in a small landing pad formation to ensure the low-precision assembly design is valid. This assembly will take place beneath the Griffin Lander so that the second test, thermal testing, can be completed. This test will consist of the Griffin Lander firing its engines directly on the face of the mock landing pad to ensure that thermal considerations are valid

This cycle is repeated throughout the mission, enabling gradual production of surface tiles as the rover alternates between regolith collection and battery recharging. By keeping the primary processing equipment stationary and assigning mobility to the rover, the system maintains operational simplicity while supporting scalable construction of landing pad infrastructure.

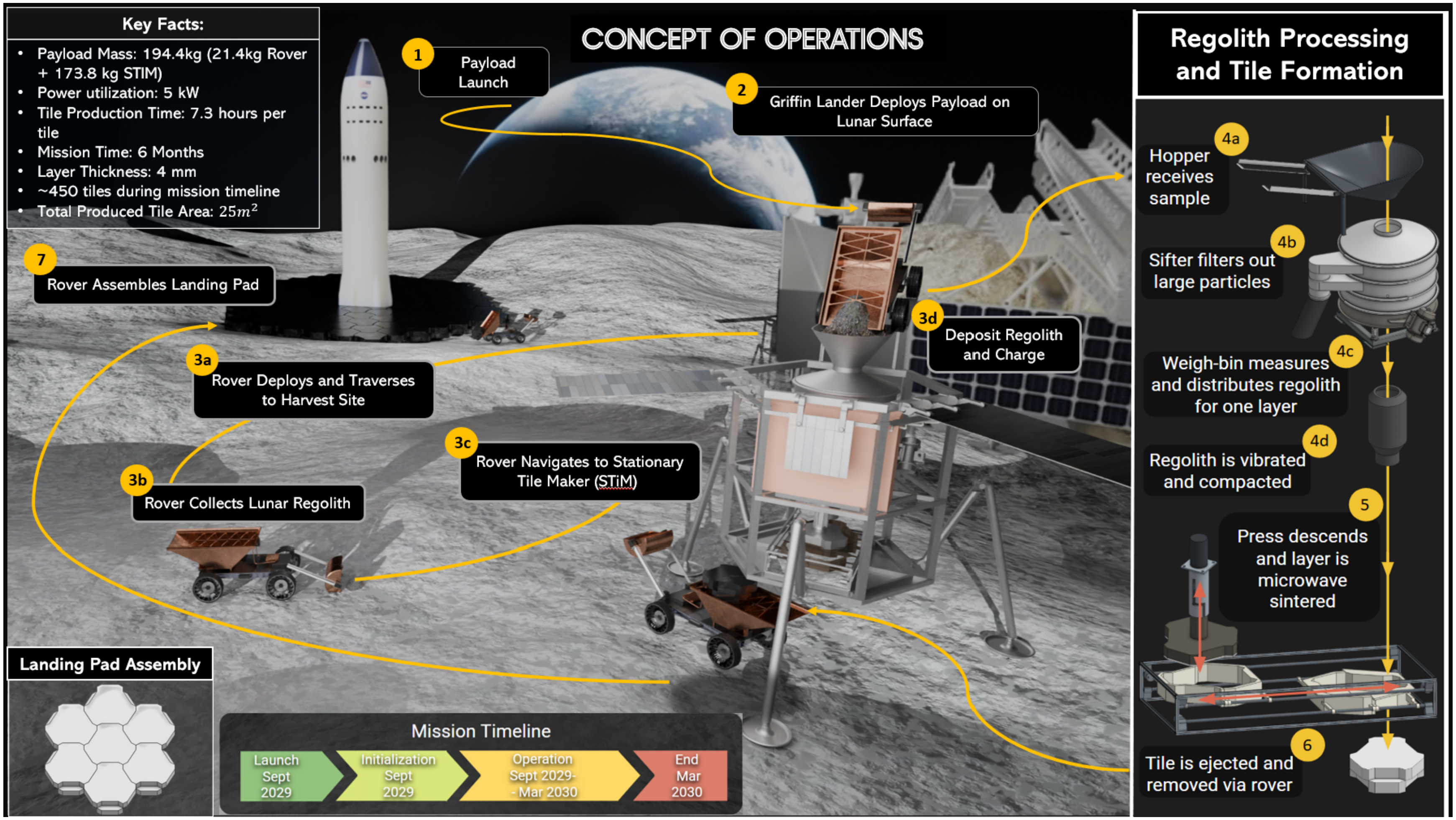


Figure 6: **Project Hestia concept of operations.** Project Hestia will deliver a two-system architecture payload to the lunar surface. Following landing, a rover will deploy from the Lander, collect lunar regolith, and return it to a stationary tile maker. Inside the STiM, the regolith is processed and sintered to fabricate tiles for lunar surface infrastructure.

3.3 Trade Studies

3.3.1 Regolith Manufacturing Trade Study

After deciding on the mission objective of constructing lunar landing pads, VT Luna needed to select a fabrication method capable of producing landing pads within the power and mass limits outlined by the RFP. The most significant design constraint was the 200 kg payload mass budget, which became the primary driving factor in fabrication method selection. After researching multiple approaches for lunar landing pad construction, the fabrication methods were narrowed down into two major categories: binder-based construction and thermal sintering.

Binder-based methods were ultimately ruled out because they require Earth-supplied materials that typically rely on sulfur, water, or other binding agents mixed with regolith. This would significantly impact system scalability, as the mass of these binders would consume a large portion of the available payload budget [10].

Following this decision, VT Luna focused on sintering as the primary tile fabrication method. Further research identified solar sintering and microwave sintering as the two most viable options. Solar sintering uses concentrated solar energy to heat regolith above its melting point, forming a rigid glass-like material. Similarly, microwave sintering uses Radio Frequency (RF) energy transmitted into the regolith, heating the material internally until it forms solid structures.

VT Luna concluded that microwave sintering was the best option for meeting the mission objectives because it allows for better control of the heating area through a mounted antenna system, can operate independently of direct sunlight, and integrates more effectively with the mission's required electrical power systems. This reduced overall system complexity compared to large optical solar concentrator systems while also improving operational flexibility [11].

Table 4: **Fabrication Method Trade Study Summary.** This trade study compares methods of tile fabrication based upon characteristics relevant to the mission requirements. This justifies the usage of microwave sintering as the missions method of tile fabrication.

Criterion	Binder-Based Construction	Solar Sintering	Microwave Sintering
Imported Material Mass	High — requires Earth-derived binders	Low — no binders required	Low — no binders required
Compatibility with 200 kg Mass Budget	Poor — consumables occupy a large payload fraction	Moderate — large optical hardware required	Good — RF hardware is relatively mass-efficient
Electrical Power Requirement	Low–Moderate	Low electrical, high thermal	Moderate (∼1–4 kW)
Dependence on Sunlight	None	High — requires direct solar illumination	None
Operational Flexibility	Limited by binder supply	Limited by sun angle and shadowing	High — operates independently of illumination
System Complexity	Moderate — mixing, handling, and curing	High — optics, pointing, and dust protection	Moderate — RF generation and antenna system
Dust Sensitivity	Low	High — optical surface contamination	Low — sealed RF aperture
Scalability	Poor — limited by binder mass	Moderate — limited by illumination conditions	High — limited primarily by available time and power
Tile Geometry Control	Moderate	Low–Moderate	High — electronically controlled heating
Overall Suitability	Low	Moderate	**High (Selected)**

After comparing the qualitative characteristics of each fabrication method, a weighted quantitative trade study was conducted to validate the selected approach. The evaluation criteria were weighted according to the mission requirements outlined in the RFP, with the greatest emphasis placed on payload mass, scalability, and produced tile strength.

As shown in Table 5, microwave sintering achieved the highest overall score due to its relatively low system mass, high scalability, and improved control over the sintering process. Although solar sintering offered lower electrical power requirements, its increased system complexity, dependence on illumination conditions, and lower tile strength resulted in a lower overall ranking.

Table 5: **Comparison of Lunar Regolith Tile Manufacturing Technologies.** This is a numerical evaluation to determine the suitability of tile manufacturing technologies based on mass, power, scalability, and tile strength.

Technology	Mass (kg)	Power (kW)	Scalability	Tile Strength	Total
Weights	0.30	0.20	0.25	0.25	1.00
Microwave Sintering	35	4.00	0.45	0.36	**0.32**
Solar Sinter	95	0.15	0.20	0.12	0.22
Chemical Binders	310	0.25	0.10	0.28	0.19
Heating Plates	1355	5.00	0.15	0.16	0.16
Pressed Brick	250	2.00	0.10	0.08	0.11

3.3.2 General Architecture Trade Study

In order to complete the mission objective outlined and use ISRU to create lunar landing tiles, several general architectures were considered during the design process. In this section of the report, VT Luna will outline these different design options as well as their strengths and weaknesses, and give a justification for the final design before diving into the details of the chosen design. These architectures will fall into the following categories: Independent Rover, Umbilical Cord, Dock to Charge, and Stationary.

Independent Rover

In the independent rover architecture, the construction rover carries its own primary power system and operates autonomously within the local mission area. Power is typically provided by body-mounted or deployable solar arrays coupled with onboard energy storage (batteries and/or capacitors). The rover is responsible for executing the full construction task cycle—regolith collection, transportation, tile fabrication support, and placement—without requiring a physical tether to the Lander during normal operation.

This architecture closely mirrors many historical planetary rover missions, such as the Opportunity and Perseverance. This heritage is important as many technological hurdles are well understood and have been overcome in previous missions, giving the design a layer of security. The rover is designed as a self-contained asset: in addition to power, it carries its own computing, communication, mobility, and construction tooling. Mission operations are typically constrained by the rover's daily energy budget, local illumination conditions, and thermal environment.

Table 6: **Strengths and Weaknesses of the Independent Rover Architecture.** This highlights the shortfalls of employing an independent rover system for this mission as the complications added by having to protect a moving system from the lunar environment and the requirement of carrying the vehicles own power supply would require excessive mass and system complexity.

Strengths	Weaknesses
High mobility and range: Operations are limited primarily by solar power availability and energy margins, allowing access to multiple regolith sources and construction sites.	Mass and complexity: The rover must carry solar arrays and large battery systems, increasing mass, volume, and potential failure points.
Heritage: This architecture aligns with many conventional lunar and Mars rover missions that rely on solar power and batteries, allowing the design to benefit from existing mission heritage.	Energy-limited operations: Construction activities and mobility are constrained by onboard power generation and storage. High-power processes such as sintering, excavation, and conditioning may require extended charging periods or may be infeasible.
Scalability: Multiple rovers can operate in parallel to increase construction throughput.	Reduced peak power capability compared to Lander-connected architectures.

Ultimately, the mass and, more pressing, power concerns drove VT Luna away from this design. It had its merits, no doubt, but this architecture was far more suited for the exploration and science missions it was initially designed for, not for energy-intensive construction processes. The idea of on-board solar panels seemed redundant when the Griffin Lander had its own power generation capabilities, far greater than deployable panels could provide. Finding a way to harness that was essential for the completion of the mission objective. This is not, however, the end of the independent rover concept, as it will make an appearance in the final design of VT Luna.

Umbilical Cord

In the Umbilical Cord architecture, the rover remains physically tethered to the Lander via a power cable during operation. The solar array on the Lander hosts the primary power generation system, and the rover draws continuous power through the umbilical. The operation is similar to a high-power tool on a long extension cord. A spool mechanism may manage the tether, or it may be laid on the lunar surface.

This architecture is well-suited to tasks that occur in a relatively compact work zone near the Lander. Since power is provided continuously, the rover can support energy-intensive tools and processes without needing to stop for a recharge.

Table 7: **Strengths and Weaknesses of the Umbilical Cord Architecture.** This highlights the shortfalls of employing an umbilical cord system for this mission as the complications added by having to protect the moving system and tether from the lunar environment would require excessive mass and system complexity.

Strengths	**Weaknesses**
High continuous available power: Access to a large Lander-based power array enables sustained operation of high-power tools, including sintering units, conditioning equipment, and excavation implements.	Limited mobility: The operational radius is constrained by tether length, while complex terrain and obstacles may further reduce the usable work area.
Simplified rover power system: The tether reduces the need for a large onboard battery, significantly reducing rover mass.	Tether management: The cable is exposed to abrasion from regolith, mechanical damage from rocks or wheels, and potential entanglement with the rover chassis, Lander legs, or other obstacles. Damage to the tether could result in mission failure.
Limited onboard energy storage requirements: Continuous external power allows the rover to operate without relying heavily on stored battery energy.	Interference with other assets: The tether concept reduces scalability and may interfere with future lunar operations as additional rovers, Landers, or crew must navigate around cables.

The umbilical cord design provides a much-needed solution to the problem of rover power draw. However, the added complexity from the tether system provides a logistical hurdle that VT Luna deemed too difficult to overcome.

Dock to Charge

The Dock-to-Charge architecture removes the need for deployable solar panels by making the primary power plant the Lander itself, and makes the rover act as a 'sortie' vehicle. The rover now carries a medium-sized battery capable of performing short trips and discrete construction tasks before returning to the Lander to dock and charge.

In this concept, operations are structured as sortie cycles. The rover departs from the Lander fully charged, travels to the work site, executes a defined construction sequence (i.e., gather regolith, tile placement, sintering), and then returns to the Lander to recharge.

While docked, the rover can take advantage of higher continuous power from the Lander's solar array for higher power draw tasks such as sintering.

Table 8: **Strengths and Weaknesses of the Dock-to-Charge Rover Architecture.** This highlights the reasoning behind not choosing a dock charging rover architecture as a large and massive battery would be required and mission duration would have to accommodate repeated traversal to charging and target locations.

Strengths	Weaknesses
High power availability: Access to the Lander's large solar array enables energy-intensive activities when the rover battery is fully charged.	Limited range: The operational radius is constrained by the need to maintain sufficient energy reserves to return to the Lander, limiting construction to short sortie-style operations.
No tether dependency: Eliminates the risk of dragging a power tether over rough terrain and removes failure modes associated with tethered systems.	Charging downtime: While docked and charging, the rover is unable to perform construction tasks.
Scalability: Multiple rovers can operate in parallel while others recharge, improving productivity and reducing operational downtime.	Docking complexity: Requires a robust mechanical and electrical interface with dust-tolerant connectors, alignment tolerances, and guidance systems.

Stationary Tile Maker

In the stationary tile maker architecture, the "rover" is no longer a rover and becomes a fixed manufacturing plant attached to the Lander. It possesses no mobility capabilities and is instead optimized for regolith conditioning and tile fabrication. Power and data are provided through a short harness from the Lander's primary power system.

Removes navigational capabilities and mechanical degrees of freedom, and focuses solely on producing structurally sound tiles with high power availability

Table 9: **Strengths and Weaknesses of the Stationary Tile Maker Architecture.** This highlights the reasoning behind to employing a stationary tile maker architecture as regolith harvesting would be restricted to a very local and potentially contaminated landing site area.

Strengths	Weaknesses
Simplified mechanical design: Eliminating mobility capabilities allows the design to focus entirely on the manufacturing process.	Limited access to regolith: A stationary system cannot relocate to new regolith sources and relies on excavation from nearby deposits. If local regolith is unusable, mission objectives may be compromised.
High power availability and duty cycle: Direct connection to the Lander's solar array enables near-continuous operation and higher tile throughput compared to mobile platforms.	Local congestion near the Lander: Concentrating manufacturing operations near the landing site may create a cluttered operational environment.
Scalability: Multiple stationary tile makers can operate in parallel without interfering with one another.	Limited operational flexibility compared to mobile manufacturing systems.

This design greatly simplifies the completion of mission objectives by focusing on tile fabrication and benefits greatly from continuous access to high-power. However, without any mobility capabilities, serious concerns arise.

STiM and Rover Compromise

Ultimately, the design constraints of the above-mentioned considered concepts led VT Luna into making the payload a combination of several aspects from these concepts. The final system architecture was decided to be a two-machine design that works independently but in conjunction with one another. The stationary tile maker design solves many complexity issues and allows for near constant access to high power, making it perfectly suited for a manufacturing station. The major drawback that could not be overlooked, however, is the lack of mobility and limited access to regolith. This concern was considerable. What if, upon landing, the regolith in the direct vicinity of the Griffin Lander was unusable for one reason or another? This would result in a mission failure and was considered an unacceptable risk to incur for the mission. There needed to be a way for the system to have access to regolith that was deemed acceptable for sintering standards. To remedy this, VT Luna came up with the STiM and Rover design, which uses both the high power and manufacturing qualities of the stationary tile maker design as well as the long-range exploratory capabilities of the independent rover system. The rover would be deployed independently in search of optimal regolith, and upon returning and depositing the regolith, would be able to charge while the stationary tile maker(STiM) underwent the conditioning and sintering cycle necessary to complete the mission. This unity of two system offers mobility and freedom without compromising any of the necessary steps to manufacture structurally and thermally sound tiles efficiently.

3.3.3 Landing Site Selection

Selecting a landing site requires careful consideration of several factors such as sunlight, safety, communications access, thermal conditions, and surface smoothness. The ideal site is one that offers continuous solar illumination, safe and accessible terrain, and proximity to valuable resources such as water ice. Landing pads primarily require stable, flat, and accessible terrain with continuous sunlight for power generation.

For the Project Hestia mission, five primary criteria were used in evaluating potential landing locations:

1. Maximum solar illumination: High levels of sunlight are required to support continuous power generation. Polar regions are preferred due to their extended periods of illumination, with some areas receiving up to 85–90% annual sunlight [12].

2. Terrain Safety and Accessibility: The landing site must be relatively flat and free of hazards such as large rocks, craters, or steep slopes to ensure safe landing and rover mobility.

3. Reliable communication access with Earth: A direct line-of-sight to Earth is necessary for reliable data transmission and command operations.

4. Thermal Stability: Regions with consistent solar exposure experience reduced temperature extremes, improving system reliability.

5. Regolith Quality: The surface material must be sufficiently thick and stable to support sintering processes and structural integrity.

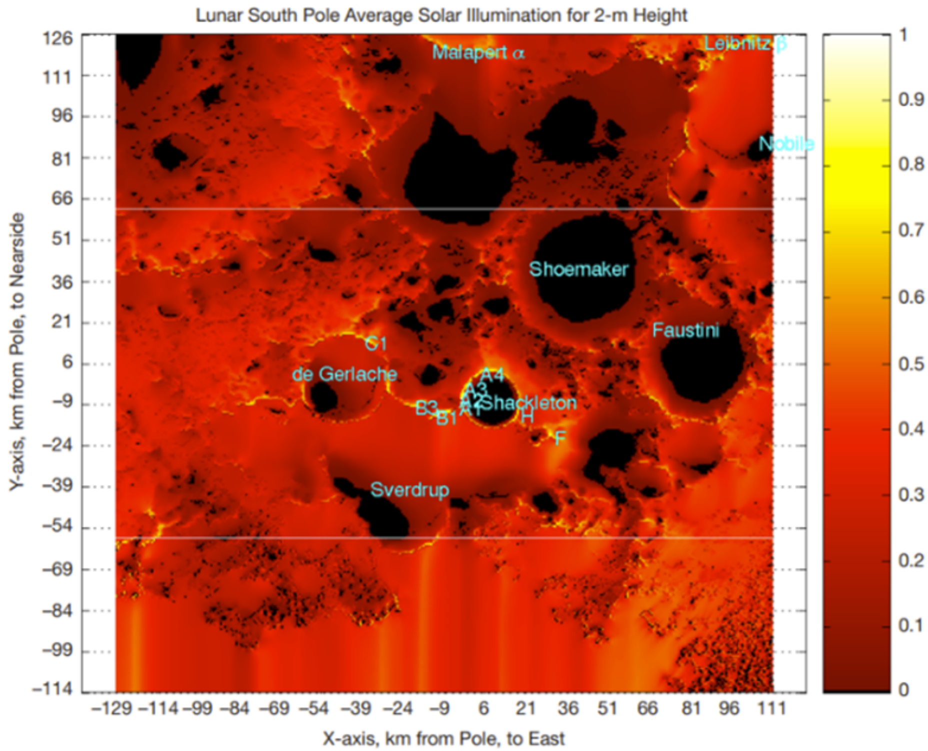


Figure 7: **Lunar south pole yearly solar illumination.** The figure shows the average annual illumination near the lunar South Pole, highlighting low-illumination crater interiors and higher-illumination regions along crater rims. This informs the landing site selection for the mission as high lighting regions enable continuous solar power [12].

In the selection of a lunar landing and construction site, sunlight remains one of the most important factors. Illumination varies dramatically across the lunar surface, making the poles particularly valuable for solar energy access. Figure 7 shows the average yearly solar illumination near the lunar south pole at a height of 2 m above the surface. The brightest regions are concentrated around elevated crater rims and ridges, including areas near Shackleton Crater, de Gerlache Crater, and the Connecting Ridge. These areas can remain illuminated for a large portion of the lunar year, making them attractive candidates for solar-powered missions [12].

According to lunar scientists, the Moon's south pole represents a completely different environment than the equatorial sites visited during the Apollo mission. This new location offers longer illumination, colder shadowed regions, and far greater terrain variability. Sunlight at the Moon's poles is unique, as some areas have near-permanent sunlight, while others experience long periods of darkness. These luminous regions, often called peaks of eternal light, provide nearly continuous exposure to sunlight, making power generation easier and reducing the need for large batteries. They also help maintain more stable temperatures, which is critical for both equipment and human operations.[13]

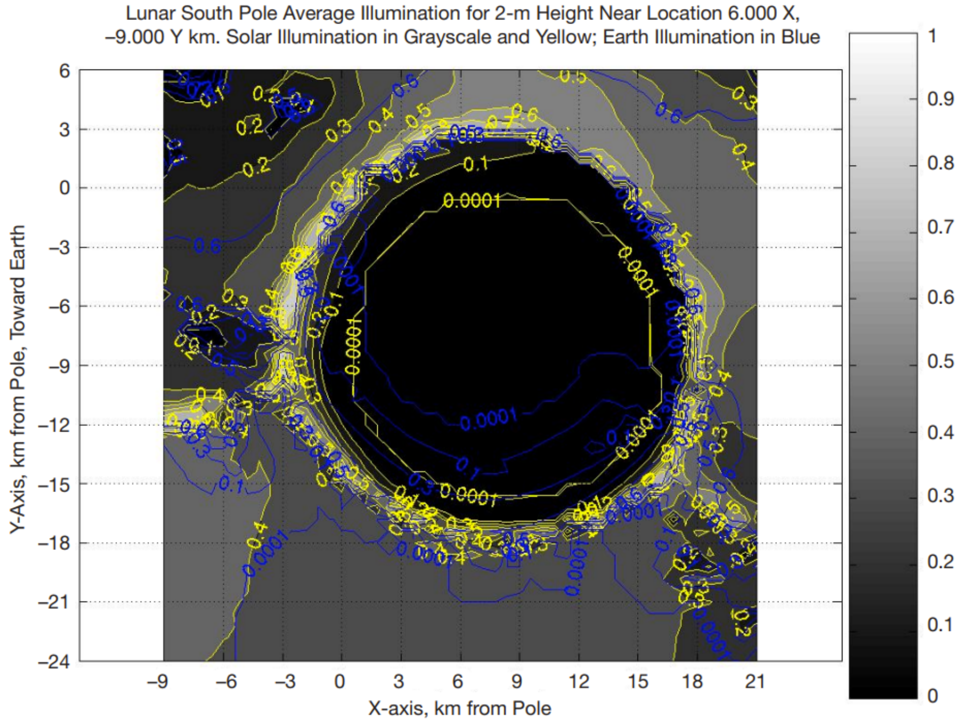


Figure 8: **Shackleton crater yearly average illumination.** The figure shows the yearly average illumination around Shackleton Crater, illustrating the contrast between highly illuminated crater rims and permanently shadowed crater interiors. This informs landing site selection with mission operations best suited to high illumination enviroments [12].

Figure 8 further illustrates the illumination variation around Shackleton Crater. The crater rim receives sunlight more than 80% of the year, while the interior remains in permanent shadow. However, despite its exceptional illumination, landing on Shackleton's rim or adjacent ridges is operationally very risky. The slopes around the crater are steep and uneven, with elevations varying from -3 km to +1.5 km relative to the local mean. So, even small errors in descent trajectory could cause a Lander to crash on a slope or in shadowed terrain.[12]

In August 2022, NASA announced thirteen candidate landing regions near the lunar south pole for the Artemis III mission, the first mission to return American astronauts to the lunar surface since Apollo 17 [14]. These initial regions are shown in Figure 9. In October 2024, NASA refined its analysis and narrowed the list to nine potential regions, shown in Figure 10, reflecting more detailed studies of illumination, communications visibility, and terrain accessibility.[15].

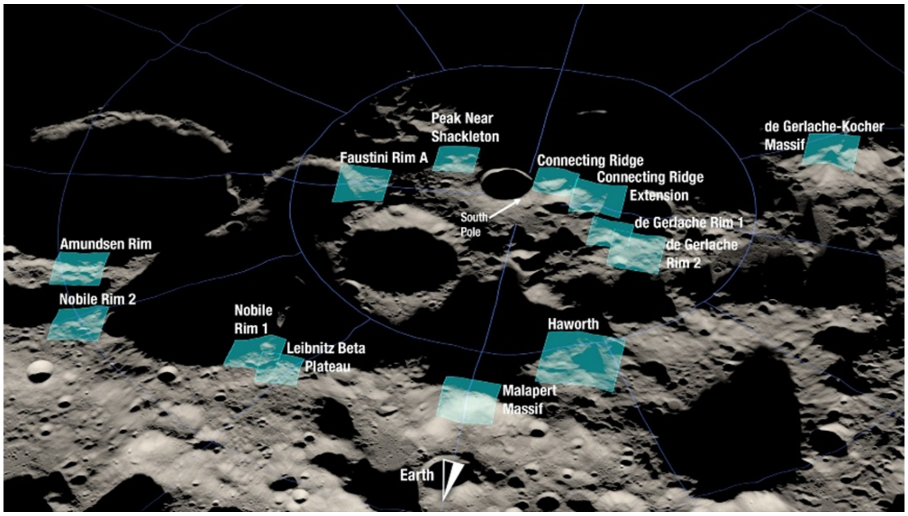


Figure 9: **NASA's 13 candidate landing regions for Artemis III.** The original set of candidate landing regions near the lunar South Pole identified by NASA for the Artemis III mission. These regions provided the initial basis for evaluating potential landing sites for Project Hestia.[14].

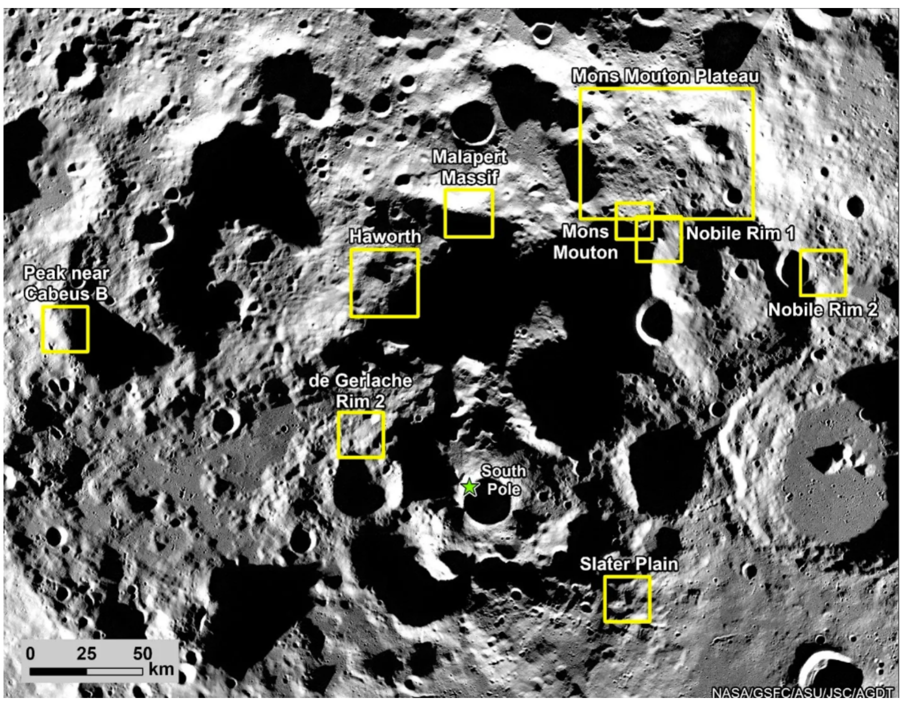


Figure 10: **NASA's redefined candidate landing regions for Artemis III** NASA's refined set of nine candidate landing regions for Artemis III. The updated regions were used to narrow the landing site trade space and compare locations based on mission-relevant factors. [15].

Based on the candidate landing regions identified near the lunar south pole, Mons Mouton Plateau, Peak Near Shackleton, and De Gerlache Rim 2 were selected for the final trade study. These sites were chosen because they represent relevant options for Project Hestia when considering illumination, terrain accessibility, proximity to permanently shadowed regions, and overall mission feasibility. The remaining candidate regions were not included in the final trade study because they were less favorable for the specific needs of this mission or did not provide the same combination of power availability, operational accessibility, and construction potential [14, 15, 16, 17].

The selected sites were compared using the criteria most relevant to Project Hestia: terrain slope, proximity to permanently shadowed regions, illumination, regolith composition, and Earth visibility. The results of this comparison are shown in Table 10. Raw literature values are shown where available, while qualitative or mission-specific factors were converted into normalized scores before calculating the final weighted result..

Table 10: **Landing Site Trade Study Results.** This shows the justification behind the selection of the Mons Mouton Plateau as the location has a regolith composition suited to microwave sintering and greater Earth visibility than other potential landing sites.

Candidate Site	**Terrain Slope (deg)**	**Proximity to PSRs**	**Illumination (%)**	**Regolith Comp. Score**	**Earth Vis. (%)**	**Result**
Weights	0.20	0.05	0.25	0.40	0.10	1.00
Mons Mouton Plateau	7.4 ± 2.3	0.5	75	0.7	74 ± 19	**0.44**
Peak Near Shackleton	~ 7.5	0.4	85.5	0.6	~ 50	0.28
De Gerlache Rims 2	~ 8	0.3	88	0.6	~ 50	0.28

As shown in Table 10, Mons Mouton Plateau received the highest overall score and was selected as the preferred landing site for Project Hestia. This result reflects the site's strong balance across the major mission criteria, including terrain accessibility, solar illumination, Earth visibility, proximity to permanently shadowed regions, and most importantly regolith composition. For this mission, the most important site is not necessarily the one with the highest individual value in a single category, but the one that provides the most reliable overall environment for landing, rover mobility, regolith collection, and tile construction [16, 18, 19]. Mons Mouton Plateau is particularly well suited for Project Hestia because it provides a broad plateau environment with useful solar illumination and favorable operational accessibility. Its terrain characteristics support safer Lander touchdown and more predictable rover operations, while its regional context near the lunar south pole preserves access to scientifically and operationally valuable terrain. These factors make the site a practical location for demonstrating early lunar infrastructure construction [16, 18, 15].

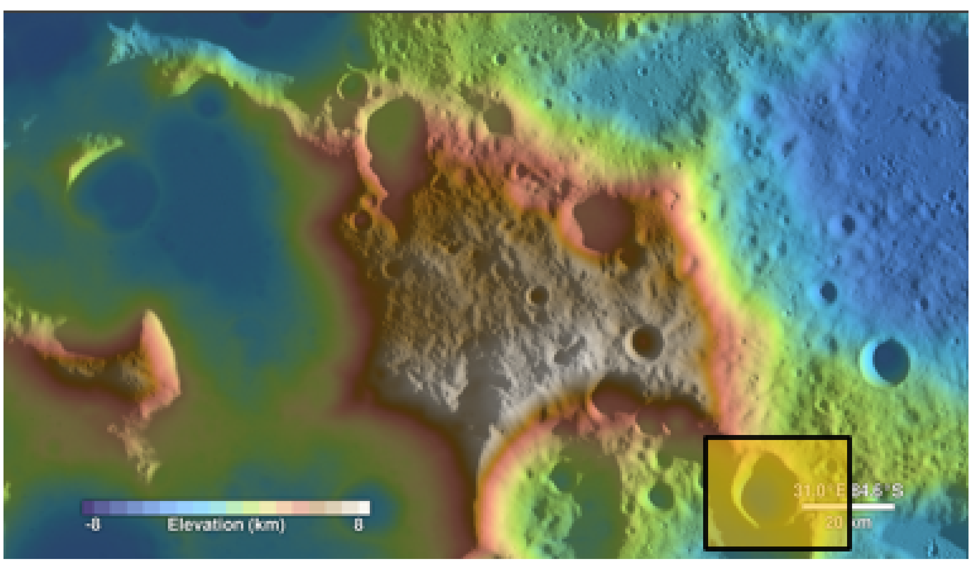


Figure 11: **Mons Mouton Plateau topographic view.** Overhead view of the Mons Mouton Plateau with color-coded elevation. The image highlights the relatively broad plateau region and surrounding topographic variation, which were considered when evaluating the site for Lander touchdown, rover mobility, and lunar landing pad construction.[18]

In short, the best sites for solar power are those near the poles, at higher elevations, where sunlight is almost constant but still accessible and safe for landing. Although the Peak Near Shackleton region remains scientifically valuable due to its illumination and proximity to potential water-ice deposits, its rugged topography makes it unsuitable for initial construction efforts. By contrast, selecting the Mons Mouton Plateau provides a balanced and operationally realistic solution for this mission: it maximizes landing safety, supports efficient construction with great regolith composition, and ensures reliable power and communications.[16, 18, 15]

4 Core Technology

4.1 Lunar Regolith

The Moon's topography is very complex, with the topological features varying from -8 km to 10 km, where the altitude is measured from the elevation of the reference ellipsoid. There are four main geographical mare characters to consider, being Basalt plains, Volcanic origin, High Ti, and Low Ti mare basalts. As the lunar topography varies, so does the chemical composition of the lunar regolith.

Table 11: **Lunar Regolith Compound Concentration.** This table shows the variation in and composition of lunar regolith across the lunar surface [20].

Chemical Compound	Concentration (%)
Silicon Dioxide (SiO_2)	42–48
Titanium Dioxide (TiO_2)	1–7
Aluminum Oxide (Al_2O_3)	12–27
Iron Oxide (FeO)	4–18
Magnesium Oxide (MgO)	4–11
Calcium Oxide (CaO)	10–17
Sodium Oxide (Na_2O)	0.4–0.7
Potassium Oxide (K_2O)	0.1–0.6
Manganous Oxide (MnO)	0.1–0.2
Chromium Oxide (Cr_2O_3)	0.2–0.4

4.1.1 Lunar Topography

The different mares can be found by their respective altitudes and can be easily distinguished using a topological map. Topological maps color-code the altitudes and distinctly show different terrain features.

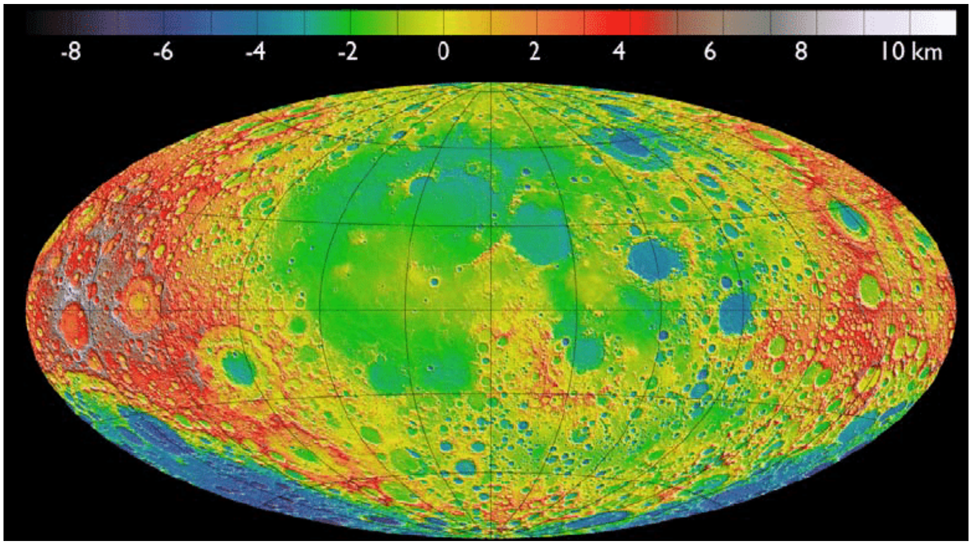


Figure 12: **Topographical map of the Moon.** Topographical map of the Moon detailing average altitude deviation. From this areas of highland, basalt plains, and mares where potential areas of interest are located can be visualized [19].

The areas of interest are Highland, Basalts plains, High Ti mare, and Low Ti mares. Basalt plains are seen in Figure 12 in the flat blue areas and are about -6 km in altitude. Highlands are seen in the white side of the spectrum and mostly seen at the south pole and are 10 km in altitude, and the high Ti and low Ti mares range from -4 to -1 km in altitude and are seen in the light blue and green areas.

The basalt plains were formed by impacts that left deep impressions in the lunar mantle, melting it in the process. These impacts created extensive, flat, dark plains like the Tranquillitatis and Serenitatis, and cover roughly 16% of the lunar surface and have a mantle thickness from 30-100 km. These areas are high in Iron Oxide (FeO) and titanium oxide. Because of the thin crust, there is significant surface variation due to exposed magma tubes and flow fronts. Due to the way they are formed, the consistency of the lunar regolith varies widely depending on the impact crater and how far away it is.

High-ti mare is exceptionally high in titanium dioxide and iron oxide. Varying in altitudes from -4 to -1 km. But low in Silicon Dioxide in comparison to Low Ti. High-Ti maria regolith also has excellent properties, being extremely absorbent of microwaves and extremely uniform, sand-like regolith.

Low Ti mare are formed from thick lava sheets and, as a result, are extremely flat without any large terrain features. They are one of the more stable areas of the Moon and are low in titanium content. Low Ti mare is most comparable to Hawaiian basalt, with large stable rocks and large granular size, and less abundant loose regolith.

4.1.2 Regolith Chemical Makeup

Table 12: **Lunar Regolith Compound Concentrations by Mare.** This table shows the variation in regolith composition by biome with High-Ti Mares being ideal due to the large presence of Titanium Dioxide which assists in microwave sintering [20].

Chemical Compound	Highland	High-Ti Mare	Low-Ti Mare
Silicon Dioxide (SiO_2)	44.9	41.0	45.1
Titanium Dioxide (TiO_2)	0.5	8.5	2.9
Aluminum Oxide (Al_2O_3)	25.1	12.4	13.1
Iron(II) Oxide (FeO)	6.2	16.6	17.0
Magnesium Oxide (MgO)	7.5	8.9	9.9
Calcium Oxide (CaO)	14.9	11.4	11.7
Other	0.9	1.2	0.3

The chemical makeup of the regolith will affect the melting point. Because the majority of the regolith is metal oxides, they absorb energy very well, having high dielectric loss, and will lose heat much faster than other materials like silica or aluminum. Because the exact mixture depends on the chosen landing location, the exact temperature and method of sintering will be affected [20]. The particle density will also affect how well the sintering process works. Studies using regolith sintering show that an average grain size of $< 600\,\mu$m work best with the sintering process. This grain size allows the density to be adequate, so the sintered material will flow into gaps, forming a solid part and improving layer quality. The sintering process melts the regolith, and it forms small drops of molten regolith. If the grain size is too big, there will be gaps between the formed drops, and it will fail at making one cohesive tile.

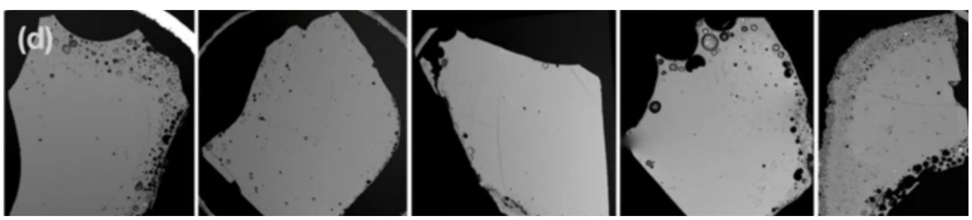
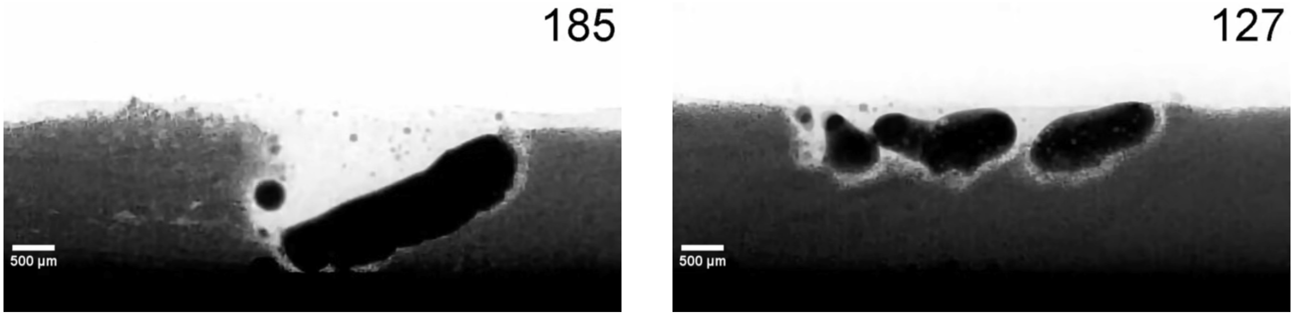


Figure 13: **Void holes in sintering.** Images depicting void holes in sintered products. Such deformations greatly reduce the structural integrity of the sintered products and as such precautions must be made to mitigate their formation [21].

4.1.3 Regolith Sintering

Heat shrinkage is an issue with lunar regolith [20]. When sintering the regolith, there will be mass loss and volume shrinkage. The shrinkage varies due to the material makeup, time under heat, boiling points, and sublimation temperatures. During sintering, boiling points are essential because if the regolith is sintered at too high a temperature, the tile would become porous due to off-gassing, weakening the tile. That is why it is important to keep the total temperature under the sublimation temperature. Sintering at 1200 °C will lead to 10.1% of the mass being lost, and having the total volume at 1150°C for over 96 hours, 50% of the mass will be lost [20]. In addition, overheating the regolith will break down the chemical composition of the regolith [22]. When sintering above 1000°C, pyrolysis and thermal decomposition of silicate compounds can occur. These processes release gases that form pores seen in Figure 13 within the material, creating voids that weaken the final structure. To mitigate this, a multi-step temperature profile is best, tempering the material and allowing for the pores to fill themselves in [22].



(a) Particle sintering shrinkage. (b) Particles bubbling and not connecting.

Figure 14: **Sintering failure from shrinkage and bubbling.** Visual representation of sintering particles where shrinkage and bubbling have occurred respectively. These undesired processes result in sintered products not maintaining desired shape and causes points of structural weakness and potentiates cavitation [23].

Because of the chemical makeup of the regolith, the base will be silicon dioxide. Silicon dioxide is not very reactive to microwave sintering, so the other compounds are important to increase the efficiency of the sintering process and absorb the heat to speed up the heating. Of the chemical compounds, Iron Oxide and Titanium Dioxide are the most important because they are the most reactive to microwaves.

The lower the percentages of other contaminants, the more homogeneous a tile will be. The chosen landing site has the highest percentage of Titanium Dioxide and a high Iron oxide concentration. Boiling points are important because if they were to be sintered at too high a temperature, the tile would become porous due to off-gassing, and weaken the tile. That is why it is important to keep the total temperature under the sublimation temperature. Sintering at 1200 °C will lead to 10.1% of the mass being lost, and having the total volume at 1150 °C for over 96 hours, 50% of the mass will be lost.

Figure 14 shows the challenges of the sintering process. When the regolith is heated up, the overall volume will decrease due to the increase in density between the regolith in particle form and in a liquid form. To mitigate the effect of this, the sinter process will make it layer by layer. This will allow for empty volume from shrinkage to be filled in and any gaps from an uneven layer seen in Figure 14 to be filled in. To further mitigate any issues with a porous tile, preventing the bubbling issue from forming voids in the completed tile is critical. The bubbling is due to the surface tension of the liquid regolith, which causes the regolith to attract to each other.

Table 13: **Boiling Points and Vacuum Sublimation Temperatures of Lunar-Relevant Oxides.** This table shows the temperatures at which regolith compounds will boil or sublimate providing insight into the heating ranges that must be employed to sublimate unwanted compounds and maintain the ideal concentrations of desirable compounds [24].

Chemical Compound	**Boiling Point @ 1 atm (K)**	**Vacuum Sublimation Temp (K)**
Silicon Dioxide (SiO_2)	3223	2073
Titanium Dioxide (TiO_2)	3245	2173
Aluminum Oxide (Al_2O_3)	3250	2073
Iron Oxide (FeO)	3273	1900
Magnesium Oxide (MgO)	3873	2173
Calcium Oxide (CaO)	3123	1723
Sodium Oxide (Na_2O)	2223	973
Potassium Oxide (K_2O)	1273	572
Manganous Oxide (MnO)	3073	1773
Chromium(III) Oxide (Cr_2O_3)	4273	2273

Lunar regolith, being mostly silicon dioxide, provides a great opportunity for future lunar infrastructure.

Silicon dioxide has high compressive strength and is inert to most chemicals. It is very stable under thermal loads with low thermal expansion. Silicon dioxide can be found in uses like concrete, silica reinforcement fibers, and sandstone. Concrete is also used to make landing and launch pad structures and is perfect for the use case of landing pad tiles. The issue with lunar regolith is that other contaminants will weaken the tile, so the smaller the other contaminants' percentages, the stronger and more homogeneous the tile.

Because of the chemical makeup of the regolith, the base will be silicon dioxide [20]. Silicon Dioxide is not very reactive to direct sintering methods or any magnetic fields, so the other compounds are important to increase the efficiency of the sintering process and absorb the heat to speed up the heating. Of the chemical compounds, Iron Oxide and Titanium Dioxide are the most important because they are the most reactive to different forms of sintering methods. The chosen landing site has the highest percentage of Titanium Dioxide and a high Iron oxide concentration. This is great for methods of sintering, like microwave sintering, because it allows for a greater absorption of the energy that is put into it. The high Iron content helps with polarization for the dielectric response, making the regolith in a sintered state more attracted to each other, helping with creating a uniform layer height [25].

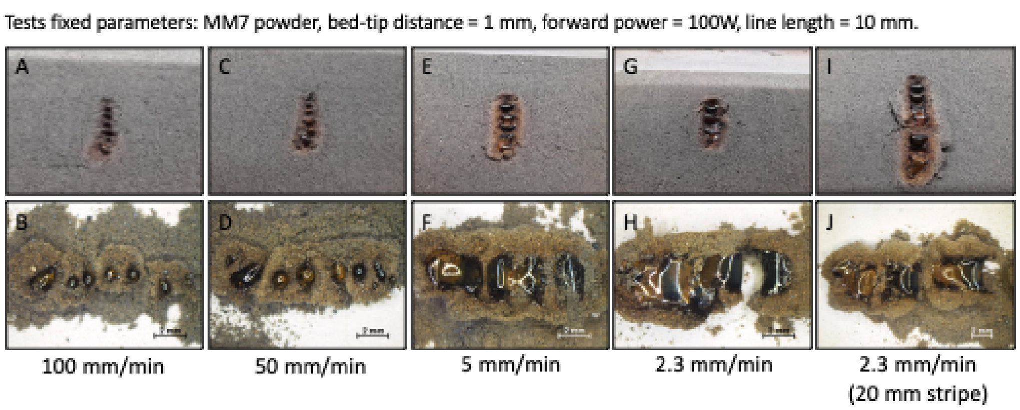


Figure 15: **Effect of sintering speed of lunar regolith simulant under microwave heating.** This depicts the necessary speed of sintering to avoid bubbles and separation to result in a continuous sintered result [26].

Figure 15 shows the sweeping speeds used in a microwave sintering experiment performed by the European Space Agency (ESA) [27]. At 1 kW of power, lunar regolith will be sintered, and the bubbling issue can be solved. It also shows the need to sinter in layers and the shrinking due to the disruption around the sintered material.

Mission targeted landing sites will need to be selected for the best lunar regolith for the sintering case, lying in the High Ti mare zone. It provides the highest concentrations of both Iron oxide and Titanium Dioxide to increase the effectiveness of the sintering process and lower the levels of other contaminants in the tile. The Iron will allow for an increase in dielectric response of the regolith and allow for the liquid sintered regolith to react with the magnetic field and attract to each other [25]. The High Ti mare has the best loose regolith to collect, leading to an excess to sort and generate an average grain size of $< 600\,\mu$m. With a bulk density that would allow for a sintered product without any air voids to weaken the tile. To best sinter the tile, a temperature profile is designed to temper the regolith to allow for gas to escape.

4.2 Microwave Sintering Analysis

4.2.1 Justification

Understanding the required power of all the subsystems is an integral part of ensuring the feasibility of this mission and its architecture. The most power-intensive and complex subsystem is the microwave sinterer. The sinterer must be able to heat 6,200 cm^3 of regolith to 1,200 °C. The power required to heat the entire tile simultaneously would exceed the Griffin Lander's power output, rendering it impossible. To enable the creation of a tile, sintering will be done in thin layers. With this approach, the instantaneous power draw remains low enough to allow the power to be sourced exclusively from the Griffin Lander.

Table 14: **Required Nomenclature for Thermal Model Equations.** This table provides the nomenclature used within the equations that were used to create the thermal model that proved the feasibility of microwave sintering in the missions application and the ideal tile layer height [28, 29, 30, 31]

Variable	Symbol	Unit	Type	Dependencies	Value
Power Density	P	$\frac{W}{m^3}$	Found	$K, f, E, k', \tan(\delta)$	N/A
Power Draw	P_s	W	Found	P, η, V	4000
Energy Required	J	kWh	Found	P_s, t	29.16
Heating Rate	ΔT	°C/s	Found	P, ρ, C_P	N/A
Heating Time	t	s	Found	$T_0, T_t, \Delta T$	26240
Instant Temperature	T	°C	Found	$T_0, \Delta T$	N/A
Initial Temperature	T_0	°C	Defined	N/A	0
Target Temperature	T_t	°C	Defined	N/A	1200
Dielectric Constant	k'	N/A	Material Property	T, f	N/A
Loss Tangent	$\tan(\delta)$	N/A	Material Property	T, f	N/A
Density	ρ	$\frac{kg}{m^3}$	Material Property	T	3000
Specific Heat Capacity	C_P	$\frac{J}{kg \cdot K}$	Material Property	T	700
Tile Area	A_t	m^2	Tile Property	N/A	0.058
Layer Thickness	h_L	m	Tile Property	N/A	0.0044
Layer Count	n_L	N/A	Tile Property	h	24
Tile Thickness	h	m	Tile Property	h_L, n_L	0.106
Tile Volume	V	m^3	Tile Property	A, h	0.0062
Microwave Frequency	f	Hz	Hardware Property	N/A	915×10^6
Electric Field Strength	E	$\frac{V}{m}$	Hardware Property	A_a, P_s	22404.96
System Efficiency	η	N/A	Hardware Property	N/A	0.25
Free Space Permittivity	ε_0	$\frac{F}{m}$	Constant	N/A	8.854×10^{-12}
Equation Constant	K	$\frac{F}{m}$	Constant	ε_0	N/A

Table 15: **Core Modeling Equations Relating the Sintering Process to Lunar Regolith Material Properties.** This table provides the equations that were used to construct the thermal models utilized by this project [29, 28]

Eq.	Expression
(1)	$P = KfE^2k' \tan(\delta)$
(2)	$\Delta T = \frac{P}{\rho C_P}$
(3)	$P_s = \frac{PV}{\eta}$
(4)	$t = \frac{T_t - T_0}{\Delta T}$
(5)	$J = P_s t$
(6)	$K = K' - jK'' = K_\infty + \frac{K_0 - K_\infty}{1 + (j\omega\tau)^{1-\alpha}}$

4.2.2 Method

To model the power required and the time required to sinter a single tile, several key inputs must be defined, including the tile geometry, microwave hardware properties, and regolith material properties. The power absorbed by the regolith is given by Equation (1), and the heating rate is given by Equation (2)[29]. These two equations form the foundation of the model, as they directly relate the hardware properties of the microwave system to the material properties of the regolith. Using these equations, the model can be expanded to determine the instantaneous power draw, the time required to sinter a layer or full tile, and the total energy required, using equations (3),(4), and (5), respectively.

The modeling process becomes more complex when the temperature dependence of regolith material properties is considered. While most variables in the core equations are either constant or can be controlled, several regolith properties depend on temperature, including density, specific heat capacity, dielectric constant, and loss tangent. For the purposes of this model, the temperature dependence of density and specific heat capacity is considered negligible, as these properties change by relatively small amounts over the temperature range of interest. In contrast, the dielectric constant and loss tangent are strongly temperature dependent, and accurately modeling these relationships is critical to producing a useful model[29].

The temperature dependence of the dielectric constant and loss tangent is specific to lunar regolith and must be obtained from experimental data. A NASA study conducted in 1973 measured these properties as functions of temperature and frequency using Apollo lunar samples[28]. This study provides the functional form used to describe these relationships, specifically the Cole-Cole frequency distribution model, shown as equation (6), but does not provide closed-form equations[28]. Instead, the relationships are presented graphically. To obtain usable multi-variable functions, data points were extracted from these figures using the free online tool WebPlotDigitizer and exported as `.csv` files[32]. These data were then processed using a custom MATLAB script to perform curve fitting and obtain a total of sixteen coefficients: eight for the dielectric constant function and eight for the loss tangent function. Comparison of the fitted functions to the extracted data showed a relative accuracy of approximately 90%.

All remaining material properties, as well as the microwave operating frequency, were obtained from experimental literature and treated as constant values in the model/[30][31]. Any uncertainties in the model were intentionally treated conservatively to ensure that predicted power, energy, and time requirements were not underestimated. The electric field strength was treated as a function of the microwave applicator area, defined as the area being sintered simultaneously, and this value was optimized later in the modeling process.

Once a constant value or temperature-dependent function was defined for each variable, the rate of temperature change became a function of temperature alone, resulting in a differential equation. To numerically solve this relationship while allowing temperature-dependent properties to influence the heating rate, a time-stepping loop was implemented in the MATLAB script. The model calculates the temperature increase and power consumed at each one-second interval and updates the current temperature and cumulative energy until the target temperature is reached. Because the available experimental dielectric data only extends to 500 °C, a linear interpolation approach was used beyond this temperature, treating the final values obtained from the numerical loop as constants[28]. The initial iteration of the model produced estimates for peak power draw, total energy required, and total sintering time.

A second iteration of the code was then created to optimize the total time to sinter one tile. This was done by treating power draw as a known value and implementing the relationship between layer volume and electric field strength. This allowed for the creation of a layer thickness vector spanning the feasible range in integer increments as layer area was held constant. Due to the electric field strength being dependent on the simultaneous sintering volume, it is recalculated for each thickness value in the vector. The model was run for each thickness and then plotted each value against its resulting total time. This provides both an estimate for the time it takes to create a tile and the layer thickness to meet those requirements. After debugging the logic, adding the delay time between layers, and adding a heating model to ensure dissipation and conduction between individual layers and the mold were properly modeled, the following results were obtained.

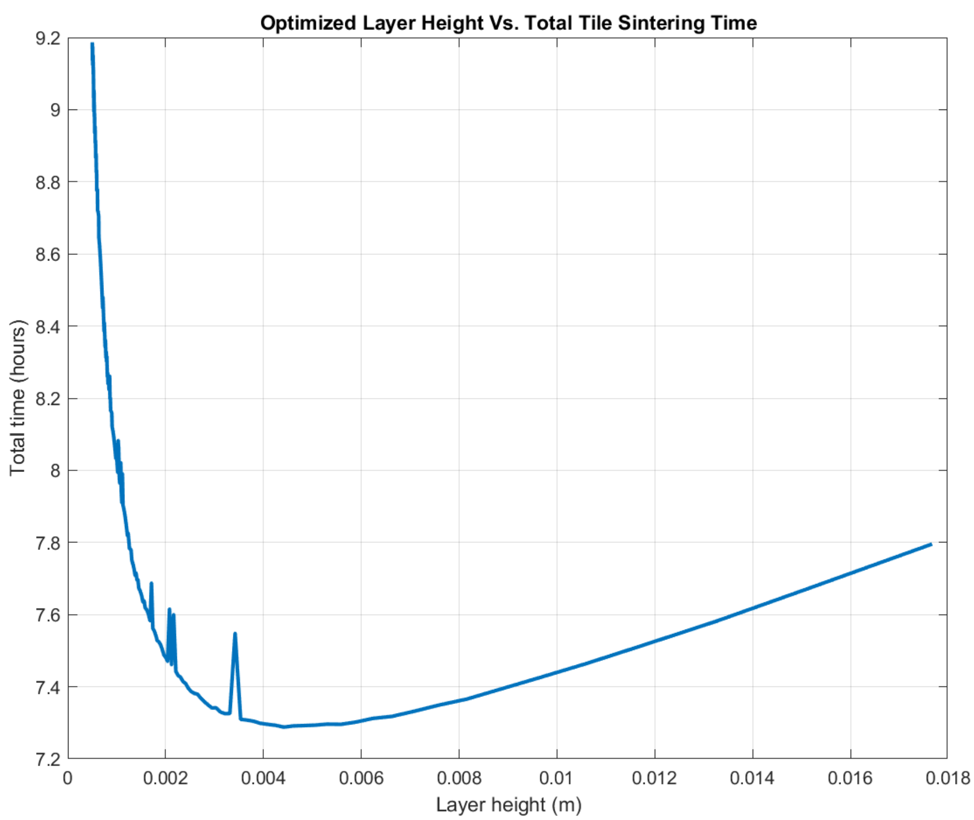


Figure 16: **Relationship of layer height and tile production time** Graph produced by the analysis model relating the total time to sinter one tile against layer height. The global minimum is the optimal layer height that allows for a minimized time of tile production.

4.2.3 Results

For a tile with a face area of 0.058 m^2 and a depth of 0.106 m. The optimal layer thickness is 4.42 mm as shown in Figure 16. The figure depicts a curve with a global minimum at 0.0044 m. The optimal thickness results in a sintering time of 18 minutes per layer. When extrapolated across the 24 layers, the total time to create one tile is 7.29 hours. These estimates are controlling for power draw at 4 kW, which is about 80% of the available system power. The relationship between power draw and time elapsed is linearly inversely proportional, meaning that if double the power were to be allotted to the microwave, the time estimate would be halved. Importantly, these estimates provide both a reasonable timeline for the sintering process and useful hardware requirements. Given that the sintering is the least proven aspect of the system architecture, these estimations suggest not only that the system is possible, but it is also well within the weight, energy, and time constraints of this mission. The model also helps to illustrate the scalability of the design, with access to power grids or multiple systems drastically reducing the timeline of the construction process.

4.2.4 Effects of Regolith Composition on Heating

Because the chemical makeup of the regolith varies widely, the exact time needed under sintering needs to be calculated. Each individual chemical changes the specific heat , Cp, of the regolith sample.

To accurately model and determine how long the lunar regolith will take to sinter the Dielectric constant and Cp will need to be found. A Materials Dielectric Constant, ϵ_r, describes the materials ability to store heat as well as its electrical energy stores when exposed to a strong electromagnetic field. This value is characterized by a complex permittivity, $\epsilon_r = \epsilon_r' - j\epsilon_r''$. where ϵ_r' is the real part, measuring how much of the microwave energy is captured and stored and ϵ_r" is the loss factor, how much is converted into heat. By taking the tangent of $\arctan \epsilon_r'/\epsilon_r$", a new value is found known as the loss tangent, $\tan\delta$, which represents the microwave heating efficiency.

Table 16: **Lunar Regolith Oxide Microwave Properties at 915 MHz.** This table details the properties of the prevalent compounds found in lunar regolith that are relevant to the microwave sintering process and these values were used within the thermal model [33, 34, 35, 36, 37, 38, 39, 40]

Compound	**Formula**	$\tan\delta$ **@ 915 MHz**	**Absorption Class**	C_p **(J/kg·K)**
Silicon Dioxide	SiO_2	0.0003	Very Low	730
Titanium Dioxide	TiO_2	0.025	High	686
Aluminum Oxide	Al_2O_3	0.0005	Very Low	880
Iron Oxide	FeO	0.200	Very High	670
Magnesium Oxide	MgO	0.00005	Extremely Low	1020
Calcium Oxide	CaO	0.003	Low	653
Sodium Oxide	Na_2O	0.006	Low-Medium	1040
Potassium Oxide	K_2O	0.006	Low-Medium	750
Manganese Oxide	MnO	0.025	High	620
Chromium(III) Oxide	Cr_2O_3	0.020	High	770

The higher loss tangent materials such as iron oxide and Titanium Dioxide are what the mission targets to increase the effect of the microwave sintering process. Having more of these high loss tangent materials will increase the amount of usable energy produced by the regolith during the trial making process.

To determine the Cp of the lunar regolith a weighted average can be calculated. Using onboard sensors the exact chemical makeup of the gathered regolith can be determined and the exact Cp can be calculated. By multiplying the percentage weights by their respective Cp values and finding the summation of the aggregate.

After finding these two important values, the only other input needed is the power and frequency of the microwaves used then the total time needed to sinter the regolith can be calculated. From the inputs given the following can be found:

- Refractive index: ϵ_e
- Attenuation coefficient: α
- Single-pass absorption: $P(x)$
- Multi-reflection absorption: $P_{\text{abs,total}}$
- Total absorbed microwave power: P_{abs}
- Required heating energy: Q
- Radiative losses: q_{rad}
- Conductive losses: q_{cond}
- Net heating power: P_{net}
- Penetration depth: δ
- Penetration efficiency: η_{pen}
- Volumetric microwave heating: P_{vol}

The next factor that needs to be considered is how the regolith properties change with respect to its temperature, which can be modeled and predicted by the Cole-Cole equations, allowing us to model how the temperature will rise with respect to time of the regolith sample

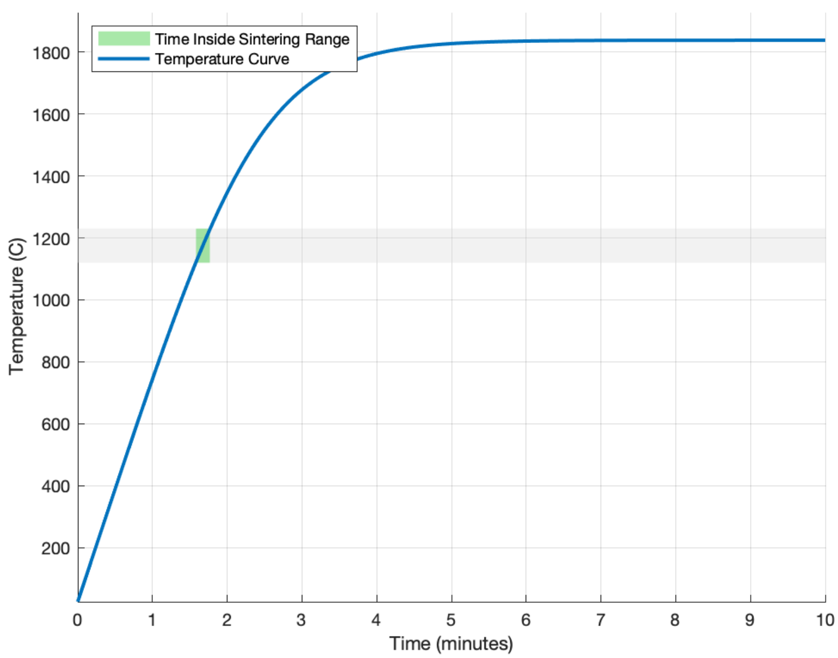


Figure 17: **Regolith temperature over time graph** The temperature rise over time under microwave sintering. This shows cooling cycles must be undergone to maintain ideal temperature for microwave sintering.

This graph shows the regolith temperature over time, with the gray line being the optimal sintering temperature and the green area being the time where the regolith is in that range. By taking into account the Cole-Cole equations the rise in temperature is not linear.

With the temperature now being able to be modeled and the exact regolith composition made up the exact time under sintering can be calculated with any given makeup for a targeted temperature

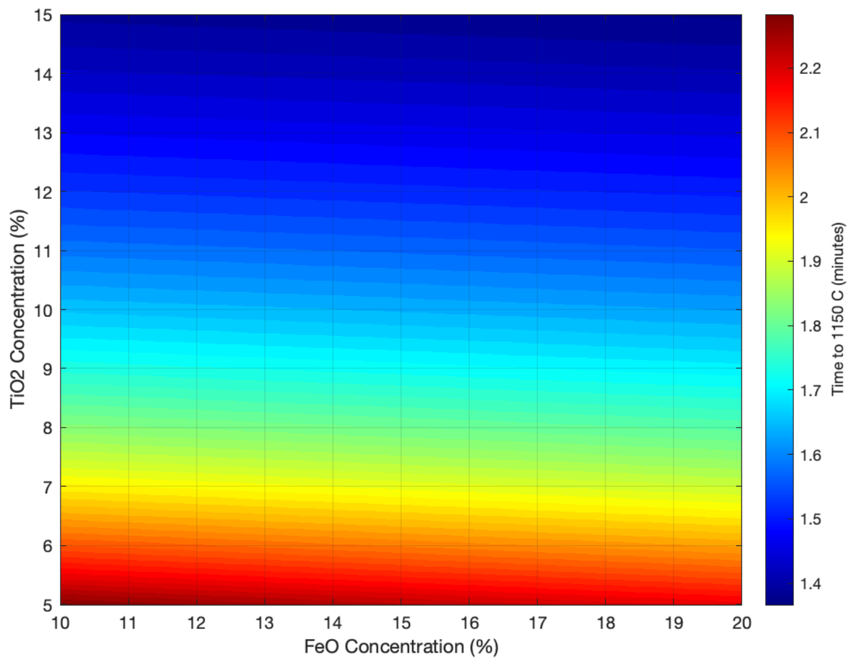


Figure 18: **Chemical composition sintering time** A heat map of the affect of Iron Oxide and Titanium Dioxide concentrations on the time it takes to sinter. This shows ideal concentration ranges to minimize heating times are high in both Titanium Dioxide and Iron Oxide. All other parameters kept constant.

The exact time it takes to properly sinter a tile can be calculated with the input being the exact chemical composition. In Figure 18 it shows the effect of the 2 components of regolith that have the highest Loss tangent and their affect on the microwave sintering time, this calculation can be done with the exact percentages for each component giving an exact time for the process.

By calculating the time to sinter while on site, the system does not rely on an exact type and make up of the regolith introducing flexibility into the system and allowing for more possibilities for landing sites and further testing. In addition the experimentally determined values through research and further testing will be validated and confirmed on site with the conditions on the Moon.

Chemical analysis can be done on site by many different means such as the use of Alpha Particle X-ray spectrometer (Alpha Particle X-ray Spectrometer (APXS)), Laser induced breakdown spectroscopy (LIBS), Near-Infrared spectroscopy (NIR), and The regolith and Ice Drill for Exploring New Terrain (The Regolith and Ice Drill for Exploring New Terrain (TRIDENT)). all these methods can be attached to the mission payload to be able to determine what makes up the regolith to calculate the proper sintering temperature.

4.3 Tile Design and Properties

The requirements for the regolith tiles are derived from two main constraints: the material properties to withstand the applied loads that occur during a launch or landing from a targeted 50,000 kg HLS; and the manufacturing constraints of the stationary tile maker. To meet the goals of the project, the following requirements were created.

Table 17: **Regolith Tile Requirements.** This table shows the requirements of the manufacture tiles, primarily derived from expected loads produced by potential lunar landing vehicles in current development.

Regolith Tile Requirements (RTR)	
RTR 1.0	The tile must survive the landing of a 50,000 kg HLS.
RTR 1.1	The tiles shall withstand a sustained vertical pressure of 35.4 kPa from the HLS landing plume.
RTR 1.2	The tiles shall withstand a 424 kPa impact load from HLS landing.
RTR 1.3	The tiles shall withstand a thermal flux of 17,800 kW/m^2 generated by the HLS landing plume.
RTR 1.4	The tiles shall withstand 25.8 kPa of static pressure from the HLS after landing.
RTR 2.0	The tiles shall be manufactured using the stationary tile maker and microwave sintering process without post-processing.
RTR 2.1	The tiles shall be vertically homogeneous.
RTR 2.2	The tiles shall not contain internal angles.
RTR 2.3	The tiles shall use a tessellating geometry.

A 50,000 kg HLS vehicle was chosen, however further research into larger HLS vehicles such as the approximately 100,000 kg SpaceX Starship should be conducted once these companies have finalized designs and released technical specifications for their systems [41].

4.3.1 Material Properties

To determine the necessary strength of the tiles, previous research was assessed in which a Finite Element Analysis (FEA) study simulating a landing of a 50,000 kg HLS was conducted [42].

Table 18: **Loading Conditions Experienced by Landing Pad During a Simulated Landing.** This table shows the forces a tile must resist imposed by a 50,000 kg HLS during decent and impact of a simulated landing [42].

Timestamp (s)	**Max Plume Vertical Pressure (kPa)**	**Max Plume Heat Flux (kW/m^2)**	**Leg Stress (kPa)**
0	0.00	0.00	0.00
1	8.85	4,446.58	0.00
2	17.70	8,893.15	0.00
3	26.55	13,339.73	0.00
4	35.40	17,786.30	0.00
5 (impact)	0.00	0.00	423.99 (one leg)
6 (static)	0.00	0.00	25.78 (four legs)

From this data the material requirements of the tile were determined (RTR 1.1-1.4). However, research into the strength of sintered tiles shows results that vary widely depending on the types of regolith used and the exact process used to sinter the regolith. Therefore, to fulfill these requirements, it was necessary to isolate and examine the properties that affect tile strength and determine the optimal characteristics for the tile fabrication process. The primary properties that affect the strength of the tiles are:

Porosity

The porosity of the tile has been shown to have an extreme effect on the compressive strength of the tiles. An increase in porosity by 10% can cut the compressive strength and failure load in half. [43] A major cause of porosity in the tiles is the particle size of the regolith. When sifted to a particle size below 212 μm the average porosity should be $\approx$ 1.44%.

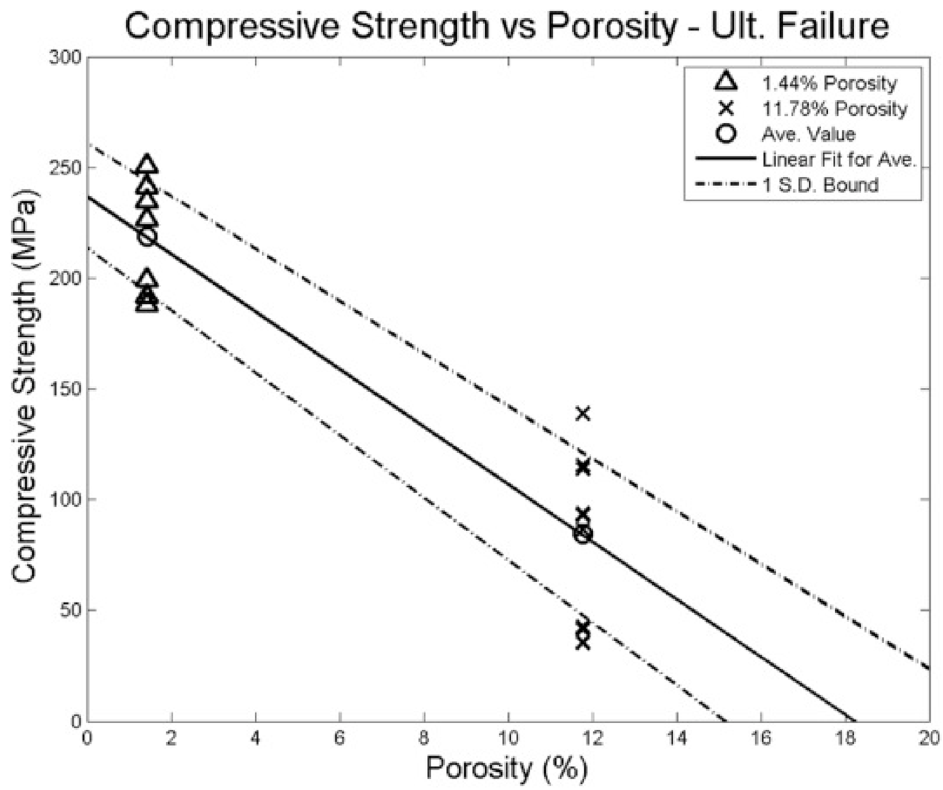


Figure 19: **Relationship of porosity and compressive strength.** This graph details the inverse relationship between compressive strength and porosity. This highlights the necessary minimization of porosity to maintain the necessary compressive strength of the tiles [43].

Sintering Temperature and Regolith Composition

Table 19: **Material Properties of Sintered Lunar Regolith Simulant Samples by Temperature.** This table shows the shrinkage characteristics of sintered regolith at various temperatures, for the microwave sintering process this mission employs a balance between shrinkage and compressive strength must be made [44].

1000°C Radial Shrinkage Percentage	0.84%
1000°C Height Shrinkage Percentage	1.16%
1100°C Radial Shrinkage Percentage	1.85%
1100°C Height Shrinkage Percentage	3.15%
1200°C Radial Shrinkage Percentage	6.17%
1200°C Height Shrinkage Percentage	18.13%
1200°C Compression Modulus (MPa)	55.73
1200°C Compressive Strength (MPa)	21.73

Sintering at a temperature higher than the melting point of most or all of the components of the regolith results in a higher compressive strength for the tile. [45] The melting point for these components tends to be below 1200 °C. [46]. Therefore, for this project, the tiles will be sintered at 1200 °C. Sintering at this temperature can allow molten components to fill voids in the tile, slightly increasing density and decreasing porosity.

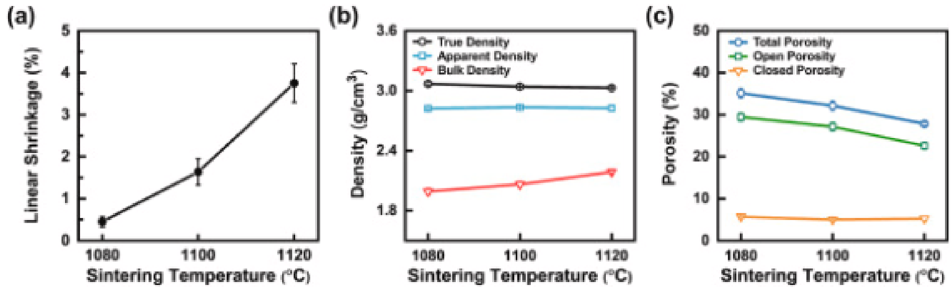


Figure 20: **Sintering temperature relationships.** Graphs with relationships between sintering temperature density, shrinkage, and porosity. Therefore, sintering must be undergone at a temperature range which minimizes porosity and shrinkage while maintaining ideal density [45].

However, this does create unique challenges in tile design as this high temperature has a side effect of significantly more material shrinkage as seen in Table 19 [**?**]. Due to some components of the regolith evaporating and the molten regolith becoming denser and filling the voids between the particles, the volume of the regolith reduces rather significantly. Sintering regolith from highland or mare regolith can give vastly different results. When sintering highlands regolith, the temperature has to be 50 to 100 °C to achieve the same results as when sintering mare.

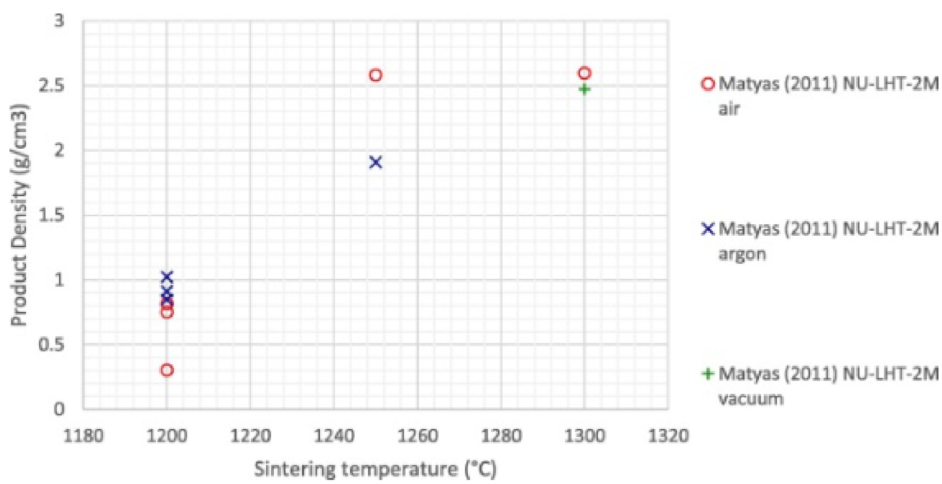


Figure 21: **Lunar highland sintering relations.** A graph detailing the relationship between highland sintering temperature and density. This highlights the high sintering temperatures required to reach optimal product density [46].

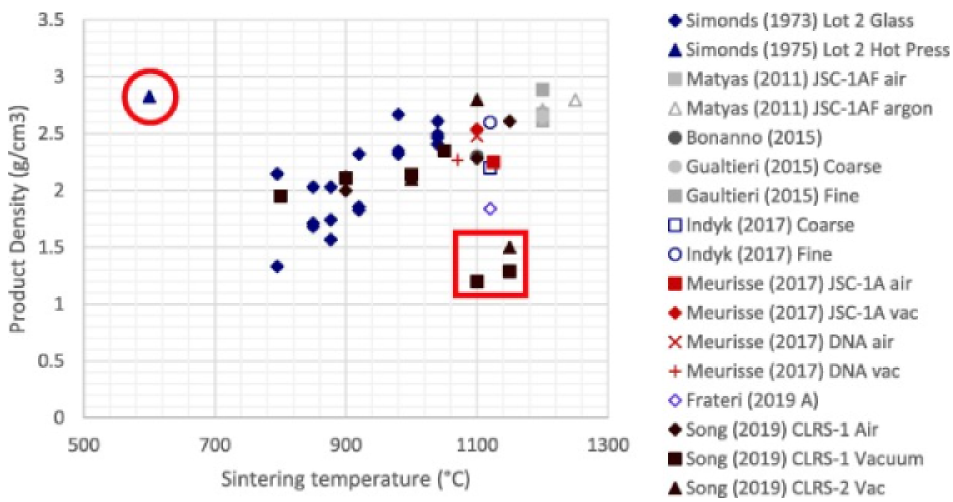


Figure 22: **Lunar mare sintering relations.** A graph detailing the relationship between mare sintering temperature and density. This shows a relative lower required sintering temperature compared to highlands [46].

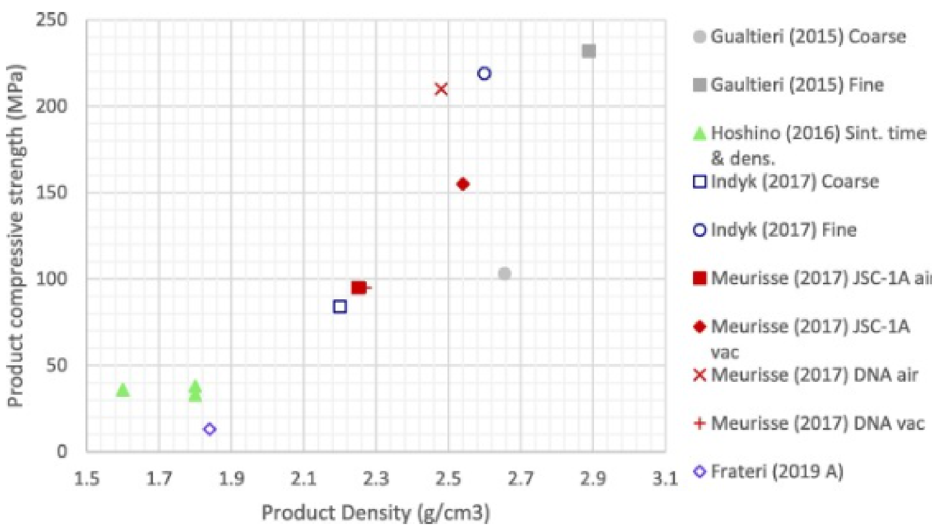


Figure 23: **Relationship of density and compressive strength.** A graph detailing the relationship between regolith density and compressive strength. This shows increasing product density result in a higher product compressive strength [46].

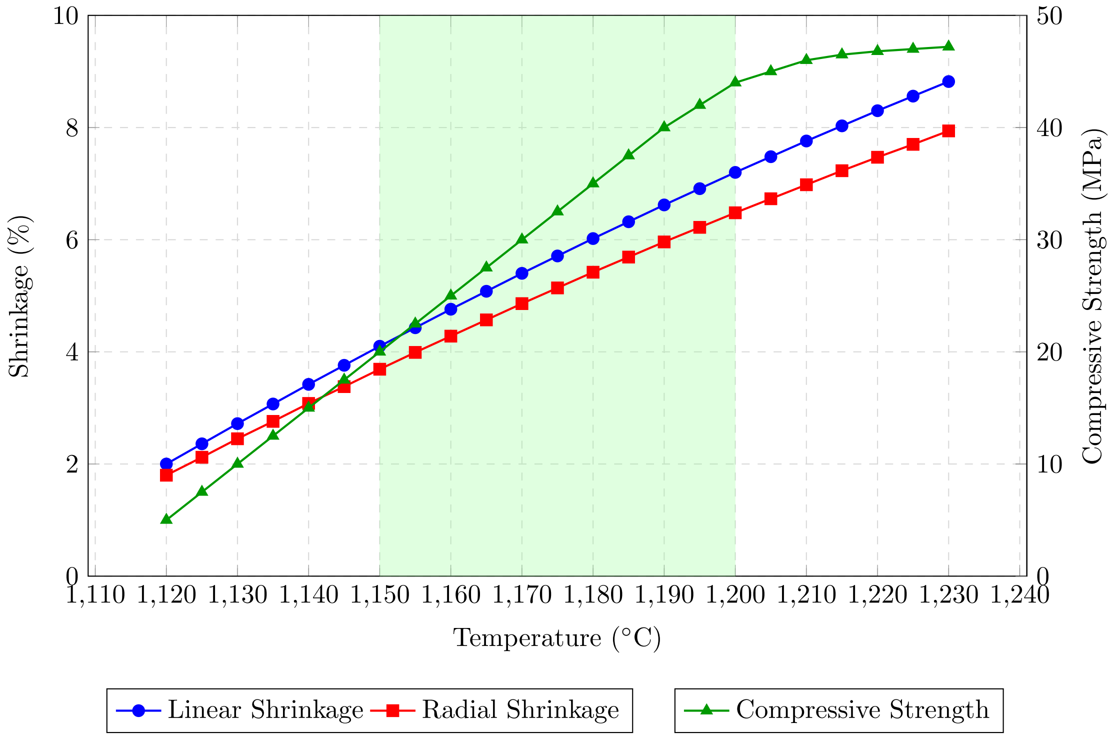


Figure 24: **Relationships of tile properties and sintering temperature.** Effect of sintering temperature on linear shrinkage, radial shrinkage, and compressive strength of lunar regolith simulant. The highlighted region between 1150°C and 1200°C represents the optimal sintering temperature range where strength increases significantly while shrinkage remains moderate.

The compressive strength of regolith when sintered at 1200 °C can vary [44] greatly when considering the environment that the regolith will be collected from; thus, when determining how to satisfy the minimum requirements for the tiles, the minimum possible properties of the tiles should be considered. Throughout all the experiments covered by research conducted by VT Luna, the low end of the compressive strength of the tiles is about 20 MPa. The bending strength tends to be about 28% of the compressive strength[47], so it will be 5.67 MPa. At this compressive strength, the tile will easily withstand the expected compression forces at any reasonable thickness; thus, the tile's thickness should be determined based on its ability to withstand bending stress due to imperfections in the ground supporting it.

4.3.2 Design Properties

The design constraints of the tile design are largely determined by the tile manufacturing process, which imposes unique constraints due to the characteristics of microwave sintering and of the specific tile maker's design, resulting in RTR 2.0. As the design is building tiles through compressing layers and subsequently sintering small splotched areas to form layers, vertical homogeneity is therefore necessary, as any non-uniform designs in the mold would cause difficulty in this compression and microwave sintering process, resulting in RTR 2.1. The aforementioned radial shrinkage of the tiles by approximately 6.17% during the cooling process imposes additional constraints on tile design [44]. As the tiles will shrink inwards horizontally, any internal angles of the mold would therefore be put under pressure by these shrinking tiles, which would result in damage to either the mold or the tile. Therefore, the tiles shall not have internal angles resulting in RTR 2.2.

As there is only one tile maker with a singular mold, only one tile design will be able to be produced. Therefore, the tile produced should be capable of independently forming a solid surface, so the tile shall be a geometry that tessellates, resulting in RTR 2.3.

Tile Design

Based upon these requirements, the tile design is based on a hexagon, but has 0.015 m overhangs and insets with 30°chamfers.

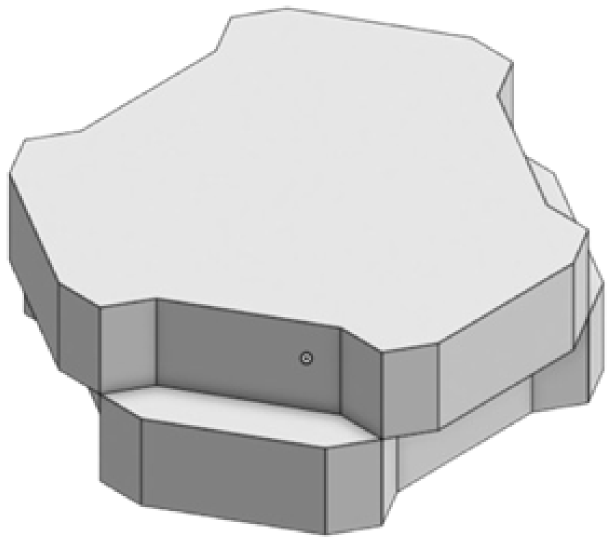


Figure 25: **Project Hestia tile design.** The hexagonal structure of the tile design uses a circumscribed radius of 0.15 m, and a depth of 0.106 m. These parameters allow for the tile to resit failure from heat flux and compressive loads respectively.

The design utilizes a 0.015 m overhang that sits over a 0.015 m inset for a total overlap distance of 0.03 m. This is done to prevent exhaust gasses from getting underneath the landing pad assembly and compromising the structure. Overhangs prevent direct paths for the gas to travel through mitigating the risk and allow the tiles to secure each other in place. The 30°chamfer on each overhang prevents direct radial overhangs. Radial overhangs can lead to cracks forming in the tile during cooling due to radial shrinkage. The elimination of radial overhangs also allows for tiles to be slid together on a flat plane preventing the need for a high precision manipulator.

This design satisfies RTR 1.0 as it can withstand the 420.0 kPa impact outlined in RTR 1.2 as calculated in equation $t = \sqrt{\frac{K\, p_{max}\, \pi r^2}{\sigma_{\text{bending}}}} * safetyfactor$[48] with K being 0.398 and the safety factor being 2.3 3, which is the maximum expected load and therefore satisfies RTR 1.1 and RTR 1.4 as well as there being less loading conditions. This design also fulfills RTR 1.3 as its surface area is approximately the same as the tiles used in a research paper where thermal analysis was conducted showing that sintered regolith tiles are capable of withstanding intense heating as expected from a rocket plume where "the maximum surface temperature (700 K) is confined within a central circular area of about 0.5 m in diameter" [49]. This design also fulfills RTR 2.0 as it does not vary vertically, satisfying RTR 2.1; does not have any internal angles, so it satisfies RTR 2.2; and it is based upon a normal hexagon, which is a shape that tessellates, so it satisfies RTR 2.3.

Aspects of the design that the design was unable to accurately quantify for this report include the lateral forces and stresses the tile will undergo, and the thermal expansion characteristics that will occur during landings. Furthermore, "acoustic waves are known to propagate within rocket plumes. However, no publications currently provide data on these waves in the lunar environment," which is another unknown factor that could disrupt tiles [42].To mitigate these unknowns, the design incorporated fillets that reduce stress concentrations, and by using a hexagon rather than a square or triangle, which also tessellate but have more acute angles, these stress concentrations are further reduced, which will help mitigate the risk of failure from both thermal expansion and lateral forces imposing stress on the tiles.

4.3.3 Tile Testing

To determine the success of the tiles, the design will subject them to a series of tests. the design will first leverage access to a surface astronaut and utilize them to collect and return the tiles to Earth, where they can undergo in-depth testing. the design will test for the tile requirements outlined earlier in table 17. The key measurements the design will need to test for are the impact pressure, the static hold pressure, and the thermal capabilities of the tiles. the design will test for the impact pressure using a drop weight tester. Drop weight testers are a great way to measure impact pressure as long as the impact area is noted. Large drop weight testers are capable of dropping a 2,700 kg object from 4 m. This is more than enough to produce the 424 kPa pressure that the tile must be able to withstand [50]. the design will also use a hydraulic press to determine the maximum static pressure the tile can withstand to replicate a large Lander's weight on the tile, as well as the sustained vertical pressure from the landing plume. A hydraulic press is more than capable of generating the required force. If the design press on the full area of the tile, which is 0.078 m^2, the design will need to generate 27,600 kg of force, which is achievable using only a small hydraulic press [51]. This test is key in understanding the static strength of the landing pad tiles. Finally, the design will use an arc jet to test the thermal capabilities of the tile. The arc jet can be used to mimic the plume of a rocket, at least thermally, and thus evaluate the thermal capabilities of the tile. The NASA Ames arc jet is capable of producing 70 Microwave (MW) of power for 30 minutes, which is more than enough to mimic the tile requirement of 17,800 kW/m^2 [52]. These tests will evaluate the basic properties required to render the tile functional as part of a lunar landing pad. Additionally, the team is aware that conducting tests on the lunar surface is important, and the team is working on designing these tests. More on this can be found later in this report under future work.

5 Architecture Overview

5.1 Regolith Collection System (Rover)

5.1.1 Rover life cycle overview

The collection system relies on a small autonomous rover designed to retrieve regolith and transport it to the stationary processing plant. The rover operates in a continuous loop, and there will be four major steps to this process.

The first step is the deployment; the rover will leave the dock/Griffin Lander in search of regolith. The navigation system will be initialized as soon as the rover deploys to guide the rover to the best location to retrieve the regolith. The navigation system chosen consists of Wheel Odometry and Inertial Measurement Unit (IMU), which will be used for obstacle detection and alignment. The rover will only be traveling short distances, between the collection site and the Lander, which is why this navigation system was chosen. It is lightweight and provides sufficient accuracy while consuming low amounts of power compared to other navigation methods. The tires of the rover were also chosen with the location in mind. Rigid aluminum wheels with grousers are the tires chosen, because the landing site will have loose lunar soil. These types of tires will ensure traction and allow the rover to travel between operational locations.

The second step is collecting the regolith. Once the rover reaches the excavation area, it will begin the collection process. This will be done via a two-joint robotic arm, consisting of an elbow pitch joint and a wrist tilt joint and an Alpha Particle X-ray Spectrometer. These joints will supply 8–10 Newton-meter, and 2–3 Newton-meter of torque, respectively, which will allow the arm to scoop loose regolith and transfer it to the hopper. The hopper will be constructed of aluminum alloy, similar to the two-joint robotic arm. The hopper will hold 16 kilograms of regolith, but the rover will return to the stationary tile maker when there are 14.5 kilograms of regolith stored. This accounts for regolith spillage in the rough terrain. The arm will have to have multiple scoops per trip, because the scoop can only hold up to 3 kilograms of regolith at a time.

The third step in the rover cycle is the return to the stationary tile maker. The path will be short and repeatable, which will allow the rover's navigation to be as accurate as possible. Once the rover reaches the stationary tile maker, it will align itself with a telescoping ramp. The ramp will be designed to guide the rover in the exact position needed to dump the regolith into the stationary tile maker's hopper.

Once the rover is aligned to the ramp, it will travel up to the stationary tile maker's hopper. It will start the dumping procedure, starting with the actuation of the hopper's tilt motor. The hoppers will invert, causing the collected regolith to be released in the reception hopper. After the regolith is received, the material-processing sequence starts, which includes screening, weighing, and ultimately microwave sintering inside the stationary tile maker. While this sequence is taking place, the rover will remove itself from the ramp to begin the next step of the cycle.

The last step is dependent on the stage in the tile-making process. The rover will either go into dock-to-charge mode, where it will interface with a charging node integrated into the Lander, and the 6S Li-ion Battery Pack [53] will have the energy consumed during the previous cycle replenished, or it will continue to gather regolith. The stationary tile maker can hold enough regolith to create multiple tiles, so depending on the rover's charge level, it could be beneficial to continue to gather regolith. The rover will also run basic diagnostic checks while charging to make sure the rover stays operational. Once the charging is complete, the rover will resume autonomous operations. It's also important to note in the lunar night, the rover will return to the charging node, not to charge, but to enter night mode. This will prevent the damage that could be done to the rover over cold nights.

After the tile is made, which will take 7.29 hours, the rover will move itself directly under the STiM to catch the tile. The rover has a mesh strong enough to catch the tile, allowing it to rest on top of the bucket. After the tile falls onto the rover, the tile clamp will lower, preventing the tile from moving as the rover starts driving forward. The rover will then move to the desired location to place the tile and dump the bucket. This will place the tile on the ground where it can be manipulated to a certain degree. It is only a two-jointed arm, though, which means it has limits on where and how it can place the tile.

5.1.2 Rover design

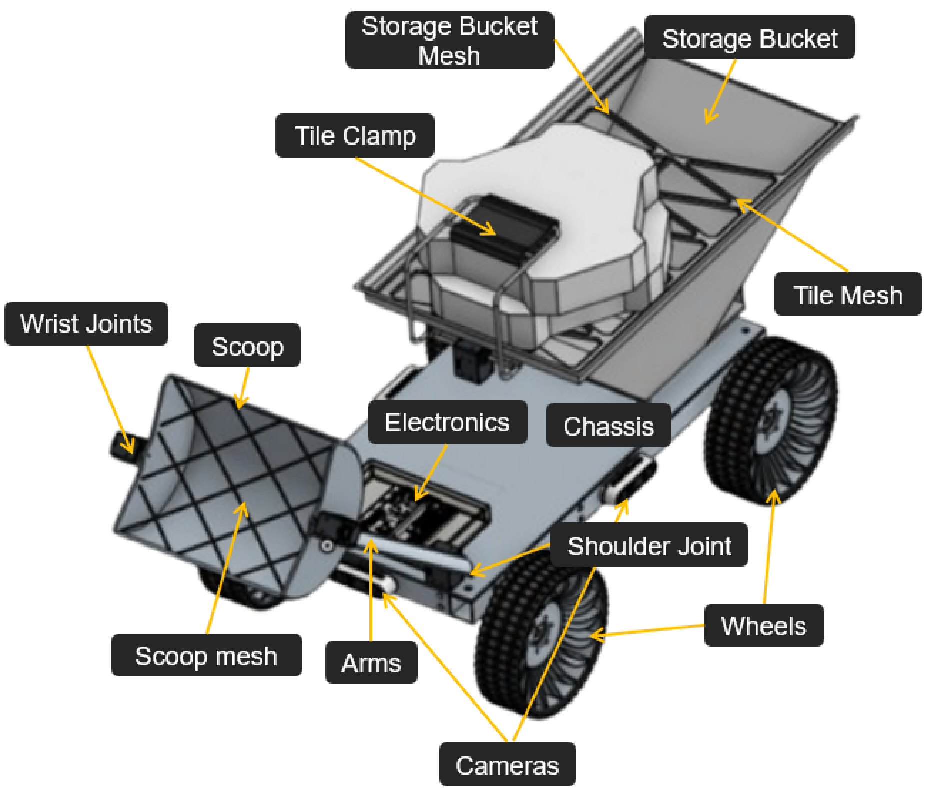


Figure 26: **Rover Diagram.** Labeled rover subsystem layout showing the major structural, mobility, sensing, and material-handling components. Key systems labeled include the chassis, wheels, camera system, robotic arm joints, scoop assembly, onboard electronics, tile clamp, storage bucket, and tile handling mesh systems.

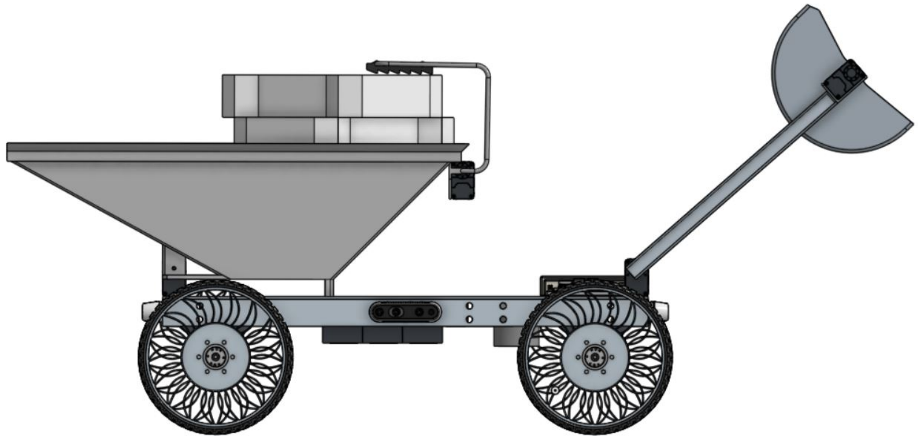


Figure 27: **Rover side view.** Side view of the rover design showing the overall profile of the chassis, wheel assembly, robotic arm system, storage bucket, and tile-handling mechanisms. The side layout highlights the rover's mobility configuration and the positioning of the material collection and transport systems used during autonomous regolith collection and tile placement operations.

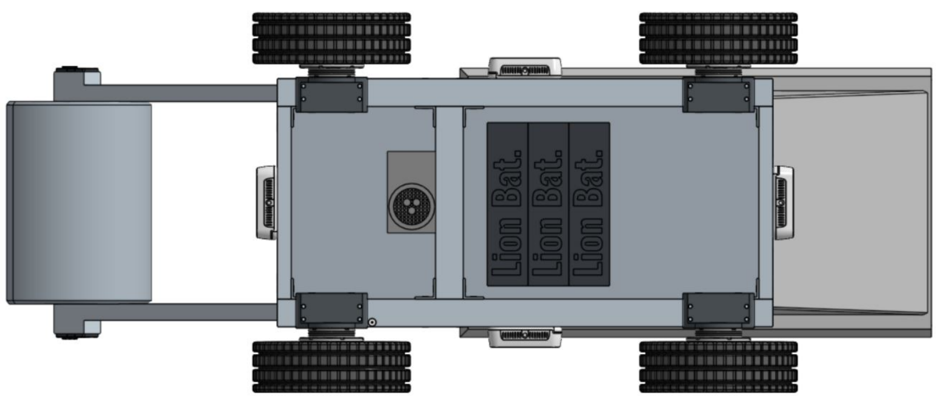


Figure 28: **Rover bottom view.** Bottom view of the rover design showing the chassis structure, wheel placement, drivetrain mounting locations, battery compartment, and underside component layout. The underside configuration was designed to provide structural support, balanced weight distribution, and stable mobility across uneven lunar terrain during regolith collection and tile transport operations.

The figures above show the design of the rover. The rover is 1.20 m by 0.50 m by 0.40 m, which is small relative to other rovers built for lunar travel. This is due to the mass limit, because of this restriction, the rover was designed to be as streamlined and lightweight as possible. The material used is primarily 6061-T6-Aluminum alloy throughout the rover and is designed to be as thin as possible while maintaining structural integrity with an adequate safety margin.

Rover Chassis

The chassis functions as the central structure for the mounting of other components and provides rigidity to the rover. It is made of aluminum alloy and has several beams on the bottom to provide structural support. The majority of the electronics will be mounted to the chassis.

Rover Scoop and Arm

The rover scoop and arm are designed to effectively harvest regolith from the lunar surface and deposit this regolith into the bucket. The scoop has a mesh on the open face to prevent the collection of regolith large enough to potentially damage the STiM. The scoop is 0.24 m by 0.20 m, and the mesh is designed to prevent the collection of regolith greater than 0.052 meters by 0.052 m. This is well below the size of potential rocks that the STiM can handle. The arm has two degrees of freedom and is 0.39 m long. While this does restrict the placement and manipulation of the tile due to limited range of motion, it such a design is capable of collecting regolith and performing rough manipulation of placed tiles. A more complex manipulator was considered to allow more precise tile placement, but due to the weight limit, a manipulator with only two degrees of freedom has been chosen.

Rover Storage Bucket

The storage bucket is also connected to the chassis. It is 0.33 by 0.58 by 0.17 m and also made from aluminum alloy . It also has a mesh, but this is slightly different because it has to hold the tile after it is made. The storage bucket can hold 5.65 liters of regolith at a time, which is more than enough to fulfill the missions goals. The bucket also has a motorized tile clamp attached that can secure one tile at a time to move a tile placed on top of the bucket to the intended drop of location of the tile.

Rover Wheels

The wheels selected enable the traversal of the rover in the conditions expected the missions landing site. Rigid aluminum wheels with grousers have been chosen for the wheels as the landing site will have loose lunar soil. These types of tires will ensure traction is maintained in the loose lunar regolith and allow the rover to travel between operational locations without sinking into the regolith or slipping uncontrollably.

5.1.3 Electronics list

With the rover having to fulfill its responsibilities for any tile that is produced, it is important to confirm that the rover has all of the components needed. The chart below shows all of the electronic components the rover is expected to need.

Table 20: **Rover Electronics and Actuation Components.** This table shows rover electronic components the justification behind the component selection as well as the power requirements for each component.

Part Name	Purpose	Power Requirements	Ref.
Rad-Hard 32-bit ARM Cortex-M7 MCU	Main autonomy computer responsible for rover navigation, control, and autonomous decision making	3.3 V I/O, 1.8 V core	[54]
maxon EC 32 flat SPACE + GPX 32 UP (3-stage)	Controls wrist joint movement for the robotic arm	24 V, 0.421 A, ∼15 W	[55]
maxon EC 45 flat SPACE + GPX 42 UP (3-stage)	Controls elbow joint movement for the robotic arm	12 V, ∼50 W	[56]
Motor Control MCU	Controls and manages drive motor operation	5 V logic, up to 60 V motor supply	[57]
2.4 GHz Relay Communication Module + Low-Gain Antenna	Handles wireless communication between rover and Griffin Lander relay	0.5 W max	[58]
SAMRH71 Rad-Hard Cortex-M7	Controls robotic arm and hopper systems	3.0–3.6 V I/O, ∼1.8 V core	[54]
Microchip LX7720 Motor Driver	Provides safety monitoring and power management for motor systems	5 V logic, up to 60 V motor supply	[59]
CubeMars RI50 Frameless BLDC	Tilts hopper to dump collected regolith or tiles	12–48 V, ∼500 W	[60]
CubeMars AK60-6 Actuator (×4)	Provides wheel actuation and rover mobility	24 V, ∼50 W each	[61]
LN-200S IMU	Provides rover orientation and navigation data	∼8–15 W	[62]
IM200 Space Camera (×2)	Stereo vision system for navigation and obstacle detection	∼1.4 W total	[63]
EKF + Visual Odometry	Navigation software used for localization and autonomous movement	∼5–15 W (shared compute)	[64]
6S Li-ion Battery Pack	Primary rover power storage and distribution system	∼400 Wh, 22.2 V, ∼18 Ah	[65]
Polyimide (Kapton) Etched-Foil Heater	Maintains electronics temperature during lunar night and cold conditions	∼5–10 W	[66]
Pragyan Alpha Particle X-Ray Spectrometer	Accurate regolith chemical composition measurement	∼4 W	[67]

Each component was carefully considered in relation to other parts, and the value they would add as well as their power requirements and weight requirements, were considered. The navigation equipment was chosen first because they are on of the most important parts of this mission; with the wrong navigation system, the rover could become lost or stuck and retrieve no regolith. The Navigation chosen will be talked about more in the navigation section. After the equipment for the navigation was picked the motors, MCUs, and heater were chosen. Finally, the battery was chosen.

Many different types of batteries were considered, but this specific battery was one of the lowest-weight ones that still provided more than enough power needed for this mission. This was proven through a MATLAB code. The electronic requirements were taken into account in each stage of the process, as well as what the power requirements would be when the rover was in sleep mode. It was found that there will still be excess power even after the rover completes one cycle, which provides enough regolith for the STiM to produce one tile.

5.1.4 Rover navigation

The rover uses a navigation system based on an Inertial Measurement Unit (IMU), stereo cameras, an Extended Kalman Filter (EKF), and visual odometry because this approach provides reliable autonomous navigation in the lunar environment without depending on GPS. GPS satellites on the Moon will not be available during the mission, so the rover has to determine its position and orientation using its own sensors. This is where the IMU comes in; it provides acceleration and rotational data. Stereo cameras will also be working with the IMU by letting the rover detect obstacles. This will let the rover avoid unpredictable obstacles. Visual odometry tracks changes in camera images over time to estimate the rover's motion and position.

An EKF-based sensor-fusion system was also selected. It was selected because it combines data from the sensors to improve navigation accuracy and reduce error buildup over time. This was needed due to the many dust-related issues that regolith cases cause; it will help fix temporary camera-tracking issues caused by dust or lighting conditions. The system was also chosen because it is well-tested in robotics and aerospace applications, indicating it is a valid system.

5.1.5 Prototype

Mechanical Design and Manufacturing

The rover's chassis was designed with a focus on structural integrity, weight efficiency, and adaptability to support subsystem integration. Measuring 0.46 m in length and 0.28 m in width, the primary frame is constructed from square aluminum tubing. Each portion of tubing was precisely cut to length to ensure proper alignment during assembly. To further enhance rigidity, a centrally positioned crossbar was incorporated into the frame design. This beam plays a critical role in distributing loads more evenly across the chassis, reducing localized stress concentrations and improving the rover's ability to withstand dynamic forces encountered during operation, such as uneven terrain traversal and material handling.

The chassis assembly process utilized aluminum brackets and riveted joints, creating a semi-permanent yet lightweight fastening method that balances durability with ease of fabrication. Riveting was chosen over welding to simplify construction and avoid heat-induced material distortion while still providing sufficient joint strength for the rover's expected operational loads. After assembly, the chassis underwent basic load testing to verify its ability to support the combined mass of onboard electronics, power systems, and mechanical subsystems. This validation step ensured that the frame would maintain structural integrity without excessive flexing or failure under anticipated working conditions.

To facilitate the integration of electronic components, a Plexiglass mounting plate was installed on the top portion of the chassis and securely riveted to the aluminum frame. This plate serves multiple important functions as it provides a flat and stable mounting surface, electrically isolates sensitive components from the conductive aluminum structure, and simplifies component layout and cable management. The use of a transparent material also aids in visual inspection and troubleshooting, allowing for quick identification of wiring issues or component placement without disassembly.

Additionally, the plate contributes minimally to overall weight while maintaining sufficient stiffness to support the mounted electronics. The rover's external mechanical components, including the arms, scoop, bucket, wheels, tires, and dust cover, were designed using Computer-aided design (CAD) software to ensure precise geometry and compatibility with the chassis and actuation systems. These components were subsequently fabricated using additive manufacturing (3D printing) with an approximate infill density of 40%. This infill level was selected as a balance between reducing mass and maintaining adequate structural strength, particularly for components subjected to moderate mechanical loads. The use of 3D printing also enabled rapid prototyping and iterative design improvements, allowing for refined geometries and optimized performance without significant material waste or manufacturing delays.

Figure 29: **Rover prototype.** Fully assembled prototype rover. Key systems such as the scoop, bucket, electronics cover, and wheels are visible in this image.

Electronics

The rover electrical system was designed to provide reliable power delivery, precise actuation, and centralized control for both mobility and material-handling operations under varying load conditions. The architecture integrates four 12 V DC drive motors for locomotion, each paired with an independent motor controller to enable fine-tuned speed and torque regulation. In addition to the drive system, the rover incorporates two high-torque 25 kg servo motors and three 125 kg servo motors, which are responsible for articulated functions such as material collection, manipulation, and deposition. At the core of the control system are an Arduino microcontroller, which handles real-time motor and servo control, and a Raspberry Pi, which provides higher-level processing capabilities such as sensor integration, communication, and potential autonomy functions. Power is supplied by two lithium polymer (LiPo) batteries, a primary 6700 mAh pack and a secondary 2200 mAh pack, distributed through a dual-bus architecture supported by two power distribution boards to ensure organized and stable electrical routing.

The physical implementation of the system reflects a deliberate emphasis on modularity, accessibility, and efficiency. Components are arranged to minimize wire length for high-current paths while maintaining clear separation between power and signal lines. The layout also facilitates rapid maintenance and troubleshooting by grouping related subsystems, such as motor controllers and distribution boards, into easily identifiable sections. Structural mounting considerations ensure that sensitive electronics, including the Arduino and Raspberry Pi, are protected from dust, dirt, or any other possible dangers during rover operation.

All four drive motors are wired directly to their respective motor controllers, forming independent drive channels that allow for differential steering and precise motion control. These motor controllers are connected to the first power distribution board, where they receive a regulated 12 V supply from the primary 6700 mAh

LiPo battery. This configuration ensures that high-current demands from the drive system are isolated from more sensitive electronics.

Similarly, the two 25 kg servo motors and three 125 kg servo motors are connected to the upper power distribution bus, which is also supplied by the 6700 mAh battery. This separation of power buses allows the system to better manage transient loads caused by sudden servo actuation, reducing the risk of brownouts or interference with the drive system. Each servo is directly interfaced with the Arduino, which generates the required control signals to achieve precise angular positioning.

The secondary 2200 mAh LiPo battery is dedicated strictly to the Raspberry Pi. This separation ensures that computational and communication functions remain stable even during periods of high mechanical load. It also allows the raspberry pi to continue to function even the 6700 mAh battery dies, allowing for just a quick change of batteries to keep the rover moving during testing. Together, the dual-battery and dual-bus design enhances overall system robustness, allowing the rover to maintain operational integrity across the experiment.

Software and Automation

The rover's software was designed to function consistently in an unpredictable environment. A motion capture system was used to track the rover's position and posture from above and send that position data through WiFi to a Raspberry Pi running Robot Operating System (ROS). On the Raspberry Pi, the tracking data was received and processed to determine the rover's motion commands, which were then sent to the Arduino through a serial USB connection.

The Arduino was used to handle the low-level hardware tasks, including motor driving and servo control for the rover mechanisms. C++ was used on the Arduino because it was compatible with the required hardware libraries, while Python was used on the Raspberry Pi because it worked well with ROS. This setup allowed the rover to combine external position tracking with autonomous navigation in a simple and modular way.

The navigation logic followed a goal-oriented sequence, in which the rover moved to the sand pile, completed the scoop process, and then navigated to the next target location. Using real-time motion-capture feedback, the system updated every 0.1 s, allowing the rover to adjust its orientation, distance to target, and approach angle throughout the task. This was particularly important for low-tolerance maneuvers such as ramp traversal, where accurate positioning was required due to the lack of guide rails

The control system was also designed to be adaptable to disturbances during operation. If the rover slipped, lost course, or the target position changed, it could automatically re-orient, re-localize, and continue toward the intended location. This state-based architecture allowed the rover to make corrections and complete tasks while navigating unpredictable terrain without manual intervention, improving the reliability of the autonomous collection and transport cycle.

The code serves as the main ROS2 control node for the rover, acting as the primary bridge between the motion capture system and the physical hardware. It subscribes to position and orientation data from the Virtual Reality Peripheral Network (VRPN) motion capture system to constantly track exactly where the rover and its targets are in the room. Using these coordinates, the script calculates the distance and angle required to reach a specific destination. It then translates these calculations, and based on its distance and orientation relative to its target destination chooses direct motor commands, such as, forward, reverse, or turn and then sends them over a serial connection to an onboard Arduino that actually drives the motors.

To handle navigation, the node uses a straightforward proportional control system. It continuously measures the error between the rover's current heading and the target angle. If the rover is far away, it drives forward, but if its heading is off, it adjusts the left or right wheel speeds to steer back on track. As it gets closer to the target, it scales the speed down to prevent overshooting. There are also specific dead-bands programmed into the logic so that the rover stops when it is close enough to its destination without needing to be exact. This prevents the rover from frantically twitching back and forth if it is already facing the correct direction or sitting right on top of its goal.

Figure 30: **Prototype rover in experimental setup.** Rover in the testing area running through its autonomous sequence. This shows the terrain of the testing area and the experimental setup.

Finally, the code includes a fully autonomous state machine built specifically for the regolith collection cycle. When triggered, the rover follows a set, cyclic sequence: it navigates to an excavation site, runs a hard-coded scooping macro to pick up material, navigates to a dumping site at which it must precisely travel up a ramp, dump its contents into a bucket, and then reliably returns to its starting point. A separate background thread runs alongside all this, listening for keyboard inputs to enable manual override the autonomous sequence, set new waypoints, or trigger specific actions such as scooping while navigating.

5.1.6 Summary

As can be seen, each component of the rover was evaluated on the basis of its mass, durability, energy efficiency, and lunar environmental constraints in the component selection process. It was decided that this method of regolith collection and the design of the rover was the best option given the constraints of the project.

5.2 Manufacturing System (Stationary Tile Maker(STiM))

The STiM is designed as a material-handling and fabrication system with the autonomous capability to convert lunar regolith into dense, microwave-sintered paving tiles suitable for landing pad construction. The architecture consists of four major subsystems: 1. Collection 2. Conditioning 3. Sintering 4. Cooling. Together, these components create a controlled material pathway that ensures regolith moves from unprocessed, variably graded soil into a uniform, compacted, and sintered tile with consistent mechanical properties. A labeled diagram of the STiM is featured below.

To support autonomous operation in the lunar environment, the STiM utilizes a distributed avionics architecture centered around a ruggedized embedded compute core with heritage in space and CubeSat applications. The central processing system manages high-level autonomy, subsystem coordination, sensor fusion, thermal monitoring, and operational sequencing during tile fabrication. Critical low-level functions, including actuator control, safety interlocks, and microwave power regulation, are handled through dedicated real-time microcontrollers to improve reliability and fault tolerance. This architecture reduces single-point failures while maintaining a lightweight and power-efficient control system suitable for long-duration lunar surface operations.

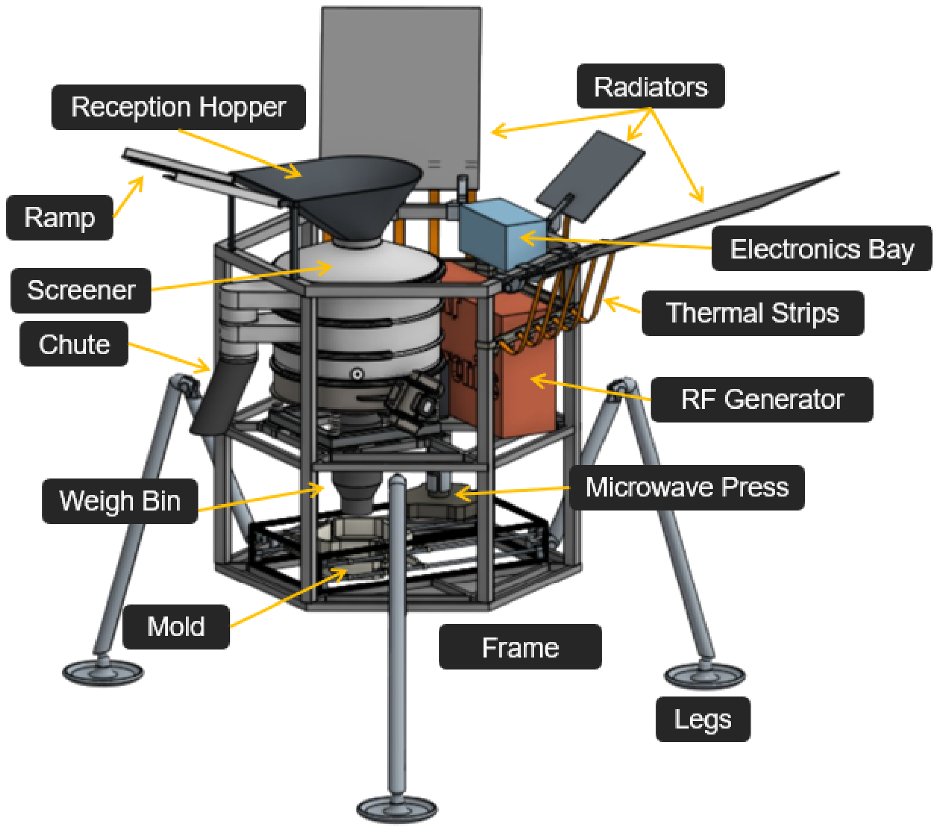


Figure 31: **STiM Diagram.** This is a labeled diagram of the major components within the STiM. Key subsystems are labeled including the reception hopper, screener, weigh bin, microwave press, and mold.

5.2.1 Collection

The first stage system, after the collection rover completes its process, is the collection system. This system is characterized by a ramp and a reception hopper. The rover will navigate up the Griffin Lander via a series of ramps provided by Astrobotic. Upon reaching the top of the Griffin Lander, the rover will then descend a small ramp mounted to the top of the STiM. The ramp must remain structurally rigid under the rover's mass while minimizing weight and stored volume. Because of this criterion, an aluminum alloy such as 6061-T6-Aluminum is an advantageous material selection. Additionally, its surface is treated to reduce electrostatic adhesion, given the dielectric charge of lunar regolith.

The reception hopper functions as the first buffering and storage unit in the system. It will receive raw regolith from the rover and then provide a controlled outlet to downstream processes. Its geometric structure will be that of an inverted pyramid/cone design. The hopper system is shown below in Figure 32 . The internal geometry of the hopper is essential to its design to prevent flow obstruction and inconsistent discharge. The upper stage, or deposit stage, of the hopper will have a wide mouth with an area of 0.4 m^2 acting to catch any regolith plume that may result from the initial deposit. This will then narrow down to an inner funnel with a mouth area of 0.15 m^2, and wall angles of 60° to prevent lunar regolith bridging and to promote mass flow under reduced lunar gravity, as this angle is above the angle of repose of lunar regolith [68]. Because bridging and arching tendencies increase in reduced lunar gravity, the hopper relies on a combination of steep interior surfaces, conductive coatings, and the vibration provided by the vibratory screener to which the hopper is mounted to maintain flow reliability. The hopper also decouples rover delivery cycles from tile production rate, thus maintaining operational continuity and reducing idle time. This is done through a slide gate system that releases when the stationary tile maker has sufficient charge for a sintering cycle and engages when the system is in idle mode, or sufficient regolith for a tile has been processed. This slide gate is attached to a flange, which is connected to the first stage of the conditioning subsystem, the vibratory screener.

Figure 32: **Reception Hopper CAD Model.** The reception hopper acts as the entry point to the STiM as well as a storage unit. As seen in the image the walls of the hopper as inclined at a high angle to mitigate regolith buildup.

5.2.2 Conditioning

Vibratory Sifter

Downstream of the hopper, the vibratory sifter serves as the first stage as well as the primary material-conditioning subsystem. The main purpose of this subsystem is to remove over-sized regolith aggregates and ensures that the regolith that makes it to the sintering stage is below the required particulate size ($< 600\mu m$) as defined in the materials properties section. Having a uniform particle size is a critical step in the conditioning process. It results in predictable compaction behavior and allows for effective microwave sintering. This is due to the fact that particle size and its distribution strongly influences the powder packing behavior, densification rates, and the resulting microstructure and mechanical properties of sintered materials. Studies show that controlled particle-size distributions facilitate consistent compaction and more uniform densification during sintering, which is critical for predictable mechanical performance [69][70].

The design also relies on a multiple stage sifter. The first stage will consist of a laser-cut 6061-T6-Aluminum plate with a cut diameter of 2.0 mm. This laser-cut plate architecture is chosen because it is more durable and abrasion-resistant than a woven mesh of aluminum. This first sifting stage exists to remove large clasts of regolith and oversize rocks from the selection process. The secondary stage will also be made from a laser-cut Aluminum plate, reducing the cut diameter to 0.8 mm. The second stage is more fine-tuned sifting while still removing larger particles and reducing the maximum load that will be incurred by the more delicate fine mesh woven final stage. The final stage will consist of a woven mesh with a cut diameter of 550 μm. This diameter is slightly below the required particle diameter of 600 μm. This design choice is to ensure that no abnormally shaped particles above the required diameter make it through to the final selection and ensure uniformity in the chosen regolith.

The nature of the vibratory screener's sinusoidal plate motion creates a whirlpool-like flow of the particulates within the screener. This flow pattern creates a centrifugal acceleration, naturally ushering particles to the cylindrical walls of the screener. At one point along this wall there will be a guided outlet which is directly mated to a chute, the vibratory sifter and chute system is shown below in Figure 33. This outlet will have a diameter of 12 cm to ensure that no clogging occurs in the vibratory screener from clasts. This diameter was chosen because due to the mesh on the scoop of the rover, no rocks larger than 5 cm should make it in to the flow of regolith. The non-optimal granularity regolith will be deposited outside of the screener and STiM system via the chute mounted to the screener outlets. The rejected regolith will then be bulldozed by the rover and disposed of in a designated 'no collection' zone. The regolith that passes the screening will make its way into the weigh bin.

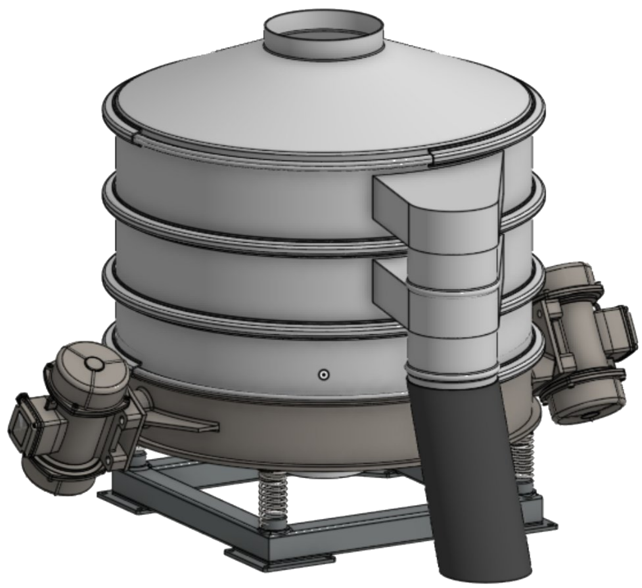


Figure 33: **Vibratory Screener.** The vibratory screener uses a three-stage approach to remove non-optimal granularity from the system. Vibratory motors on the sides drive the sifting process and regolith of undesired granularity is ejected through the chute.

It is assumed that the maximum load that the sifter will undergo at any given point is the mass of a single tile, being 12.8 kg. The deck size chosen is 0.52 m to ensure an acceptable regolith flow, as well as to ensure that the regolith can be evenly distributed for sifting. The frequency range of $25-50$ Hz is typical for many lab applications of finely tuned sieves, which typically operate between 1,400 and 3,000 cycles/min [71]. The sifter uses an Eccentric Rotating Mass (ERM) motor, operating at a frequency 50 Hz. By operating in the upper end of this range, the ERM motor ensures that the vibratory motion will overcome cohesive forces. Given the geometry of the sifter, the maximum mass load, and the required frequency, the motor specifications can be found. After research into lab-grade sieves with similar specifications as the vibratory sifter being implemented on the stationary tile maker, the following motor specifications were found [72].

These values come from 0.5 m three-stage lab sieves with a mass rating of 25 kg. This means that the power rating for the motor is conservative and will have excess power capabilities to operate as the single vibratory motor for the whole system (including reception hopper and weigh bin). The brushless DC motor is optimal for lunar conditions as it has a longer life cycle than a brushed motor, and no contact points for lunar regolith

Table 21: **Vibratory Sifter Motor Specifications.** This table shows the type, power rating, excitation frequency, and amplitude of the eccentric mass vibratory motors used on the vibratory sifter.

Type:	DC Brushless Motor
Power Rating:	25 W 12 V
Excitation Frequency:	50 Hz
Amplitude:	$1-3$ mm

to interfere with. As of yet, there are no space-grade ERM motors for industrial purposes such as driving a vibratory screener. This means that a space-grade motor with these specifications would have to be developed specifically for this mission; this will drive research and testing costs upwards. The main adjustment to make a motor of this specification space-grade is by introducing a dry lubricant such as Molybdenum Disulfide (MoS2) and ensuring that it is vacuum sealed.

Weigh Bin

Once regolith is passed through the sifter, it enters the weigh bin, where precise mass metering will take place. the top flange of the weigh bin will be mated to the vibratory screener and sealed with a slide gate. This slide gate will release, and the weighing bin will accumulate material until the required mass for a single tile is reached (12.8 kg). When this target mass is reached, the vibratory screener will shut off and the flow of regolith will be halted by sealing the hopper/screener slide gate as well as the screener/weigh bin slide gate. This is to ensure that each sintering cycle has a consistent volumetric and mass input. Four compression load cells will be mounted to the weigh bin to provide mass measurement. This subsystem plays a vital role in tile reproducibility. The bottom of the weigh bin will be equipped with an outlet rotary gate. When the target mass is reached, this rotary gate will dispense 1/24 of the total mass, or 0.535 kg at a time. This metric comes from the layered sintering process of 4.4 mm layers per sintering cycle for a total tile height of 106 mm.

Sinter Mold/Press

The final stage of the conditioning process occurs in the sinter mold itself, where the regolith undergoes both a vibration and compaction sequence. The mold material will be made from a stainless steel frame with an zirconium ceramic mold lining. The stainless steel mold is chosen due to its high rigidity and microwave reflective capabilities and with an zirconium ceramic coating to protect it from the heat of the sintered regolith. This is essential for the sintering process, in which as much microwave energy as possible must be directed and absorbed by the regolith. The geometry of the mold reflects the tile dimensions. Due to the overhangs and multilayered tile design, the mold itself is separated by a lower and upper half as shown in Figure 34 and need to have the ability to independently move each part of the mold. These upper and lower halves are also both separated down the middle to allow for tile ejection. These four quarters of the mold are all mounted to the frame via 8 space-rated MAXON AxialPower Enhanced Series (APES) linear actuators to allow for all parts to have independent freedom of motion[73]. These APES linear actuators have several important purposes. First, they allow the mold to move from the regolith reception position, i.e., directly under the weigh bin, over to the compaction/sintering position, i.e., directly under the microwave press. They also allow the mold to separate for easy tile ejection and recovery from the rover. Lastly, and most relevant to the regolith conditioning process, they are the primary drivers behind the vibration sequence. The vibration sequence ensures even distribution of regolith within the mold so that the microwave radiation will be evenly distributed, resulting in uniform heating and optimal sintering conditions. To achieve this, after receiving a layer of regolith, the linear actuators will vibrate at a frequency of 50 Hz and an amplitude of 1 mm.

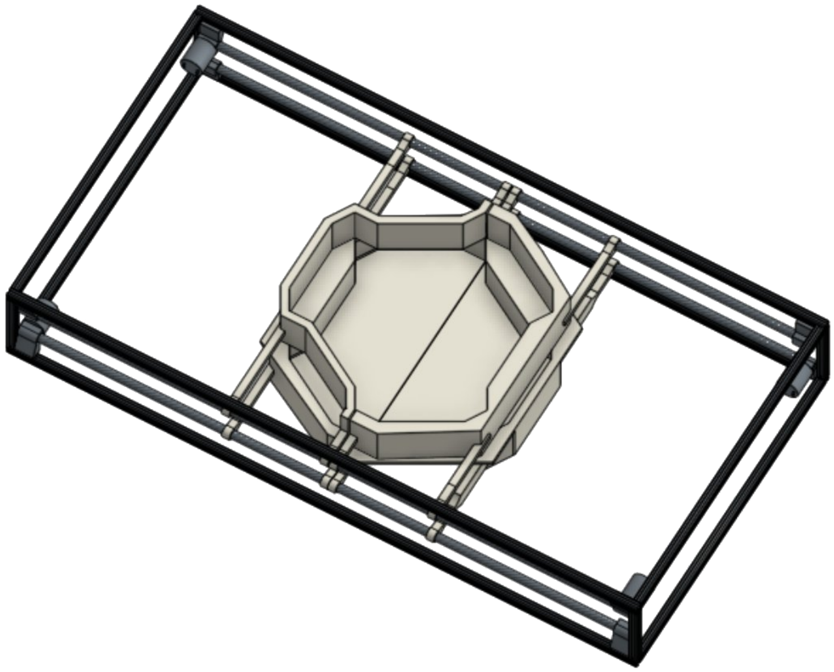


Figure 34: **Sinter mold.** The sinter mold consists of 4 independently moving parts mounted to the frame via linear actuators. This allows for the top and bottom of the mold to independently move enabling layered construction and vibration of the mold to level regolith.

At this point, the mold will move to the compaction/sintering position directly under the microwave press to undergo compaction. At this point, the microwave press will lower and provide roughly 0.4 Mpa of pressure to the regolith to densify the material to the requisite $\rho_b = 2.14\ g/cm^3$ and reduce voids. The compaction phase is essential to the conditioning of the regolith as it enhances microwave absorption uniformity and final structural integrity. The face of the microwave press will be mounted with a quartz face as shown below in Figure 35. The quartz was chosen due to the architectural requirement that the compression plate must be mounted below the waveguide antenna but also must not impede the microwave process. The microwave transparency of quartz, combined with its high compression and temperature-resistant qualities made it an optimal choice for the compression plate. [74][75].

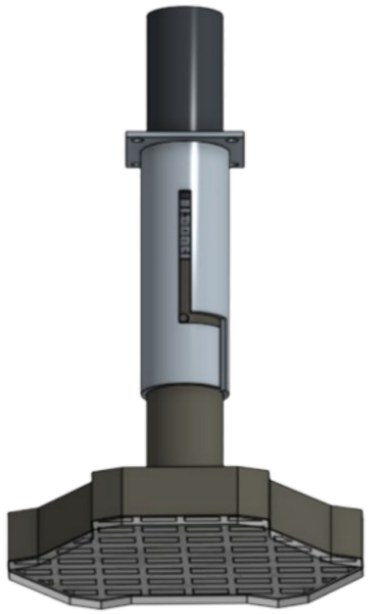


Figure 35: **Microwave Press.** The microwave press is capable of both compacting regolith and subsequently sintering regolith layers. Combining these distinct steps in the tile making process within a single system reduced moving parts and mission complexity.

5.2.3 Microwave Sintering

Microwave sintering was selected as the primary regolith fabrication method due to its ability to efficiently heat lunar regolith while operating within the mission mass and power constraints. The system uses a layer-by-layer sintering process to create dense regolith tiles capable of withstanding repeated rocket landings and takeoffs. These tiles are designed to be modular, replaceable, and repairable, allowing the landing pad system to scale over time as additional tiles are produced.

The microwave sintering process relies on volumetric dielectric heating, where microwave energy penetrates into the regolith and heats the material internally rather than only heating the surface. This creates more uniform heating throughout the material volume and promotes strong bonds both within and between layers. Mare-derived lunar regolith is particularly well suited for microwave sintering due to its iron and titanium oxide content, which improves microwave absorption and thermal coupling. The target sintering temperature for the process is approximately 1,200 °C, where the regolith transitions into a dense glass-like ceramic structure.

To support the required thermal loading and production rate, the system uses three 1.5 kW Gallium Nitride (Gallium Nitride (GaN)) RF generators based on the RIM091K5-20 architecture, providing a total peak microwave power of approximately 4.5 kW at 915 MHz. The 915 MHz operating frequency was selected due to its improved penetration depth in lunar regolith compared to higher microwave frequencies, allowing energy to be distributed more evenly throughout each layer. Additionally, operation at 915 MHz avoids interference with the Griffin Lander's 2.45 GHz WLAN communication systems while leveraging a mature and widely supported industrial microwave frequency band [76].

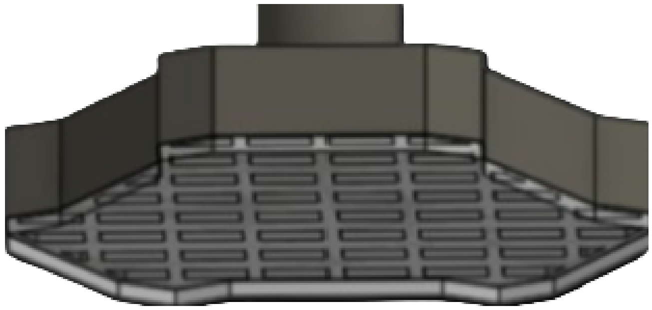


Figure 36: **Microwave Sintering Head CAD Model.** CAD model of the VT Luna microwave sintering head assembly incorporating a custom-designed waveguide applicator system. The assembly integrates the RF generation interface, waveguide feed structure, and regolith-facing aperture used for controlled layer-by-layer microwave energy deposition. The design prioritizes mechanical simplicity, dust tolerance, and reliable RF delivery in a lunar environment, enabling repeatable and uniform sintering of regolith layers for landing pad construction. The sintering head is enclosed with a microwave-transparent quartz layer that provides structural protection, dust mitigation, and controlled compression of the regolith during processing while maintaining high microwave transmission efficiency.

One of the most important design decisions within the microwave system was the selection of the microwave applicator geometry used to deliver RF energy into the regolith. Initial research considered conventional horn antennas, phased-array systems, and localized moving applicators. Traditional horn or patch antennas were ultimately determined to provide insufficient control over the electromagnetic field distribution within the regolith bed. These approaches tend to produce uneven heating regions and localized hot spots, which can lead to inconsistent tile density, thermal cracking, and poor bonding between sintered layers.

Localized moving applicators were also investigated as a possible solution. In this configuration, a small microwave applicator would physically traverse the tile surface while sequentially sintering localized regions. Although this method reduces instantaneous power requirements, it introduces significant mechanical complexity and additional failure points into the system. Any jammed actuator, dust-contaminated rail system, or damaged positioning component could critically impact mission success.

To address these limitations, VT Luna investigated both phased-array microwave systems and custom waveguide applicators. Phased-array systems provide exceptional control over electromagnetic field steering through constructive and destructive interference, allowing microwave energy to be electronically focused and moved across the regolith surface without mechanical motion [77, 78]. While phased arrays offer high adaptability and precise spatial control of the heating region, they require complex phase-shifting networks, extensive calibration, and increased electronic overhead, all of which add significant system mass and design complexity.

Based on this trade study, VT Luna selected a custom waveguide antenna system as the preferred microwave applicator architecture. The waveguide design provides a simplified but highly effective method for directing microwave energy into the regolith layer. Shaping the waveguide geometry and aperture allows for the system to achieve improved electromagnetic field confinement compared to conventional horn or patch antennas, resulting in more uniform energy deposition and reduced localized overheating. This leads to improved thermal uniformity, reduced cracking risk, and more consistent sintered layer density.

Additionally, the waveguide-based approach eliminates the need for phase control electronics required in phased-array systems while maintaining a robust, mechanically simple architecture with fewer failure points. The waveguide also improves coupling efficiency due to minimizing stray radiation and directing a larger fraction of RF power into the regolith bed.

Figure 37 illustrates a representative electromagnetic field distribution within a 915 MHz single-mode microwave waveguide applicator adapted from prior microwave sintering research [79]. The field pattern demonstrates strong spatial confinement and controlled energy deposition within the applicator region, supporting the selection of waveguide-based heating for the VT Luna sintering system. Similar field behavior is expected in the final design and is critical to achieving uniform layer-by-layer sintering performance.

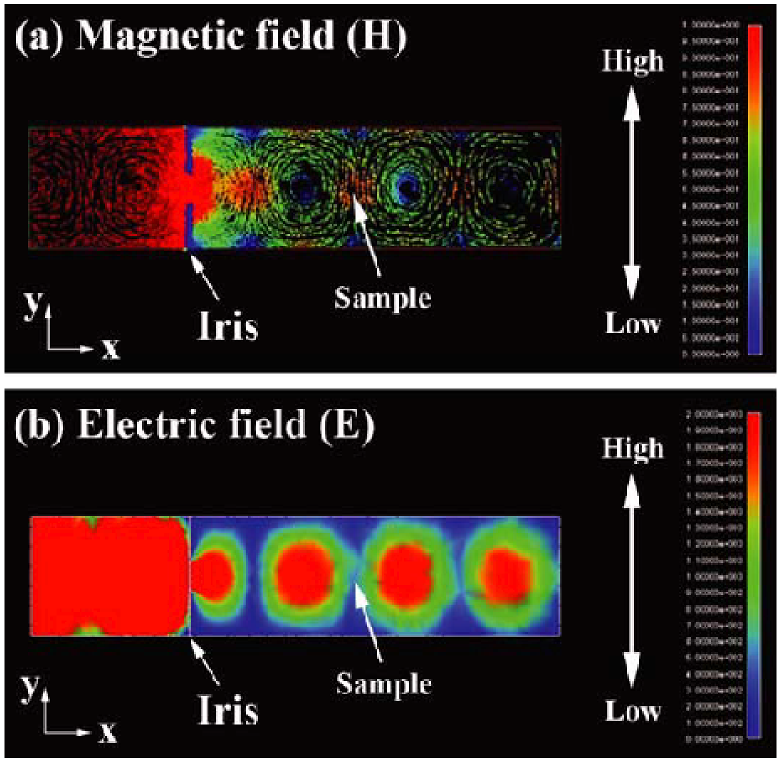


Figure 37: **915 MHz Waveguide Electric Field Distribution.** Representative electric field distribution within a 915 MHz single-mode microwave waveguide applicator adapted from Xie et al. [79]. The figure illustrates strong spatial confinement of electromagnetic energy within the waveguide structure, producing a localized heating region suitable for controlled regolith sintering. This confinement improves thermal uniformity, reduces hot spot formation, and enables more consistent layer-by-layer densification compared to conventional horn or patch antenna approaches. *Note the numbers on the right were not readable in the original paper and therefore could not be fixed for this paper without contacting the author*

5.2.4 Cooling Subsystem Architecture

The STiM payload is fundamentally power reliant with a 4 kW microwave generating 1,600 W of continuous waste heat in a lunar vacuum, the environment is extreme. In the absence of convective air cooling, flawlessly rejecting this massive heat load to deep space is the concern for mission's success. All waste heat must be transported and rejected radiatively per the Stefan-Boltzmann law.

Thermal Architecture Trade Study

To solve the 1.6 kW rejection challenge, a formal trade study was conducted comparing four candidate thermal control architectures against four weighted criteria: total system mass, parasitic power consumption, mechanical complexity, and resistance to lunar dust contamination. With the limited 200 kg total payload mass limit, mass was assigned the highest relative weight of 0.40 in the scoring matrix, nearly three times the weight given to dust mitigation or system complexity. All thermal systems researched took into account the tradeoff of each one to fine-tune the best system for Project Hestia. A weighted scoring methodology was applied and the results of this analysis are presented in Table 22.

A mechanically pumped fluid loop operates through circulating a liquid coolant, typically a water-glycol mixture or a specialized spacecraft fluid such as HFE-7100, through a network of tubing connecting the heat source to a remote radiator [80]. This approach offers significant flexibility in heat routing, as the fluid can be directed around structural obstacles and to radiators positioned at any location on the spacecraft. Pumped loops have demonstrated scalability to thermal loads well above 10 kW in large satellite programs. However, the mechanical pump that drives this circulation is the fundamental source of concern for a systems like Project Hestia's STiM. First, the pump represents a single-point failure: if it seizes or loses prime due to cavitation from dissolved gas or entrained lunar dust particles, the entire thermal control system fails immediately. Second, the pump consumes a continuous parasitic power draw of approximately 50 W, power that the STiM payload cannot spare when simultaneously operating the 4 kW microwave generator and the rover drive system. Third, the added tubing fittings, pump housing, motor controller, and coolant reservoir push the total system mass to approximately 35 kg, which consumes over 17% of the entire payload mass allocation solely for thermal control. For these reasons, this architecture received the lowest trade study score of 0.16.

Pure conduction to structural walls is simplest possible thermal control approach is to rely on direct conduction through the spacecraft's aluminum structural frame, allowing waste heat to passively spread across the vehicle and eventually radiate from whatever external surfaces are available. This architecture carries essentially zero power consumption and zero mechanical complexity, since it requires no moving parts, no working fluid, and no active control. In terrestrial and low-power applications, conduction-based cooling is highly effective. However, on the lunar surface it is fundamentally inadequate for a high-power payload. Aluminum 6061, the structural alloy used throughout the STiM frame, has a thermal conductivity of approximately 167 W/(m·K), sufficient to conduct heat across a short path, but incapable of efficiently transporting 1,600 W across the full length of the payload to a radiative surface without generating severe internal temperature gradients. The result would be extreme thermal hot-spots concentrated near the microwave generator, driving electronics temperatures well above their rated operating limits. Furthermore, since the structural walls are not optimized for thermal radiation, a prohibitively large wall area would be required to achieve 1,600 W of rejection, adding structural mass that serves no load-bearing function. Total system mass was estimated to exceed 65 kg once this additional wall area is accounted for, making this architecture the heaviest of the four candidates. A score of 0.17 was assigned, only marginally above the pumped loop due to its perfect complexity score.

The phase change material is a substance engineered to absorb large quantities of heat at a nearly constant temperature as it transitions between solid and liquid states [81]. Common spacecraft phase change materials include paraffin wax, which undergoes a solid-to-liquid phase transition in the range of 52°C to 58°C, and inorganic salts for higher-temperature applications. The appeal of this approach for the STiM is clear: during the microwave sintering cycle, the 1,600 W heat pulse could theoretically be absorbed into the melting wax, smoothing the load before it reaches the radiator. This would allow the radiator to be sized for the time-averaged load rather than the peak, potentially reducing radiator area and system mass. However, a phase change material stores heat, it does not reject it. All thermal energy absorbed during the sintering cycle must still be rejected to deep space during the recovery period before the next tile cycle can begin. A thermal analysis confirmed that achieving full recovery within the required inter-cycle window demands a radiator of essentially the same area as a direct-rejection architecture, eliminating any mass benefit on the radiator side. More critically, the quantity of paraffin wax required to absorb a 1,600 W load over 30 minutes, representing approximately 2.88 MJ of stored thermal energy, is approximately 45 kg at a latent heat of fusion near 200 kJ/kg. This mass penalty alone represents nearly a quarter of the entire mass limitation, making this architecture immediately disqualifying for the system. A score of 0.23 was assigned.

The passive heat pipe with articulated radiator architecture was selected as the winning design with a weighted score of 0.44. A heat pipe is a hermetically sealed, two-phase heat transfer device that operates on the thermodynamic principles of evaporation and condensation to transport large quantities of heat across a distance with an extremely small temperature gradient and no external power input [82]. The working fluid inside the pipe evaporates at the hot end (the evaporator, located at the heat source), travels as vapor to the cold end (the condenser, located at the radiator), condenses back to a liquid, and is returned passively to the evaporator through capillary forces in a porous wick structure. This thermodynamic cycle requires zero pumping power; the waste heat itself provides the only driving force for the fluid circulation. Because no pumps, motors, or coolant plumbing are required, the total mass for the complete 2-Loop architecture is only 17.3 kg, a mass savings of more than 17 kg relative to a pumped fluid loop, and over 47 kg relative to pure structural conduction. The articulated radiator panels are coated with AZ-93 white paint ($\varepsilon = 0.91$, $\alpha_s = 0.14$), providing highly efficient thermal emission to deep space while minimizing solar heat gain during the lunar day [83]. Motorized hinges allow the panels to be deployed 90° outward for maximum rejection during operations and stowed flat against the vehicle frame during transit and lunar night. This stow capability is what earned this architecture the highest dust mitigation score of 9 out of 10: as this system actively protects the radiator surface from the accumulation of thermally insulating lunar dust, a known long-duration mission risk for any static exposed surface on the Moon [80]. Due to this active mitigation the architecture preserves nearly full thermal rejection capacity throughout the relatively short mission lifetime [80].

Lunar Regolith Adhesion and Radiator Degradation

One of the most mission threatening risks for any long-duration surface thermal system on the Moon is the adhesive behavior of lunar regolith. Unlike terrestrial dust, lunar regolith particles are highly angular and jagged at the microscopic level, having never been subjected to the erosive rounding processes present in terrestrial environments. More critically, these particles are continuously charged by solar ultraviolet radiation and solar wind ion bombardment, generating a strong electrostatic charge that causes them to adhere aggressively and unpredictably to virtually any exposed surface [80]. For a thermal radiator, this is a direct mission risk. Regolith accumulation degrades the thermo-optical surface properties of the panel in two compounding ways. First, because lunar regolith has extremely low thermal conductivity, a deposited layer acts as a thermal blanket, physically impeding the radiative transfer of heat from the panel surface to deep space. Second, and more severely, the dark basaltic composition of regolith particles significantly increases the solar absorptivity of the panel surface while reducing its effective emissivity, worsening the critical ratio between heat absorbed from sunlight and heat rejected to space. Studies of Apollo hardware confirm that this effect is not theoretical: dust contamination of the Lunar Roving Vehicle's radiator panels caused the onboard batteries to operate continuously above their target temperature thresholds despite active cleaning attempts, demonstrating that even partial dust coverage can produce measurable and harmful thermal performance degradation [80].

Project Hestia's architecture directly addresses this risk through the motorized panel articulation system. During all non-operational periods, including nighttime survival, payload transit, and any intervals between sintering cycles, all radiator panels are commanded to stow flat against the vehicle frame. In this stowed position, the AZ-93-coated radiating surfaces face inward toward the structural frame rather than outward toward the open lunar environment, providing physical protection from regolith deposition. When the microwave generator is activated and active heat rejection is required, the panels are deployed 90° outward by the 12 V gearmotor hinge assemblies, exposing the clean radiating surfaces to deep space.

Beyond passive stow protection, the motorized hinge system is additionally leveraged as an active dust-clearing mechanism. Because the rover deposits regolith into the sintering mold at the conclusion of each collection run, a predictable deposition event occurs on a cycle-synchronized basis. Following each rover deposit and prior to the initiation of the next sintering cycle, the flight software is designed to command a rapid panel articulation sequence, in which the radiator panels are driven from the deployed position through a full stow and back to deployed. This repeated angular actuation generates sufficient inertial force at the panel surface to dislodge loosely adhered regolith particles and allow them to fall clear of the radiating surface under lunar gravity. The 12 V DC gearmotors driving the hinge assemblies are already sized for panel deployment

Table 22: **Project Hestia Thermal Cooling Trade Study Results.** This table shows the trade study of thermal cooling systems and the justification behind the missions employment of a passive heat pipe and radiator system due to the low mass and power requirements and high dust mitigation [80]

Technology	Mass (kg)	Power (W)	Complexity	Dust Mitigation	Score
Weights	*0.40*	*0.30*	*0.15*	*0.15*	*1.00*
Mechanical Pumped Fluid Loop	~ 35.0	~ 50	8	1	0.16
Pure Conduction to Walls	> 65.0	0	1	1	0.17
Phase Change Material	~ 45.0	0	2	1	0.23
Heat Pipes and Radiators	**17.3**	**~ 5 (intermittent)**	**3**	**9**	**0.44**

loads, and this clearing actuation sequence introduces no additional mass, no consumables, and no dedicated hardware. It is acknowledged that electrostatically bound regolith particles may resist purely inertial clearing, as electrostatic adhesion forces in the vacuum lunar environment can exceed the gravitational and inertial forces generated by panel actuation alone. The precise effectiveness of this strategy therefore remains difficult to predict analytically without dedicated surface test data, and it is identified as a residual risk recommended for characterization during future hardware-level validation phases.

As detailed in Table 22, the 2-Loop Hybrid Architecture, referred to throughout this paper as Project Hestia, achieves the optimal balance of mass, power, and reliability for the system. The architecture utilizes zero-power passive heat pipes paired with motorized articulated radiators, achieving a weighted trade study score of 0.44 out of a possible 1.00 while maintaining a total thermal system mass of only 17.3 kg.

Heat Pipe Operating Principles & Thermodynamics

To understand why the passive architecture is so effective, the operating principles of the heat pipes must be examined [82]. A heat pipe is a completely passive, two-phase heat transfer device. It consists of a sealed envelope, a porous wick structure on the inner wall, and a working fluid.

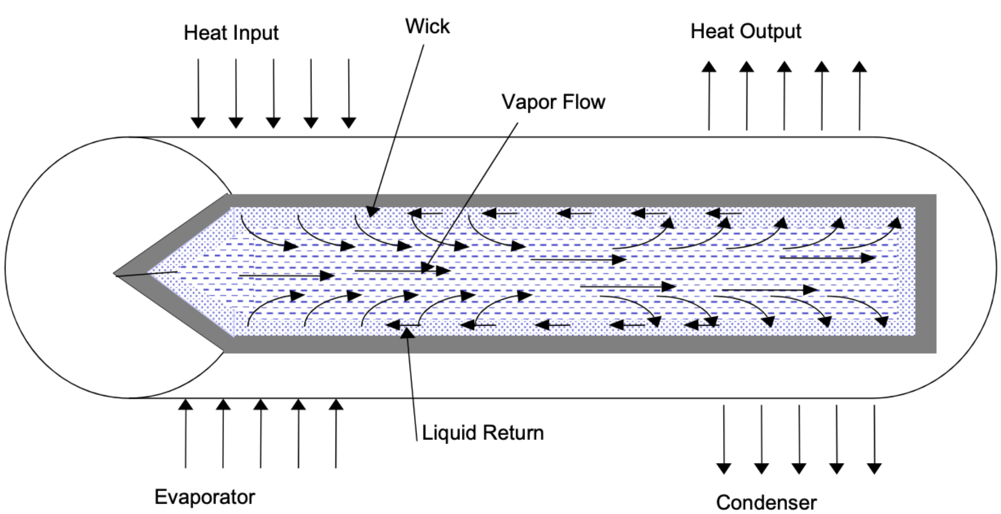


Figure 38: **Heat pipe operating mechanics.** Operating principles of a heat pipe, showing vapor flow and liquid return via capillary action. This allows for the passive transfer of heat, ideal for the mission requirements due to being low mass and low complexity

Heat is applied to the evaporator section, vaporizing the liquid working fluid. As liquid is vaporized, the vapor pressure builds up, forcing the vapor to flow axially along the center core to the colder condenser section. At the condenser, the vapor condenses back into a liquid, releasing its latent heat to the radiator panel. The liquid is then drawn back to the evaporator through the capillary force generated along the wick grooves. The continuous energy flow Q is governed by the mass flow rate $\dot{m}$ and the latent heat of vaporization λ:

$$Q_e = Q_c \equiv Q = \dot{m}\lambda \tag{5.1}$$

At any cross section of the heat pipe, a pressure differential exists between the vapor and liquid phases. This ΔP is sustained through the surface tension force developed at the liquid/vapor interface at the tip of each groove. The lowest pressure differential occurs at the very end of the condenser (zero), and the highest occurs at the very end of the evaporator. This purely passive cycle requires no external pumping power; the waste heat itself provides the driving force for the fluid flow.

Constant Conductance Heat Pipe (CCHP) Specifications and Thermal Limits

There are three basic functional types of heat pipes: Variable Conductance (VCHP), Diode, and Constant Conductance (CCHP) [82]. The STiM payload exclusively utilizes CCHPs to maximize reliability and minimize mass. The selected CCHPs feature an Aluminum 6061 envelope, an Ammonia working fluid, and a 1.27 cm outer diameter.

Given the extreme lunar environment, the thermal limits of the ammonia CCHPs are a primary design constraint:

- **Lowest Temperature Limit (-173°C):** Ammonia freezes at -77°C. During the 3-4 day lunar night at the South Pole, surface temperatures plunge to -173°C. Consequently, the ammonia fluid inside the pipes will freeze solid. **This is not a failure.** Aerospace-grade ammonia CCHPs are explicitly designed to survive thousands of freeze-thaw cycles without bursting. In fact, this freezing acts as a perfect thermal diode, safely deactivating the pipe and preventing the avionics core from bleeding its survival heat out into deep space.

- **Highest Temperature Limit (+80°C):** The critical temperature of ammonia is +132°C. The radiator sizing ensures the system operates at a safe 80°C, well below the supercritical threshold.

- **Evaporation Load Limit (1.6 kW):** Evaporating 1,600 W of continuous heat poses a potential dry-out risk for a single heat pipe. To mitigate this, the architecture utilizes a parallel array of **14 CCHPs** for the microwave generator. This divides the load to approximately 114 W per pipe, which is well within the 150–300 W capillary transport limit of a 1.27 cm aerospace CCHP.

The 2-Loop Project Hestia Architecture

Because the payload produces two thermally distinct heat loads, the thermal control system is divided into two independent loops (Fig. 39).

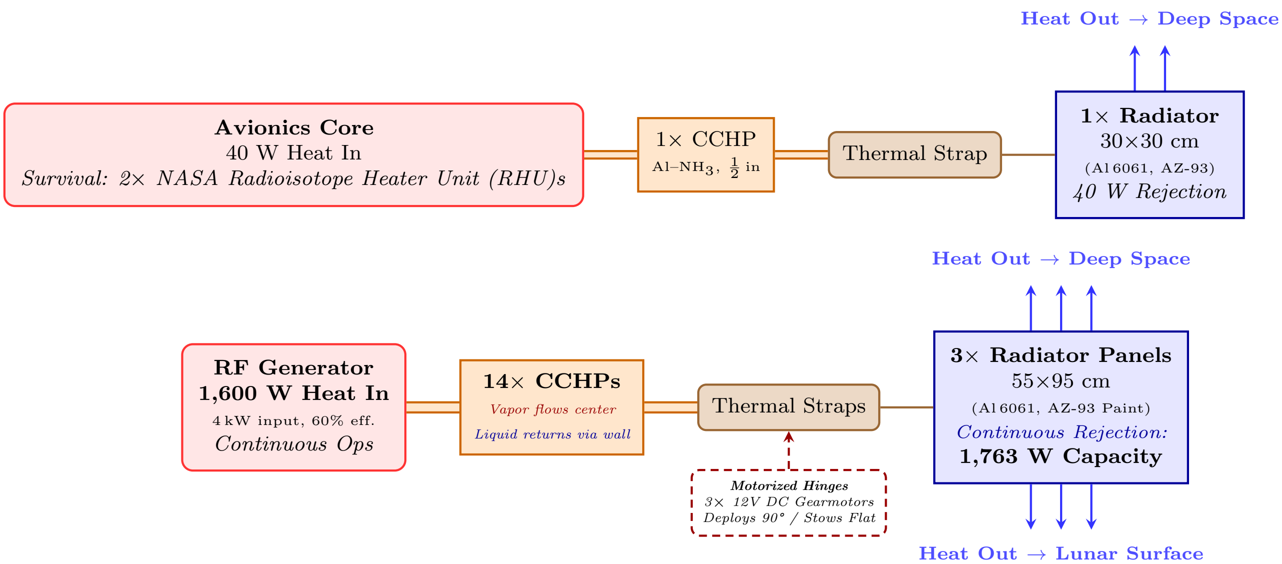


Figure 39: **Thermal architecture.** Finalized Hybrid 2-Loop thermal architecture. The primary loop utilizes motorized hinges for day and night panel articulation and therefore can adjust heat dissipation and retention to the systems needs.

Loop 1: Avionics

The rover computer, sensors, and motor controllers produce a continuous 40 W load. A single aluminum-ammonia constant conductance heat pipe transports this heat to a 30×30 cm Al 6061 radiator panel coated with AZ-93 white paint ($\varepsilon = 0.91$, $\alpha = 0.14$) [83]. A motorized hinge deploys the panel 90° outward during the sunlit phase to face deep space, and stows it flat against the frame during shadow for thermal insulation. Notably, this motorized hinge also provides a critical dust mitigation feature: the ability to physically ´´flap" the panel to shake off hazardous lunar regolith buildup.

Zero-Power Night Survival: During the unpowered 96-hour lunar night, the stowed panel minimizes the parasitic heat leak to ∼1.6 W. Two NASA RHUs [84] provide 2 W of continuous internal heat. Using just these two RHUs, the core effortlessly survives the lunar deep freeze, remaining safely above its -40°C limit without any electrical power.

Loop 2: RF Microwave Generator

The solid-state microwave generator operates at 60% wall-plug-to-RF efficiency [85], producing 1,600 W of waste heat at 4 kW input. Combined with the 40 W avionics load, the total STiM payload continuously generates 1,640 W of waste heat during sintering operations. Because uninterrupted tile production is mission-critical, Phase Change Material (PCM) thermal buffering was discarded in favor of a fully continuous cooling architecture.

Fourteen zero-power ammonia CCHPs route this heat to three deployable 55×95 cm Al 6061 radiator panels (1.50 m^2 total radiating area). Similar to the avionics loop, the motorized hinges allow these large panels to flap and clear regolith dust accumulation to maintain peak efficiency.

These massive arrays are sized to reject 1,763 W continuously to deep space and the lunar surface. As shown in Fig. 41, the 1,640 W total waste heat load is fully mitigated through the 1,763 W rejection capacity, allowing the architecture to operate with a safe +10.2% thermal margin [86]. This zero-PCM design completely eliminates microwave downtime, allowing 24/7 continuous sintering. During the lunar night, the RF generator is powered off and safely permitted to deep-freeze to -173°C without damage.

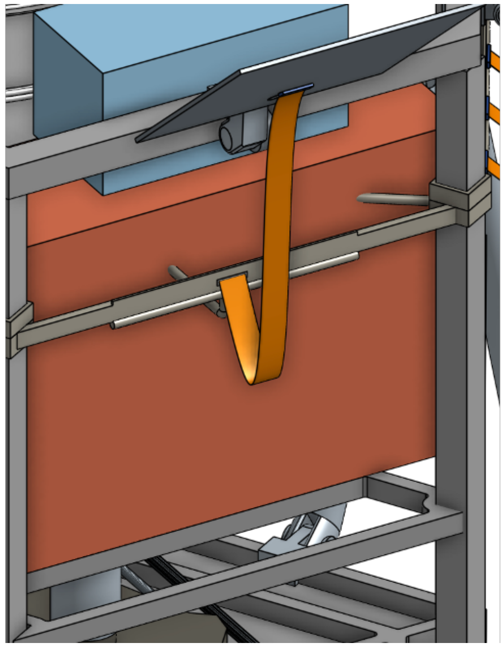


Figure 40: **Heat strap routing.** Expanded view of the Project Hestia 2-Loop cooling system, separating avionics from the RF generator. The heat strap in this image is exaggerated to visually depict the heat strap routing such the systems are thermally isolated.

Figure 40 illustrates an expanded view of the cooling system, specifically detailing the interface between the heat pipes, the flexible thermal strap, and the external radiator panel. While the CAD model depicts an elongated strap for visual clarity, the physical flight hardware utilizes significantly shorter thermal straps, with the heat pipes routed in closer proximity to the upper frame. One of the primary engineering challenges associated with passive heat pipe architectures is managing rigorous physical placement and routing constraints. This specific structural design provides the heat pipes with sufficient spatial clearance to avoid sharp bend radii, ensuring optimal capillary flow and top-tier thermal transport performance.

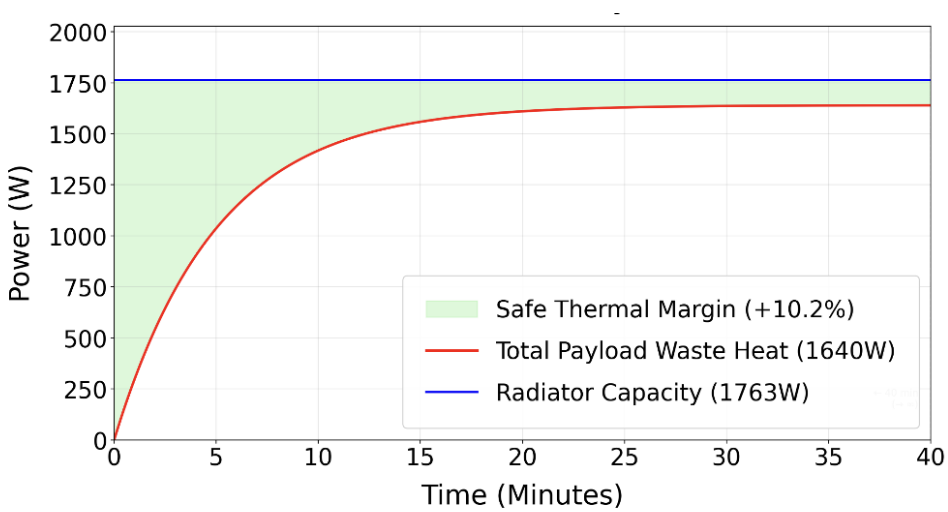


Figure 41: **Project Hestia thermal performance.** Three large passive panels provide 1,763 W of constant cooling, maintaining a continuous +10.2% margin against the 1,640 W payload heat load. Therefore, until degradation of the radiative system due to environmental exposure an regolith accumulation, there is excess dissipation to facilitate continuous full power operation.

Table 23: Thermal management system mass breakdown for the two-loop Project Hestia architecture.

Loop	Component	Mass (kg)
Avionics	Radiator Panel (30 × 30 cm)	1.36
Avionics	CCHP (×1) + Flex Strap	0.05
Avionics	Motorized Hinge + RHU Pellets (×2)	0.48
Avionics Subtotal		**1.89**
Microwave	Radiator Panels (×3, 55 × 95 cm)	12.30
Microwave	CCHPs (×14) + Flex Straps	1.06
Microwave	Motorized Hinges (×3, 12 V DC Gearmotor)	2.10
Microwave Subtotal		**15.40**
Total Thermal System		**17.30**

Sintering Chamber Insulation

The sintering chamber is thermally isolated from the cooling loops. MLI rated to 1500°C envelops the mold cavity, trapping the 2400 W of RF energy inside the regolith [87, 88]. The only connection between the generator and the chamber is the RF waveguide, which conducts electromagnetic energy but negligible thermal energy. This decoupling ensures that the cooling system removes waste heat from the electronics without diminishing the sintering temperature.

Cooling System Impact

Ultimately, the Project Hestia thermal architecture is the vital cornerstone that enables continuous STiM operations. By employing a completely passive, zero-power heat pipe infrastructure, the system successfully rejects 1.64 kW of continuous waste heat while weighing only 17.3 kg accounting for less than 10% of the total 200 kg payload mass budget. Without this capability, the high-power microwave generator would undergo catastrophic thermal runaway within minutes, crippling the entire mission. The ability to autonomously survive the -173°C lunar night via RHUs and effortlessly reject heat during the lunar day proves that this architecture is robust, highly scalable, and fully prepared to support the relentless, high-energy demands of autonomous lunar infrastructure construction.

5.2.5 Shielding

Shielding against radiation and micrometeorites allows lunar missions to be sustainable over much longer durations. Primary risks from radiation include damage to electronic subsystems and structural components that compounds over mission durations.

Radiation Damage to Electronics

Damage to electronic components as a result of radiation occurs both through the over time degradation due to "accumulation of ionizing dose deposition over a long time in insulators" and through "Single Event Effects (SEE) which are probabilistic failures induced by single high-energy ionizing particles" [89]. SEE, are the result of high energy particles that, due to their excessively high energy, are impractical to shield against [89]. These particles originate from interstellar sources, solar flares, and radiation belts. The mission traverses through the Earths Van Allen radiation belts briefly during lunar injection. However, for the duration of the lunar surface operations, solar flares and interstellar high energy particles are the greatest sources of potential mission ending SEE. Significant testing will be conducted to determine the missions specific susceptibility to SEE and modifications to the placement of electronic systems and addition of redundant systems can mitigate the risk these SEE pose such that, statistically, the mission will be capable of surviving for its intended six month surface operation.

Radiation Damage to Structural Elements

The potential for damage to structural systems requires that structural elements have effective shielding or be over-designed to accommodate the decay of elastic properties, as "metals exhibit reduced elasticity and ductility and increased hardness following irradiation" [90]. For this mission, structural elements are over-designed such that, over a six month mission duration, concerns of elastic decay are mitigated. This over design is already built into the structure to allow effective launch to a lunar injection orbit without structural failure. Therefore, the durability of these systems exceeded what will be needed for lunar operations within the scope of the mission duration due to radiative decay.

Micrometeorite Damage

Micrometeorites also impact the lunar environment with hyper-velocity collisions resulting damage that can be mission ending. However, collisions on the lunar surface occur at lower relative velocities than those in orbit. Furthermore, as the STiM only occupies approximately one square meter of lunar surface and is only expected to survive six months of operation, only particles on the order of micrograms and smaller are likely to be impacting as shown in Figure 42.

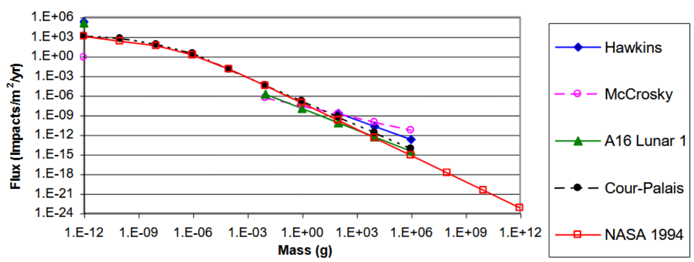


Figure 42: **Micrometeorite flux by mass.** Figure showing the flux of micrometeorites of varying mass with data collected from a variety of missions [91]. This shows that only small objects on the scale of micrograms are likely to impact a small area short duration mission such as this one.

Furthermore, as the mission is located near to the lunar south pole, it can expect to receive less micrometeorite flux than the average lunar surface [92].

Penetration from micrometeorites of the size expected to impact this mission are capable of degradation of system components over longer durations with larger impacts on the scale of millimeters being capable of single event catastrophic damage [93]. However, the accumulated damage from "flux of small particles typically results in a cratered surface area of 0.01–0.001% per year of exposure" and therefore is not relevant for the six month lunar surface operation of this mission [94].

Shielding Architecture and Justification

Due to the short mission duration, usage of rad hardened components, and low micrometeorite flux at the chosen landing site VT Luna settled on light MLI based shield. Although the incorporation of shielding solutions such as a state of the art Whipple shield would allow for a longer mission operation, the aforementioned timeline choices made due to lighting conditions and regolith accumulation make the justification of the added mass of such a shielding solution unreasonable. The MLI based shield would serve as a physical barrier to the lunar dust protecting internal components from abrasion. The Aluminized Kapton and Mylar layers in the cover will act as light radiation shielding, mitigating some mission ending SEE. These layers will also help insure efficient thermal control[95].

5.2.6 Structural Analysis

A structural analysis of the STiM was run using ANSYS in order to ensure that the design could survive the acceleration that it would experience during launch and landing. The model used for the analysis was a simplified version of the full STiM model, as the full model was too complex for the program to handle within an acceptable time period. The only parts that were simplified were the sifter and mold area, with the simplified versions having the same weight and location as in the full model. This simplified model also had the STiM's radiator panels and legs folded inwards to more resemble its configuration during launch.

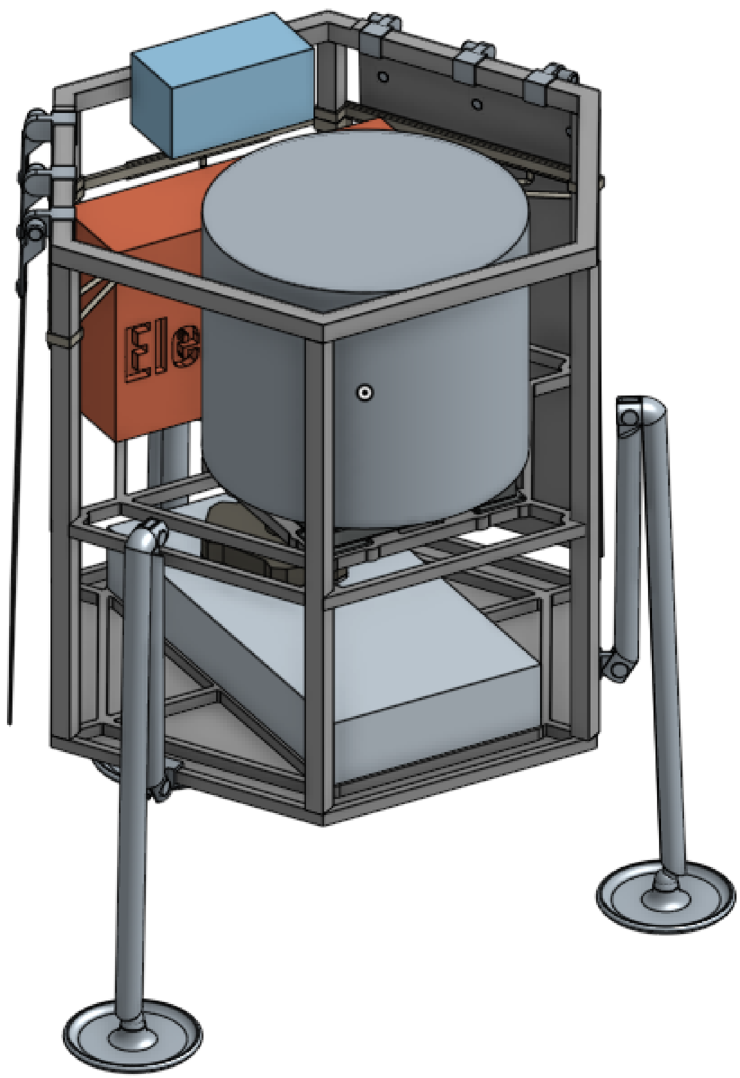


Figure 43: The Simplified model of the STiM used for structural analysis

The structural analysis was run with the STiM experiencing an acceleration of 100 m/s^2 while recording the Stress and strain as well as the deformation.

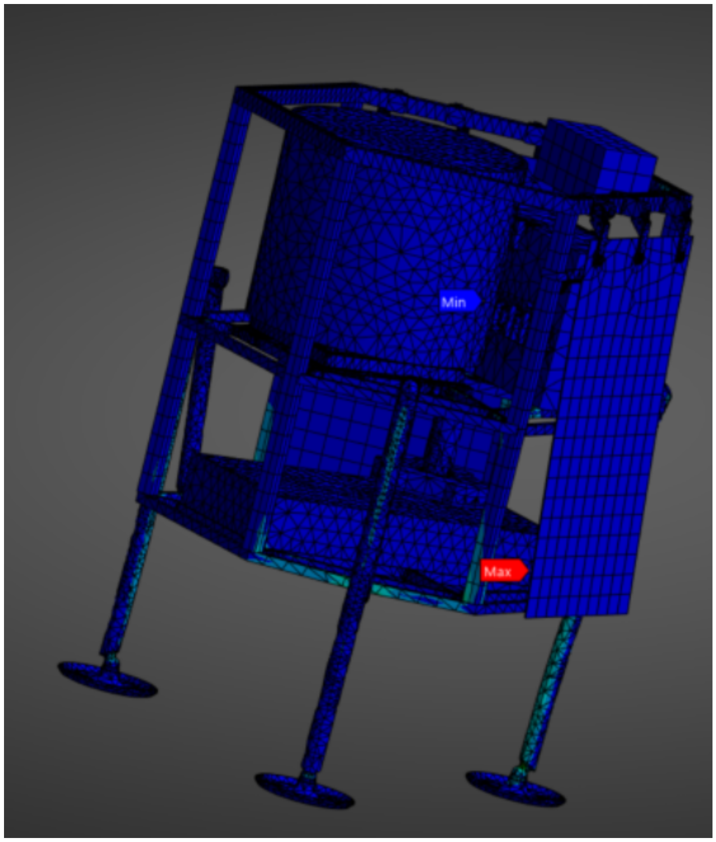


Figure 44: The model of the STiM after an analysis of stress at a 100 m/s^2 acceleration

The stresses the STiM experienced at this acceleration were well below the limit of what the material could withstand. The maximum stress experienced at any point in the body was 100 MPA. The majority of the STiM, including the legs and frame, will be made of aluminum T6, which has a yield strength of about 380 MPA. The highest levels of stress were experienced within the legs and in the frame around the legs, this means that the stress experiences during the test was under 1/3 of the yield strength of the material. Furthermore, The simulation was run with the STiM standing on its legs, however during the launch the Lander will likely be more supported, so the launch should cause even less stress on the STiM than the simulation predicts. The simulation shows that the STiM can easily withstand the acceleration involved in launching and landing.

5.3 Communication System

The communication architecture prioritizes low mass and reliability, utilizing the Lander as a relay rather than Direct-to-Earth communication. The communication system is an essential part of the mission to relay information from the mission to Earth. The design will use the Griffin Lunar Lander's communication system to directly relay information from the stationary tile maker and a separate system that will use the Lander as a relay for the rover system. The following systems are gone into depth below.

5.3.1 Griffin Lander Communication System

The Griffin Lunar Lander communication system enables Lander commanding and telemetry. The communication system relays information from both the Lander itself as well as the payload to Earth. To do this the Lander uses a high-powered and flight-heritage transponder, as well as multiple antennas. The Lander has multiple low gain antennas to maximize coverage as well as a medium and high gain antenna as well for long distance communication with Earth. The high and low gain antennas can be seen below in a diagram of the Lander in Figure 45. The Lander-Earth communication operates in the X-Band frequency range. The connection of the Lander and payload is achieved through a Serial RS-422 or SpaceWire for wired communication throughout the mission. This method is reliable and will be viable for the stationary tile maker part of the payload.

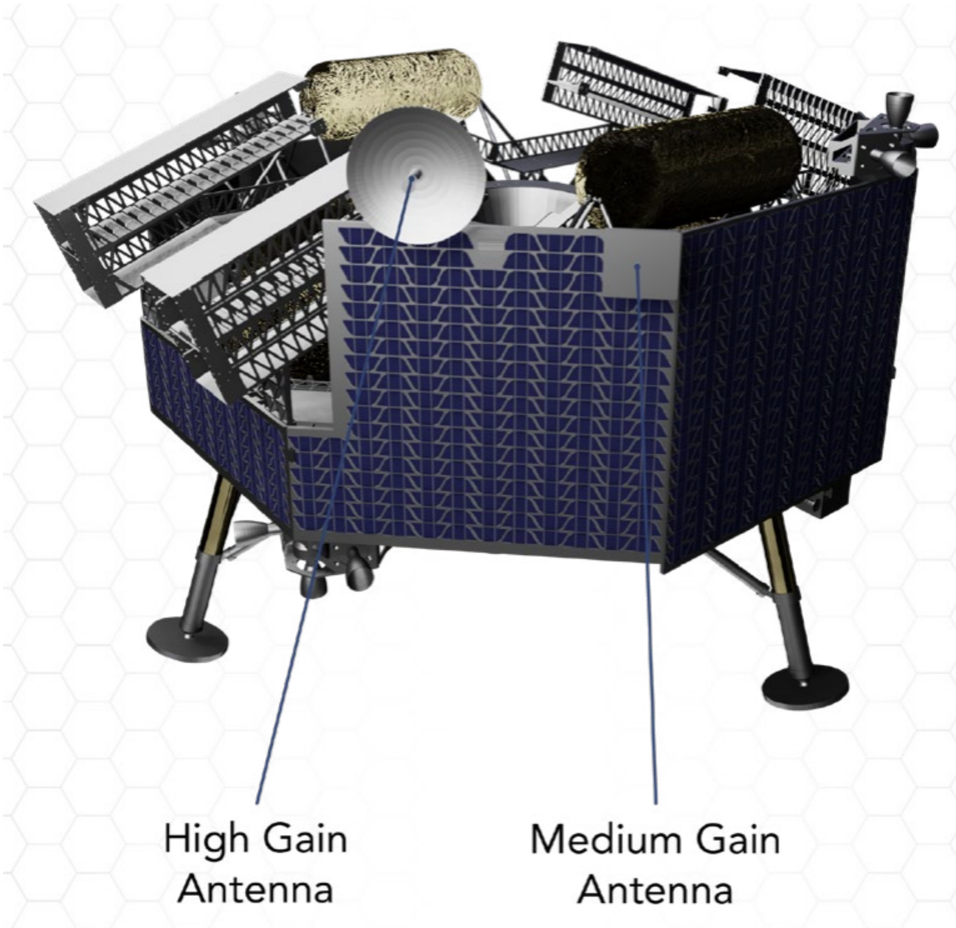


Figure 45: **Griffin Lander antennas.** Diagram of the Griffin Lunar Lander highlighting high and medium gain antennas which allow the Lander to communicate to Earth in the X-band. This allows for mission monitoring and modification from on Earth stations [9].

The Lander does also have wireless connection capabilities through a 2.4 GHz IEEE 802.11n compliant wire local area network (WLAN) modem. Payloads compatible with 2.4 GHz IEEE 802.11n can communicate with the rover, and the Lander relays payload telecommands and telemetry in near real-time. This feature was leveraged for the rover system part of the payload and is discussed more in the next section. The process can be seen in the diagram below produced by Astrobotic. The in-place communication system that the Griffin Lunar Lander uses is a key part of the mission and heavily relied on.

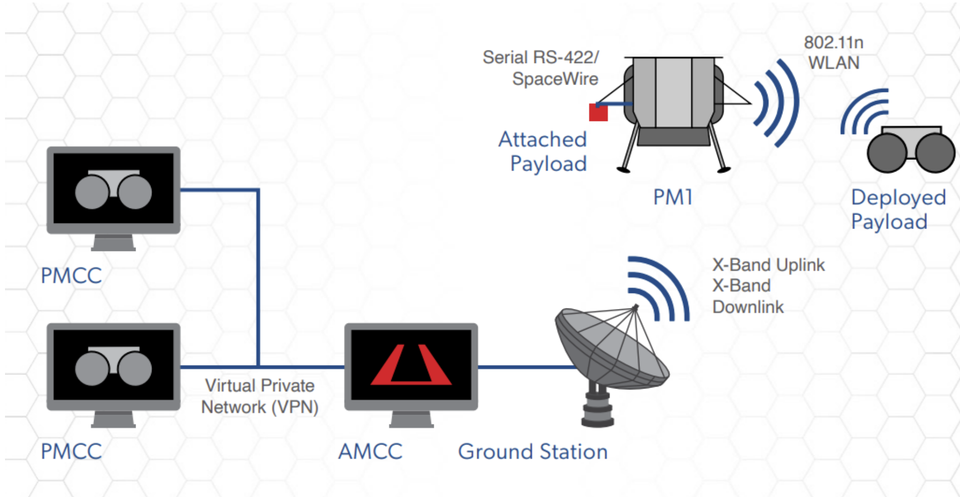


Figure 46: **Mission flow of communications.** Diagram of communication process between rover, Lander, and ground station. This shows the usage of the X-Band for ground station to Lander communication and the usage of WLAN for communication in the lunar environment between the STiM, Griffin Lander, and rover [9].

5.3.2 Rover Communication System

As the rover will completely detach from the Griffin Lunar Lander, it will need a communication system of its own. Establishing a robust and reliable form of communication system for the rover is a fundamental requirement for ensuring the success of the rover and achievement of the goals. The communication system serves as the rover's connection to the rest of the mission as it is deployed. It allows the rover to receive commands from the Lander, report its health and status, transmit sensor data, and coordinate the regolith-collection workflow in real time. This link is how the rover stays synchronized with overall mission objectives, updates its tasks as the tile-maker requires more material, and lets operators monitor progress and intervene if needed. In short, without a reliable communication system, the rover cannot safely navigate, cannot deliver materials on demand, and cannot function as an integrated part of the landing-pad construction architecture.

In theory the simplest method of communication is to set up a link directly between the rover and a ground station on Earth. This method while effective and proven, is not without drawbacks. To start, a high-gain antenna is required to relay information all the way to Earth from the Moon as the distance between Earth and the Moon is so great, approximately 384,000 miles [96]. A high-gain antenna adds both mass and complexity to the communications design. Gain is the term used to describe the antenna's ability to amplify signal power. In essence, how much the antenna can focus the radio beam and in turn the distance it can transmit it. As the radio beam is narrowed, the sweep of directions it covers decreases. This means whenever the rover moves and turns, the high gain signal runs the risk of missing Earth [97]. Additionally, high gain antennas in general have more mass than low gain antennas as larger parabolic dishes are good at focusing radio beams are great high gain antennas [97]. For example the NASA Perseverance rover uses a high gain antenna system that weighs approximately 8 kg [98]. The rover is simply too small to handle the complexities as well as mass concerns that using a high gain antenna would cause.

The Griffin Lunar Lander can serve as a relay between the rover and Earth, which opens the door for the use of low-gain antennas that do not have the distance range to send radio waves to Earth directly. The lander uses a 2.4 GHz IEEE 802.11n-compliant WLAN modem to connect with deployable payloads wirelessly [9]. This narrows down the options for communication systems as the design must be compatible with the Griffin Lunar Lander modem.

The design will use a WGM160P module to establish the rover's primary wireless communications link with the lander. This module provides a compact and low-power Wi-Fi radio that interfaces directly with the rover's onboard computer to maintain a continuous data link with the Griffin Lander [99]. The WGM160P operates in the 2.4 GHz band and supports IEEE 802.11b/g/n protocols, making it directly compatible with the Griffin Lander's onboard WLAN modem. The module draws a maximum of 347 mW during active transmission, which is well within the rover's available power budget and represents a significant advantage over higher-power radio alternatives [99]. Additionally, power to the module will be conditioned through regulated and filtered electronics. A dedicated DC-DC converter will step down the rover's bus voltage to the 3.3 V supply required to power the module, and bypass capacitors will filter transient noise that could otherwise disrupt the radio frequency circuitry. Because these electronics are not independently rated for the lunar environment, they will be housed in a protected electronics box. The design will also incorporate thermal insulation methods discussed later in this section.

For the antennas, the design will use two 2.4 GHz Pagoda antennas. While the distance range is lower for these antennas than a high-gain or medium-gain antenna, the rover will not travel very far from the Griffin Lunar Lander at all, making it so that the antennas do not need to have a great distance range. These low-gain antennas have a gain of 0.6 to 1.3 dBi, which is very low and will ensure a wider range of coverage, which also reduces complexity in how the design orients the antennas. The Pagoda antenna achieves this near-omnidirectional pattern through a stacked, symmetrical radiating element geometry that produces a toroidal radiation pattern, covering nearly the full hemisphere above and around the rover without any mechanical pointing requirement. The wide range of these antennas can be seen in Figure 47 below, which compares the ranges of different dBi (the measurement of gain) antennas. These antennas are also built for lunar communications and are compatible with the 2.4 GHz IEEE 802.11n WLAN modem and thus will work with the communication architecture. The design will use two on opposite sides of the rover to ensure full coverage as well as add redundancy to the system in the case that one is damaged. This is a common practice in spacecraft missions to prevent single points of failure [97]. With both antennas active, any orientation of the rover relative to the lander will keep at least one antenna within line of sight, maintaining an uninterrupted link throughout the regolith-collection operation.

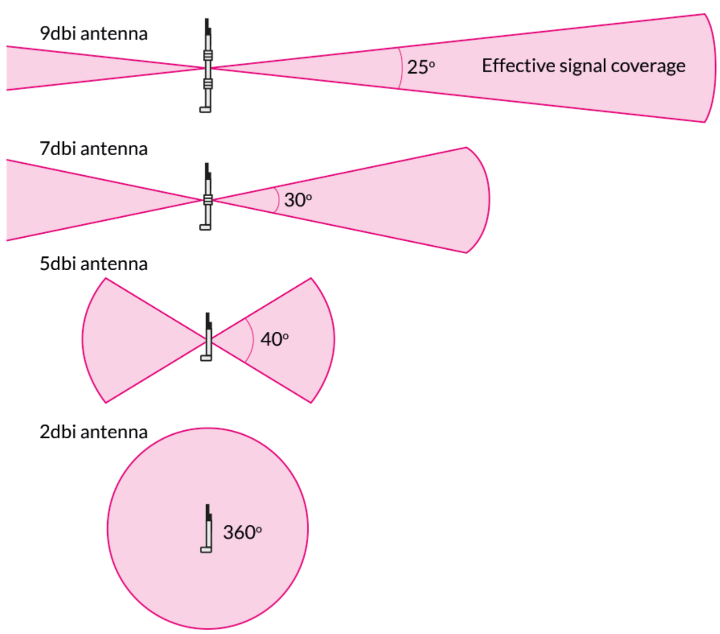


Figure 47: **Antenna gain range in relation to dBi.** The figure shows the approximate antenna gain ranges used to classify low-gain, medium-gain, and high-gain antennas, with low gain ranging from 0–3 dBi, medium gain from 3–8 dBi, and high gain from 9–15 dBi. This informs the missions profile due to restrictions in communication to prevent communication loss with the rover during regolith harvesting operations [100].

Finally, the design will need to have an insulation system for these electronics, as they are not all designed to survive the lunar environment. Extreme temperatures, large temperature swings, and solar radiation are among the biggest concerns for these electronic components. For this reason, the design will use MLI. The MLI will consist of aluminized Kapton, Kapton, Mylar, and aerogel. This layered architecture is designed to combat radiation, thermal, and dust concerns. The aluminized Kapton is reflective and can reflect solar thermal radiation that would otherwise damage the electronics. This material is also functional between -250°C to 290°C, which encompasses the expected lunar surface thermal range [101]. Kapton laminate has a very low absorptance of under 0.14, which means the material does not heat up as much under the sun. Kapton also has a very low emittance of under 0.04, meaning heat loss is slowed during cold lunar nights [102]. This combination makes Kapton a fantastic way of regulating the extreme temperatures experienced on the Moon. The Mylar layers in the MLI stack serve as additional low-emittance radiation shields between the Kapton layers, further reducing the net heat flux across the blanket. Because each reflective layer intercepts and re-radiates energy rather than conducting it, the addition of even a small number of Mylar sheets can meaningfully reduce the effective thermal conductance of the overall blanket [102]. Finally, the design will use aerogel layers in the MLI. Aerogel is extremely resilient to low and high temperatures, up to 1100°C, incredibly light as the material is mostly air, and has very low thermal conductivity, making aerogel a great insulation material. The thermal conductivity of aerogel is approximately 0.015 W/m·K at ambient conditions, which is lower than that of still air and roughly an order of magnitude below that of conventional foam insulation [103]. In the past, aerogel has been too brittle; however, advancements in the technology have made the material more flexible and stronger. Aerogel is about one-third the mass of state-of-the-art insulation materials [103]. There is no doubt that aerogel is the future of spacecraft aviation and will serve the insulation requirements well. These materials will be layered together with the aluminized Kapton on the outside, as this layer is reflective, and the aerogel on the inside, as aerogel is still the most fragile of the materials. There will be approximately 10 to 20 layers, only accumulating to a few millimeters thick, as this is standard [104].

The rover communication system is designed to be lightweight, effective over short ranges, robust, and environmentally protected. The design will accomplish this using the Griffin Lunar Lander as a relay, a Wi-Fi module, low-gain antennas, and MLI of proven materials. The low mass of each chosen component is crucial, as the weight of the rover is so small. The relay capabilities of the Griffin Lunar Lander open the door for the use of lower distance range antennas, which is also a large factor in keeping the mass low. This communication system will help transmit and receive critical information from the rover as it gathers regolith for the construction of tiles.

6 System Summary

6.1 Mass Budget

The system mass budget was developed using a bottom-up estimation approach based on preliminary subsystem sizing and component-level design estimates, rather than relying solely on historical percentage allocations commonly used during conceptual design phases. Separate detailed mass budgets were generated for both the rover mobility platform and the STiM processing system. Major subsystems included structural components, power systems, actuation hardware, thermal control, microwave sintering hardware, regolith processing systems, electronics, and harnessing.

For the rover subsystem, the estimated dry mass prior to growth allowance was 21.4 kg, with major contributors including the wheels and motors (4.00 kg), electronics components (3.99 kg), battery pack (3.56 kg), chassis (3.25 kg), and excavation bucket assembly (3.27 kg). After applying subsystem-specific Mass Growth Allowance (MGA), the rover total mass increased to 22.9 kg. Growth margins ranged from 2% for mature structural and mobility hardware to 15% for less mature excavation and collection mechanisms such as the bucket assembly. The rover system carried an overall average growth allowance of approximately 7%.

The STiM processing subsystem was developed with the same methodology and resulted in a significantly larger estimated mass due to the inclusion of microwave sintering hardware, thermal control systems, structural supports, and radiation protection. The initial estimated STiM subsystem mass was 174 kg prior to growth allowances. Major contributors included the aluminum structural frame (38.9 kg), battery pack (30.0 kg), mold and sintering system (20.4 kg), cooling system (17.3 kg), regolith processing system (16.6 kg), and environmental/radiation shielding (17.0 kg). After applying MGAs, the STiM subsystem mass increased to 184 kg, while the total integrated system mass increased from 194 kg to approximately 207 kg. Component-level growth allowances ranged from 2% for highly defined structural hardware to 15% for regolith processing systems where design maturity remains lower and operational uncertainty is greater. The average applied growth allowance across the STiM subsystem was approximately 6%.

This budgeting methodology provides a more realistic assessment of total system mass and design maturity by accounting for subsystem-level uncertainty while maintaining traceability to physical hardware elements identified during the preliminary design process.

Table 24: **Rover Mass Budget.** This table shows the rover mass budget calculated to 21.5 kg and 22.9 kg with growth allowances.

Component	Mass (kg)	GA (%)	MGA (kg)	Total MGA (kg)
Battery Pack	3.56	5	0.178	3.74
Electronics Components	3.99	10	0.399	4.39
Chassis	3.25	2	0.065	3.31
Wheels and Motors	4.00	2	0.080	4.08
Tires	0.960	5	0.048	1.01
Bucket	3.27	15	0.491	3.77
Scoop and Lift Arms	1.57	10	0.157	1.73
Harness	0.820	5	0.041	0.86
Total Mass	**21.4**	**7**	**1.50**	**22.9**

Table 25: **STiM Mass Budget.** This table shows the STiM mass budget calculated to 174 kg and 184 kg with growth allowance resulting in a total system mass of 194 kg and 207 kg after accounting for growth allowance.

Component	Mass (kg)	GA (%)	MGA (kg)	Total MGA (kg)
Aluminum Frame	38.9	2	0.78	39.7
Support Legs	13.5	5	0.68	14.2
Battery Pack	30.0	5	1.50	31.5
Vibratory Motors	6.35	5	0.32	6.67
Regolith Processing System	16.6	15	2.49	19.1
Cooling System	17.3	10	1.73	19.0
Hopper and Ramp	5.47	5	0.27	5.74
Mold and Sintering System	20.4	10	2.04	22.4
Environmental/Radiation Shielding	17.0	5	0.85	17.9
Harness	8.28	5	0.41	8.69
Total Mass	**174**	**6**	**10.4**	**184**
Total System Mass	**194**	–	–	**207**

6.2 Power Budget

The power budget provided by the Griffin Lunar Lander is 5 kW as stated in the RFP [105]. The following power budget tables represent preliminary estimates of the operational power consumption for both the rover subsystem and the stationary STiM (Sintering Tile Manufacturing) processing system. These estimates were generated during the preliminary design phase using representative subsystem loads, actuator requirements, thermal control assumptions, and expected operational duty cycles. More accurate power estimates will be developed in future design iterations once specific electronics, motors, microwave hardware, batteries, avionics, and thermal management components are selected and validated.

The rover subsystem was designed to operate within a relatively low continuous power envelope to maximize operational efficiency and battery endurance during lunar surface traverses. During nominal driving, communications, and dumping operations, the rover is expected to draw less than 200 W. Excavation and regolith collection activities, including scoop and dump operations, are estimated to require less than 175 W due to increased actuator loading from the excavation arm and bucket system. The rover power system currently incorporates a 480 Wh battery pack, which is projected to provide sufficient energy for multiple excavation and transport cycles per lunar day before recharging is required. These estimates also include baseline avionics and communication loads necessary for autonomous operation and navigation.

The STiM processing system represents the dominant power consumer within the overall architecture due to the high energy demands associated with microwave sintering and thermal processing of lunar regolith. During active sintering operations, the system is estimated to require approximately 4.50 kW of power, with 4 kW allocated specifically to the microwave RF generation hardware while the remaining power supports the press mechanisms, thermal systems, controls, and supporting electronics. Tile ejection and material handling operations are expected to consume approximately 0.40 kW, while conditioning operations, including vibratory settling and mold release mechanisms, are estimated at roughly 250 W. During idle periods, the system power demand decreases substantially to approximately 50 W during lunar daytime operations for standby electronics and control readiness, and below 10 W during nighttime survival mode when only essential thermal survival heaters and communication systems remain active.

Table 26: **Rover Power Budget.** This table shows the rover power budget calculated to utilize a 480 Wh battery which supports its operational modes of to enable multiple regolith collection trips with a single charge.

Mode	Power Draw	Description
Controls	< 200 W	Driving, dumping, and communications
Digging/Scoop/Dump	< 175 W	Regolith excavation and collection
Total Battery Capacity	480 Wh	Provides sufficient energy for multiple trips per charge

Table 27: **STiM Power Budget.** This table shows the STiM power budget utilizing the available 5 kW provided by the Griffin Lander to undergo sintering at the highest available power draw while maintaining other operational modes.

Mode	Power Draw	Description
Sintering	4.50 kW	Microwave heating, press operation, and control systems
Mold and Press Actuation	0.40 kW	Moves finished tiles away from the STiM system
Conditioning	0.25 kW	Vibration and release mechanisms for regolith preparation
Idle	Day: 50 W; Night: $<$ 10 W	Daytime electronics standby; nighttime survival heating and communications

Overall, the combined operational architecture was intentionally designed to remain within the 5 kW power allocation provided by the Lander while maintaining margin for transient startup loads, thermal regulation, communication overhead, and future subsystem growth. The current power budget therefore demonstrates the feasibility of simultaneous rover excavation and stationary regolith processing operations within the available lunar surface power constraints.

6.3 Cost Estimate

The system cost budget was developed using a subsystem-based estimation methodology that accounts for the primary hardware, software, integration, labor, and testing expenses associated with both the rover excavation platform and the stationary STiM (Sintering Tile Manufacturing) processing system. The estimates are intended to represent preliminary-order-of-magnitude development costs during the early design phase and were generated using expected subsystem complexity, manufacturing requirements, electronics integration needs, and anticipated testing campaigns. These values will continue to mature as specific components, manufacturing methods, and operational requirements are finalized in future design iterations.

The rover subsystem resulted in an estimated total development cost of approximately $83.4 million. The largest contributors to the rover budget include Software and Integration at $15.3 million, Structure and Mechanical Systems at $12.7 million, Power and Thermal systems at $10.8 million, and Testing Costs at $18.0 million. Additional major contributors include Electronics and Avionics ($8.0 million), Navigation Sensors and Cameras ($5.3 million), Actuation systems such as motors and gearboxes ($3.3 million), and Labor Costs totaling $10.0 million. The relatively high testing allocation reflects the need for environmental testing, mobility validation, thermal vacuum testing, and excavation performance verification under simulated lunar conditions. Overall, the rover budget reflects a comparatively lightweight but highly integrated robotic mobility system requiring reliable autonomous operation and regolith handling capability.

The STiM processing subsystem represents the majority of the total program cost due to the complexity of microwave sintering operations, thermal management systems, structural support hardware, and large-scale regolith processing equipment. The STiM subsystem alone is estimated to cost approximately $270 million. The largest cost contributors include Structure and Mechanical Systems at $65.6 million, Power and Thermal systems at $51.4 million, Testing Costs at $45.0 million, Software and Integration at $32.5 million, and Regolith Processing hardware at $31.7 million. Additional costs include Electronics and Avionics ($14.2 million), Actuation systems ($10.0 million), and Labor Costs ($20.0 million). The elevated testing and thermal system costs are driven by the need to validate high-power microwave sintering operations, thermal cycling durability, vibration isolation systems, and long-duration lunar surface survivability.

Table 28: **Rover Cost Budget.** This table shows the rover cost budget with cost estimates for subsystems totaling $83,400,000.

Category	**Cost (USD M)**
Power and Thermal	10.8
Navigation Sensors (IMU and Cameras)	5.3
Structure and Mechanical Systems	12.7
Electronics and Avionics	8.0
Actuation (Motors and Gearboxes)	3.3
Software and Integration	15.3
Labor Cost	10.0
Testing Cost	18.0
Rover Total Cost	**83.4**

Table 29: **STiM Cost Budget.** This table shows the STiM cost budget with cost estimates for subsystems totaling $270,000,000 resulting in a total system cost estimate of $353,000,000.

Category	**Cost (USD M)**
Structure and Mechanical Systems	65.6
Electronics and Avionics	14.2
Actuation (Motors and Gearboxes)	10.0
Regolith Processing (Microwave and Mold)	31.7
Power and Thermal	51.4
Software and Integration	32.5
Labor Cost	20.0
Testing Cost	45.0
STiM Total Cost	**270**
Total System Cost	**353**

When combined, the total integrated system cost is estimated to be approximately $353 million. This overall cost reflects the challenges associated with developing a fully operational lunar regolith excavation and tile manufacturing architecture capable of autonomous surface operations, high-power material processing, and sustained functionality within the harsh lunar environment.

6.4 Mission Risk Analysis

Project Hestia is considered a Class C mission according to NASA's Launch Services Risk Classification Fact Sheet, which corresponds to a moderate-risk-tolerance mission [106]. This classification means that not all risks are expected to be fully eliminated; but the most significant risks should be identified, tracked, and supported by realistic mitigation strategies to reduce their impact on the mission.

Based on this classification, a risk analysis was performed to identify the most critical risks for Project Hestia and to define mitigation strategies for each one. The analysis focuses on risks that could affect mission performance, schedule, safety, or the ability of the payload to demonstrate lunar infrastructure construction.

Each risk was evaluated using two criteria: consequence of failure and likelihood of failure. Both criteria were assessed on a scale from 1 to 5, with higher values indicating more severe consequences or a greater probability of occurrence. The qualitative definitions for each level are provided in Table 30. These values were then used to place each risk within the 5×5 risk matrix shown in Figure 48.

Table 30: **Risk Level Definitions for the 5 × 5 Risk Matrix.** This table shows the definition for the scale used for the risk table and mission risk analysis for both consequences of failure and likelihoods of failure.

Consequence of Failure		Likelihood of Failure	
Catastrophic (5)	Total mission failure; failure to satisfy mission requirements	**Very High (5)**	50%–100%
Critical (4)	Failure to satisfy multiple mission requirements	**High (4)**	25%–50%
Moderate (3)	Failure to satisfy a mission requirement	**Medium (3)**	12%–25%
Minor (2)	Mission requirements are met; disruption to mission timeline	**Low (2)**	5%–12%
Negligible (1)	Minimal to no impact on mission outcome	**Very Low (1)**	0%–5%

Risks were assigned unique identifiers based on their category. Table 31 defines the risk identifiers used throughout the mission risk analysis. These classifications correspond to Mission, Rover, Environmental, Power and Thermal, and Structural risks referenced in Table 32.

Table 31: **Risk Classification Legend.** This table shows the categories of risks considered based upon what part of the mission the risk is relevant to.

Risk Classification	Category
M	Mission
R	Rover
E	Environmental
P	Power and Thermal
ST	Structural

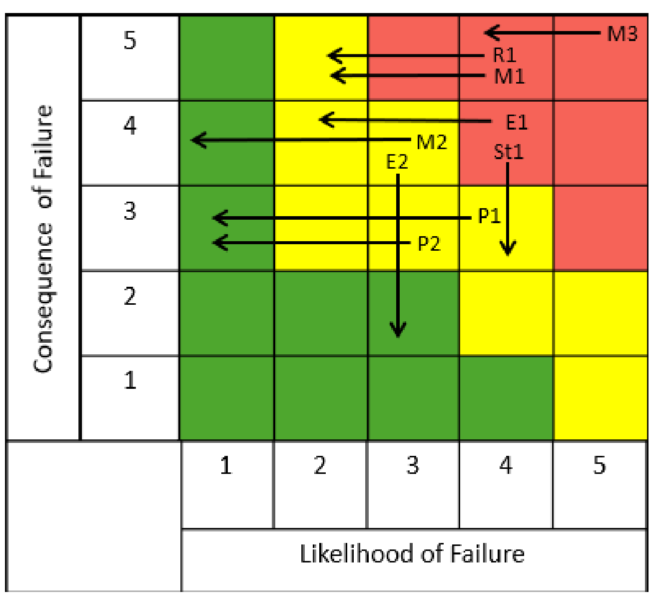


Figure 48: **Risk Matrix.** The primary risks identified for Project Hestia are organized into a 5 × 5 risk matrix. The likelihood and consequence of the risks are categorized from 1–5 on the horizontal and vertical axes, respectively. Risk statements corresponding to the identifiers in the matrix are provided in Table 32. Mitigated risks fall in the yellow and green zones showing the missions properly suited to its risk tolerance and has taken measure to reduce potential risk

The risk analysis is visualized in Figure 48 using a 5 × 5 risk matrix. The matrix compares the likelihood of failure with the consequence of that failure for each identified risk. The arrows represent the expected risk level after mitigation methods are applied.

The matrix separates risks into three categories based on the combined likelihood and consequence of failure. Risks located in the green cells represent lower-risk areas that are acceptable for the current stage of the mission. Yellow cells represent moderate risks that require continued monitoring. Red cells represent the highest-risk areas, where both likelihood and consequence are significant enough to be considered critical to mission success.

At this early stage of the design, several risks fall into moderate or critical zones. These include uncertainty in the regolith composition at the landing site, insufficient available power for sustained sintering, overheating of the sintering system, the possibility that produced tiles may not meet structural requirements, and exceeding the mass limit after growth allowance is applied. As the design becomes more refined and mitigation measures are implemented, these risks are expected to shift toward lower-risk regions. However, the mass limit remains one of the main risks for the mission and will continue to be a concern as other mitigation techniques and redesign options are explored.

Table 32 summarizes the primary mission-level risks identified for Project Hestia, their potential consequences, and the mitigation strategies proposed to reduce their likelihood or severity.

Table 32: **High-Level Mission Risks and Mitigation Strategies for Project Hestia.** This table shows the risks considered on the risk matrix detailing the risk, its consequence, and the mitigation employed to minimize the risk.

Risk ID	Risk Statement	Consequence	Mitigation Strategy
M1	If the microwave sintering system fails to operate as expected	Tiles may be weak or improperly formed, leading to cracking	Test the system on Earth using lunar regolith simulant
M2	If dust accumulates on critical infrastructure	Available power for the rover and STiM is reduced, heat dissipation worsens, mechanical components become damaged	Use controlled regolith dumping methods and dust shields
M3	If mass limit is surpassed after growth allowance is applied	Mission requirements may not be met, and the system may require redesign	Use conservative mass estimates, track mass growth throughout design, and identify lower-priority components that could be removed
P1	If microwave sintering efficiency is lower than expected	More energy is required to produce each tile	Validate sintering efficiency through ground testing
P2	If available power is insufficient for sustained sintering	Tile production is incomplete or reduced	Budget power conservatively and include operational margins
R1	If the rover cannot traverse the terrain	Regolith delivery to the STiM is interrupted	Select a flat landing site and limit rover travel distance
E1	If landing site regolith is unsuitable for tile production	Tile quality decreases and production time increases	Select a site with predictable regolith properties
E2	If the sintering system overheats during operation	Processing stops or hardware may be damaged	Add thermal monitoring and cooling controls
ST1	If produced tiles do not meet structural requirements	Tiles cannot be used for infrastructure construction	Produce demonstration tiles before full-scale operations

Taken together, the results shown in Table 32 and Figure 48 suggest that, once mitigation is applied, most of the evaluated items fall within the green area of the matrix. In practical terms, this means they are acceptable for the current stage of the mission. Although some uncertainty is unavoidable at this point in the design, the overall trend shows that the mitigation measures proposed so far are working as intended and are effective in bringing potential issues down to manageable levels.

6.5 Path to PDR

VT Luna is targeting a launch in September 2029, with the Preliminary Design Review serving as a critical milestone in 2027. The timeline to reach that point reflects both the substantial progress already made and the significant development work that remains. Each phase of the timeline builds deliberately on the last, ensuring that design decisions are validated before resources are committed to manufacturing and integration.

To date, the team has accomplished a considerable amount of foundational work. A mission concept has been established, payload requirements have been defined, and trade studies have been completed to determine the desired design approach. These trade studies evaluated competing concepts across key subsystems, weighing factors such as mass, power consumption, mechanical complexity, and compatibility with the Griffin Lunar Lander's payload constraints. From this process, the current architecture, consisting of a regolith-collecting rover paired with the STiM, emerged as the most viable path toward achieving the mission objective. In addition, 3D CAD models of both the rover and STiM have been developed, tile dimensions and design have been finalized, mass and power budgets have been estimated and refined, and the total concept cost has been estimated. The rover has also been adapted to support tile receiving and placement capabilities, and a preliminary rover prototype has been fabricated. This early prototype has already provided valuable insight into the mechanical feasibility of the design, informing decisions around motor sizing, chassis geometry, and the regolith collection interface. Taken together, these accomplishments represent a well-developed conceptual foundation from which the remaining design and testing work will proceed.

To achieve a PDR, several critical prototyping efforts remain. This includes further research into the sintering subsystem, as it represents the core of the mission's construction architecture. Limited research has been done to test and optimize the concept of thin layer microwave sintering specifically. A proof of concept for the microwave sintering of tiles using lunar regolith simulant with the composition most similar to that of the missions anticipated landing site would have to be conducted to demonstrate the feasibility of such a concept. Additional tests using regolith simulant with a slightly different composition than anticipated will also be useful as it is unlikely that the mission will collect the exact composition that would be optimal.

The filtration model, weigh bin, and press must all be prototyped and validated before the PDR as well. The filtration subsystem is responsible for processing collected regolith to remove particles of non optimal granularity before the material is distributed to the tile mold for sintering. Ensuring consistent particle size distribution is critical to producing tiles with predictable mechanical properties, making the filtration system one of the more technically sensitive components in the overall architecture. Prototyping this subsystem will allow the characterization its performance, identification failure modes, and enable refinement of the design to be completed before the PDR. Once completed, the team will be well positioned to present a mature and validated concept at PDR.

Following the PDR in 2027, development will advance into higher-fidelity testing across multiple subsystems. The microwave sintering system and mold will be prototyped and evaluated, with initial testing conducted under ambient conditions to characterize the sintering behavior of the regolith simulant before progressing to vacuum testing. Tiles manufactured from lunar regolith simulant will be produced and tested to assess their dimensional accuracy, surface finish, and structural integrity. These results will directly inform updates to tile properties and sintering parameters, and may suggest opportunities to refine tile geometry. Depending on the outcomes of this testing, there is potential that tile dimensions could be changed if sintering performance and structural testing support it, though this will be determined by the data rather than assumed. Additionally, sintering conditions may be altered if it is found that a different set of conditions yields a stronger tile, or can produce an equally strong tile faster.

Vacuum testing of the microwave sintering process represents one of the more technically demanding steps in the development timeline. The lunar vacuum environment introduces heat transfer conditions that differ significantly from ambient testing, as convective cooling is absent and thermal gradients across the system can be more severe. Validating that the sintering process produces acceptable tiles under these conditions is essential before committing to the final hardware design. This testing will also provide an opportunity to assess the performance of the thermal management systems protecting the electronics and the sintering hardware itself.

In parallel with sintering development, the alpha particle spectrometer will undergo testing to validate its ability to characterize regolith composition in the field. Understanding the variability in regolith properties across the landing site is important for predicting tile quality and adjusting sintering parameters accordingly. The spectrometer will be mounted on the prototype rover to mimic the actual case it will be used in when on the Moon. Data gathered from spectrometer testing will feed directly into the refinement of the sintering process model and help the team establish acceptable bounds on regolith input variability. Additionally, analysis on the effectiveness on the spectrometer will help determine the effectiveness of the technology for the missions needs, and whether or not such a device is necessary to perform the goals of this proof of concept mission.

A key area of development following PDR will be systems integration between the rover and STiM. These two systems must operate in close coordination throughout the construction workflow, and understanding how they interact physically and operationally is essential to validating the overall mission concept. This includes characterizing the regolith handoff interface, ensuring that material is transferred consistently and without spillage, as well as validating the positional accuracy required for tile placement. The timing and sequencing of the full construction cycle, from regolith collection through filtration, pressing, sintering, and tile deployment, will be exercised end to end to identify any bottlenecks or failure points in the workflow. Results from this integration testing will drive updates to the concept of operations, ensuring that the operational procedures reflect the actual behavior of the integrated system rather than idealized assumptions.

Finalization of the rover's regolith collection path will also occur during this period. The specific trajectory the rover follows across the landing site affects both collection efficiency and the risk of disturbing areas critical to landing operations. Simulation and testing will be used to refine this path, taking into account the terrain, the rover's turning radius, and the capacity of the collection and filtration systems. Payload integration planning with the Griffin Lunar Lander will be completed in parallel, confirming that the combined mass, volume, and power requirements of the rover and STiM are compatible with the Lander's deployment constraints.

Final pre-launch steps will include manufacturing all components, encompassing both commercial off-the-shelf hardware and custom-fabricated parts, followed by individual component testing to verify performance against requirements. The concept of operations will be updated one final time to incorporate all findings gathered throughout the development process, ensuring that the operational procedures are grounded in tested hardware behavior. A full system-level test campaign will be conducted to validate the integrated payload prior to delivery for launch.

Looking beyond the launch in 2029, the team sees an opportunity to scale the concept for future missions. Larger rover and STiM configurations could process greater volumes of regolith per cycle, and deploying multiple units in parallel could dramatically reduce the time required to construct a complete landing pad. While these possibilities remain forward-looking and contingent on the success of the initial mission, they reflect the broader potential of the architecture as a foundation for sustained lunar surface infrastructure. While the current design created landing pad tiles, there is also the potential to simply use the concept to create bricks, which could be used more broadly across lunar architecture. The concept that has been developed is designed to be scaled and has the potential to be a key foundational piece in constructing more lunar architecture.

7 Future Work and Conclusion

7.1 Future Work

VT Luna's current goal is a proof of concept mission, with the goal of proving the feasibility of microwave sintering technology to create tiles on the Moon. There will be different levels of success the team might achieve. The first level of success would be if the rover is able to collect regolith autonomously but no other systems function correctly and no tiles are produced. This result would be disappointing as it will fall short of actually creating tiles. The second level of success would be the rover and STiM working to produce only one or only a few tiles. This result would still be very valuable as the properties of the tile could be found; however, a lack of many tiles would limit the number of tests that could be run, seeing as destructive tests will be important to test the limits of the tiles. The third result would be a significant number of tiles manufactured, but an inability of the rover to actually assemble them into a landing pad. This result would still be very promising in terms of the technology feasibility and design. With many tiles, a large variety of tests would be able to be conducted to get an in-depth understanding of the tile properties. The landing pad construction is in many ways already a stretch goal, seeing as the rover has no manipulator for precise placing of tiles. The tiles were designed to be able to slide together with relatively low precision; however, this design might prove to be more effective in theory than in practice, and a more capable manipulator might need to be added to the rover in later iterations of the design. This is another feature that will be tested on Earth before the actual mission. Additionally, this result scenario also includes the possibility of the systems breaking down before the 6-month period is up. This could be a result of dust covering or damaging parts, the rover getting stuck on the lunar surface, or a mechanical part on the STiM jamming or breaking. Enough tiles would still be produced to run all the desired tests; however, solutions to failure risks would need to be found for a real product designing full-fledged landing pads. A fourth result would be the mission achieving all goals set out over the 6-month period. This is of course an ideal outcome where the feasibility and effectiveness of the design would be clearly demonstrated, assuming the tiles achieved predicted properties. Finally, a fifth result, which is likely if the fourth result is already achieved, is that the system continues working past the 6-month window. There are radiation, dust coverage, and decreased illumination concerns for the overall lifetime of the design; however, if the design proves to last beyond the 6-month period it was designed for, that would only demonstrate over-engineering and the possibility of a longer operational lifetime. Beyond the mission of the proof of concept, the goal is for the design to be implemented in the real creation of lunar landing pads. This would require a full STiM factory with multiple STiM systems individually manufacturing tiles. Additionally, more rovers would need to be sent to the surface to collect the amount of regolith needed. These rovers would also be larger in size to increase the collection rate of each one. This factory proposal for the lunar surface could produce many landing pads, rapidly expanding the possibility of lunar habitats and lunar colonization. To do this, future work that VT Luna hopes to explore includes the following:

7.1.1 Griffin Lander Interface and Ramp

Currently there is no existing interface between the Griffin Lander and the VT Luna Architecture. In the future as the mission moves forwards VT Luna would work in conjunction with Astrobotic to design the interface between the lander and the payload. This includes the structural supports responsible for uniform loading during launch and landing. An efficient structural interface is integral to a clean delivery and staying within the mass restriction. Additionally a deployment mechanism capable of placing the STiM on the lunar surface and deploying a ramp for the rover. A clean deployment is a critical step as failing to do so would result in complete mission failure. The frame of the STiM is designed to have multiple free points for deployment arms to attach to. A power and communications connection cable would also need to be designed in a way that does not interfere in the deployment process. The last part of this interface is the rover ramp and transfer system. This structure is responsible for allowing the rover to travel from the lunar surface to its charging port on the Griffin Lander, and then across the top of the Lander to reach the top of the STiM for regolith depositing. One key architecture feature that VT Luna would like to employ are toothed, interlocking ramp and wheel systems to prevent slippage when climbing and traversing the ramp and Griffin structure. This comes at both a low mass and monetary cost and mitigates the risk of the rover falling off the ramp and becoming stuck ending the mission. Another key architecture would be aligning the Griffin-STiM interface ramp in a way that allows the bucket of the rover to contact the hopper when dumping regolith. By making contact and reducing the height the regolith needs to free fall regolith getting kicked up is minimized improving the yield of each collection cycle and preventing dust accumulation on key components like the radiators and solar panels.

7.1.2 Rover Autonomy and Terra-forming

Enhancing rover autonomy is a key priority for future iterations of the design. Currently, the rover is designed primarily to support regolith collection and delivery, but a future iteration will incorporate expanded navigation, perception, and manipulation capabilities. This would enable the rover not only to collect and transport regolith but also to relocate completed tiles from the stationary tile maker to a designated testing or staging area. In order to create the landing pad, the construction area must also be properly leveled and terraformed prior to tile placement. Improvements to the scoop design and autonomy will allow the rover to prepare the construction area independently. Investigating the feasibility and reliability of such autonomous tile-handling and ground-leveling operations will be critical for reducing human oversight and enabling scalable surface construction.

A prototype autonomous rover has already been made as mentioned before. The rover in the experiment uses a motion capture system to know its position and orientation at all times. Of course this system for determining position will not be available on the lunar surface. Instead, the missions actual rover will rely on IMU data and camera scanning in order to obtain surrounding information. Additionally, this experiment, there were no obstacles included in the path the rover would take to reach the collecting sites. On the lunar surface, it will be critical that the rover has obstacle avoidance capabilities so as to avoid larger pieces of regolith it cannot clear, or small craters on the lunar surface. If the rover gets stuck at any point the mission is abruptly halted. With no collection method, the STiM receives no regolith needed to sinter tiles. This is a key aspect of future work that needs to be fully developed and tested before launch.

7.1.3 Further Tile Testing

A critical area of future development is the establishment of robust, non-destructive tile testing methods capable of evaluating tile quality directly on the lunar surface. While the missions current testing and evaluation through returning tiles Earth for analysis with destructive testing methods such as, including the drop weight tester, hydraulic press, and arc jet, are essential for characterizing the structural and thermal limits of the tiles, they consume the tile in the process and relying on return missions adds significant cost and complexity. For this mission where tile production is slow and each tile represents a significant investment of energy and time, non-destructive evaluation methods will be equally important.

One such method that has been considered for future evaluation is the employment of pulse-echo ultrasonic testing. Pulse-echo, or PE, is an ultrasonic inspection technique that uses a single transducer to both emit and receive ultrasonic signals. The time-of-flight differences between signal emission and reception indicate the presence and location of internal anomalies such as cracks, voids, or delamination [107]. The amplitude of the received signal is directly tied to the depth and severity of a detected feature, meaning that both the location and the size of an internal defect can be characterized without physically breaking the tile [107]. This is particularly relevant for the sintered regolith tiles produced by Project Hestia, as the layered sintering process introduces the possibility of inter-layer bonding defects and internal voids, both of which are known failure modes in microwave-sintered ceramics. A PE scan performed on each tile after ejection could rapidly identify tiles with internal anomalies before they are placed in the landing pad, preventing structurally compromised tiles from being incorporated into the surface. Beyond defect detection, ultrasonic methods can also be used to estimate the elastic modulus and density of a sintered tile through measurement of the acoustic wave speed, providing an additional quality metric that complements the compressive strength data gathered through destructive testing. Developing and validating a PE testing protocol for sintered regolith tiles on Earth will be a necessary precursor to any consideration of autonomous on-site tile inspection on the lunar surface.

7.1.4 Further Finite Element Analysis

Finite Element Analysis (FEA) will play a central role in optimizing future iterations of the tile design. By modeling various geometries, material distributions, and interlocking mechanisms under realistic loading conditions, the team can identify configurations that offer improved durability, thermal stability, and load-bearing performance. Of particular interest is the analysis of tiles under loading conditions from larger HLS vehicles such as the SpaceX Starship HLS, which at approximately 100,000 kg represents a significantly greater structural demand than the 50,000 kg HLS vehicle used as the design basis in this report. As commercial vehicle specifications are finalized and released, FEA will allow the tile design to be updated and validated against these requirements without the cost of physical testing at every iteration. These simulations will also guide the refinement of both the sintering process parameters and the structural layout of the tiles.

Furthermore, structure analysis of the STiM and rover using finite element analysis will allow for optimized reductions in mass through evaluations of maximum loading conditions during launch and lunar operations. Through this analysis areas designed with unnecessary margin can be reduced allowing for greater flexibility within the mass budget which will help accommodate future growth that the system will undergo.

7.1.5 Interlocking Tile Designs

If constraints in the manufacturing process are overcome, designs with interlocking geometry could be employed. Interlocking tiles would be valuable as they allow for the distribution of lateral forces between tiles, enable multiple layers of tiles to be stacked on top of each other as reinforcement, and mitigate the risk of tiles being ejected by the plume of landing rockets.

Before radial shrinkage of the tiles was acknowledged during the sintering and cooling process, the tile design was based upon the hexagonal design but incorporating dovetail joints. These dovetail joints allow tiles to transfer increasing loading with no sudden failure, and this behavior is particularly suitable in terms of the safety of the construction [108]. Therefore, if the constraints of material shrinkage are able to be overcome through post-processing, relaxing RTR 2.2, implementing this sort of design would be a major improvement over the current design.

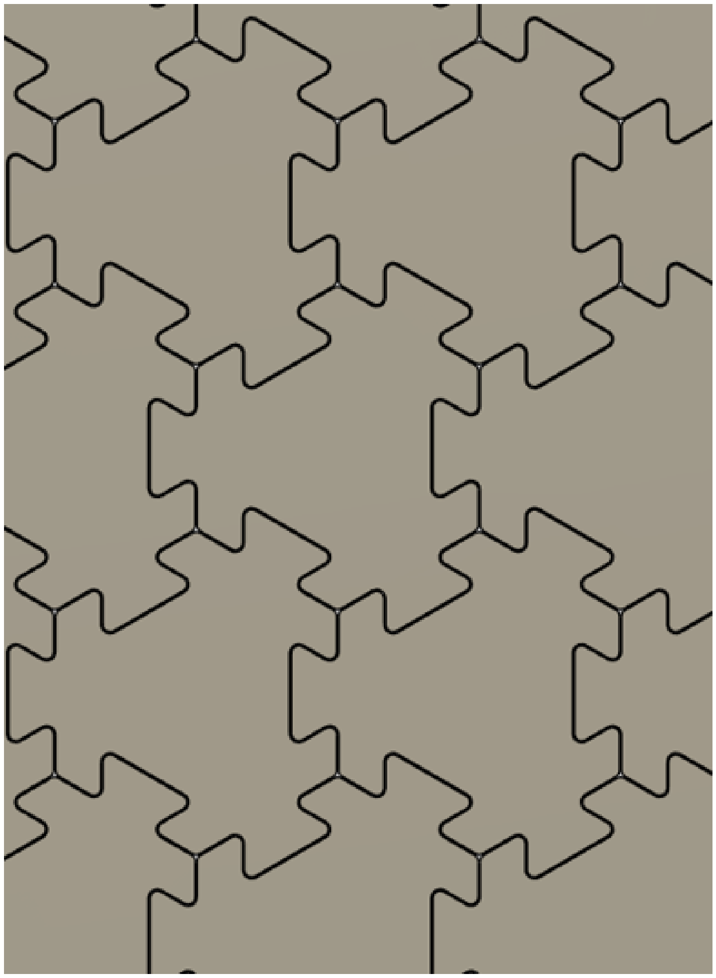


Figure 49: **Interlocked tiles.** Image of the dovetailed tile design highlighting how the tiles interlock with each other to transfer loads. This design allows for stronger interlocking than the current tile design but is not capable of being manufactured by the current system.

A vertically interlocking tile design was also identified in the literature, incorporating a geometry that allows tiles to interface with each other in the out-of-plane dimension. Stacking multiple layers of tiles in this manner increases the structure's reliability by allowing for redundancy and rapid repairs [49]. While this concept has clear advantages, the requirements imposed by the compression and sintering process under RTR 2.1 would have to change significantly to produce this design or utilize a similar concept.

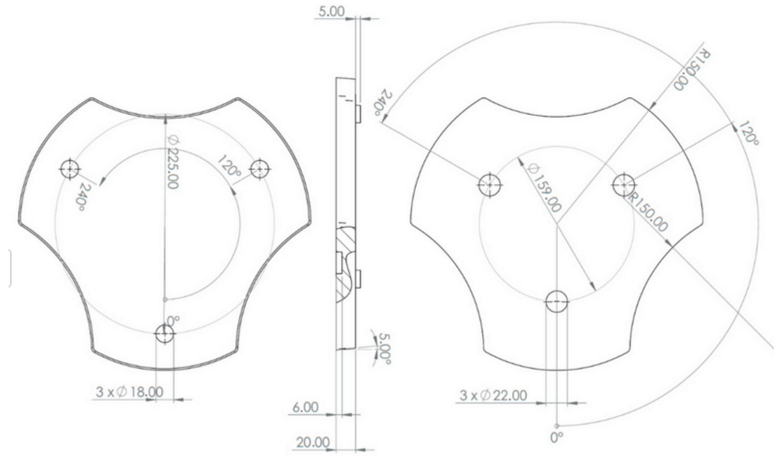


Figure 50: **Vertically interlocking tiles.** Technical drawing of the vertically interlocking tile design showing that a lack of vertical homogeneity allows for tiles to interface with each other in the out of plane dimension. Vertical layering enables construction of landing pads with strength suited to the potential client rockets. However, due to the missions restrictions and manufacturing capabilities it is not capable of producing tiles of this sort [49].

7.1.6 Shielding Improvements

To increase the mission duration a single wall aluminum shielding solution that would mitigate damage from substantial micro-meteoroid impacts and resist irradiative effects could be considered [91]. Furthermore, in-situ shielding solutions such as the creation of a hollow receptacle that the vibratory sifter could deposit the regolith of undesired granularity into would also protect the mission hardware. These systems would allow for longer duration lunar surface operations, but as other concerns such as regolith accumulation will also be restricting the mission duration, and tight mass constraints; such systems were deemed unnecessary for the current project scope.

7.2 Conclusion

The VT Luna team set out to address one of the most fundamental challenges facing sustained human presence on the Moon: the absence of stable, protected landing infrastructure. Plume ejecta from repeated rocket landings poses a serious and well-documented threat to surface hardware, and no scalable solution for constructing durable landing pads using in-situ resources had been demonstrated at the payload level. Project Hestia represents a direct response to this gap, proposing an autonomous, ISRU-driven architecture capable of fabricating sintered regolith tiles and assembling them into a functional landing pad surface, all within the mass, power, and volume constraints of the Astrobotic Griffin Lunar Lander.

The design centers on two complementary systems: a lightweight regolith-collecting rover and a stationary tile manufacturing unit, the STiM. This division of labor was the product of a deliberate trade study process that evaluated four distinct architectures, ultimately concluding that separating mobility from manufacturing was the most mass-efficient and operationally robust path forward. The rover handles excavation, filtration, and material delivery, while the STiM manages the full conditioning and sintering pipeline autonomously. Together, the two systems form a closed construction loop that requires no Earth-supplied consumables and minimal human intervention.

At the heart of the construction process is microwave sintering, selected over solar sintering and binder-based methods for its operational independence from sunlight, its compatibility with the Griffin Lander's 5 kW power supply, and its demonstrated ability to produce structurally sound tiles from lunar regolith simulant. The sintering analysis conducted by the team produced a validated power and time model for the process, yielding an optimal applicator area of 0.0143 m^2 and a total tile production time of approximately 7.3 hours at 4 kW, with the relationship between power draw and production time being linearly inversely proportional. This model provides both a credible timeline for mission operations and a clear road map for improving throughput in future, higher-power implementations.

The tile design itself emerged from a careful synthesis of structural, thermal, and manufacturing requirements. The selected custom hexagonal geometry with a circumscribed radius of 0.15 m satisfies the full set of regolith tile requirements derived from a simulated 50,000 kg HLS landing, including impact pressure, sustained plume pressure, and thermal flux. The hexagonal form also tessellates efficiently, interlocks preventing exhaust destabilization, minimizes stress concentrations at tile boundaries, and is compatible with the layer-by-layer sintering and compaction process imposed by the STiM architecture. The material properties of sintered lunar regolith at the target temperature of 1,200°C, including a compressive strength of approximately 21.73 MPa and a compression modulus of 55.73 MPa, are sufficient to meet all structural requirements under the expected loading conditions.

The landing site selection process identified the Mons Mouton Plateau as the optimal location for the mission, offering approximately 70 to 80 percent annual solar illumination, relatively flat and accessible terrain, robust Earth line-of-sight communication, and regolith composition in the High-Ti mare zone that is particularly well suited for microwave sintering due to its elevated iron oxide and titanium dioxide content. This selection reflects the team's commitment to grounding every design decision in the specific operational context of the mission rather than idealized assumptions.

The system summary confirms that the design fits within its constraints. The projected total mass of 194 kg sits within the 200 kg RFP limit despite the risk of growth allowance exceeding this value, the peak power draw of 4.5 kW during sintering is within the 5 kW supply from the Griffin Lander, and the estimated cost of approximately \$353 M reflects a realistic budget for a technology demonstration mission of this scope. The risk analysis identified the sintering subsystem and regolith variability as the highest-risk areas, both of which are addressed through the team's planned prototyping and vacuum testing campaign ahead of PDR.

It is important to be clear about what this mission is and what it is not. Project Hestia is a proof of concept. The system as designed will not produce enough tiles to cover a complete landing pad over its six-month operational window. What it will demonstrate, however, is that the fundamental process, autonomous regolith collection, particle size conditioning, layer-by-layer microwave sintering, and tile ejection, is feasible within realistic lunar mass and power constraints. That demonstration is itself the primary deliverable. If the sintering process performs as modeled, if the rover and STiM operate reliably in coordination, and if the produced tiles meet their structural and thermal requirements under testing, then the core hypothesis of the mission will have been validated. Everything that follows, larger systems, higher throughput, multi-unit deployments, full landing pad construction, is a scaling problem rather than a feasibility problem.

The scalability of the architecture is one of its most important attributes. The STiM is a self-contained manufacturing unit that can be replicated and deployed in parallel. The rover is a simple, low-mass platform that can be scaled up or duplicated. The microwave sintering process scales linearly with available power. A future deployment with access to a larger power grid and multiple STiM units could construct a complete landing pad in a fraction of the time required by the proof of concept mission. In this sense, Project Hestia is not just a one-time technology demonstration but a foundational step toward the kind of in-situ construction capability that will be required to support a permanent, self-sustaining human presence on the Moon. Landing pads are first-order infrastructure, and building them from the Moon itself, rather than shipping materials from Earth, is the only approach that scales to the level of activity envisioned by the Artemis program and beyond.

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A Appendix

A.1 ABET Considerations

The eight ABET public, global, cultural, social, environmental, and economic factors were considered for the Project Hestia's mission.

Table 33: ABET considerations for the Project Hestia mission.

ABET Value	**Consideration**
Public Health	• The 4 kW microwave sintering system operates at 915 MHz and generates significant RF energy. The sintering chamber is fully enclosed and shielded within the payload structure, preventing any RF exposure to personnel during ground testing or to astronauts operating near the landing site. • The two NASA Radioisotope Heater Units carry Plutonium-238 fuel. Their use requires nuclear launch approval and full compliance with Department of Energy radiological safety and handling requirements.
Public Safety	• STiM operates at the lunar south pole, near planned Artemis crewed landing sites. Safe standoff distances are maintained to prevent RF interference with crewed vehicle navigation systems and to avoid Lander plume contamination of the sintered tile field. • All high-power and high-voltage subsystems are fully enclosed within the payload frame. The system is designed to operate fully autonomously, requiring no human intervention and eliminating crew exposure risk during all operational phases.
Public Welfare	• By sintering landing pad tiles directly from local lunar regolith, Project Hestia eliminates the need to launch construction materials from Earth, directly reducing mission cost and enabling more sustainable long-duration human presence on the Moon.

ABET Value	Consideration
Global Factor	• Project Hestia was developed for NASA's C3 COSMIC Challenge, an open academic competition. All design methodology, results, and lessons learned will be shared with the broader international in-situ resource utilization research community. • Lunar south pole resources are a focus of multiple international space agencies. A successful ISRU sintering demonstration supports cooperative crewed exploration goals under the Artemis Accords framework.
Cultural Factors	• The mission is named Project Hestia after the Greek goddess of the hearth and home, reflecting the mission's core goal of transforming the hostile lunar environment into a stable, habitable surface for future explorers. • The VT Luna team demonstrates that student-led university programs can produce competitive mission designs, reflecting the cultural importance of investing in the next generation of aerospace engineers.
Social Factors	• Project Hestia could be funded through a NASA public challenge program, where communicating the mission's value to human exploration is essential to maintaining public and congressional support for lunar infrastructure investment. • The VT Luna team is a multidisciplinary collaboration that contributes to the social mission of STEM workforce development and inspires broader public interest in space exploration.
Environmental Effects	• All payload operations should be conducted in a lunar vacuum, with no risk to terrestrial ecosystems. All flight hardware is sterilized prior to launch in compliance with planetary protection requirements to prevent biological contamination of the lunar surface. • By utilizing regolith already present at the construction site, the payload eliminates the environmental and economic cost of mining, processing, and launching raw construction materials from Earth.
Economic Factors	• Establishing a reliable in-situ resource utilization tile production capability has the potential to significantly reduce the cost of future lunar surface infrastructure by eliminating the need to transport materials from Earth at a cost of approximately $1 million per kilogram. • The Griffin Lander's 200 kg total payload mass limit was a binding design constraint throughout the project. Every kilogram saved in the subsystems directly reduces launch cost and increases margin available for additional science payload.

A.2 Standards

Table 34: Applicable engineering standards, their discipline, and their impact on the Project Hestia and verification.

Standard	**Discipline**	**Impact on Project Hestia**
ASME Y14.5	Mechanical Design	All mechanical drawings for the payload frame, radiator panels, hinge assemblies, and rover chassis use GD&T annotations defined by this standard, ensuring tolerances are communicated without ambiguity to the manufacturing team.
MIL-STD-810H	Environmental Testing	Defines the vibration, shock, and thermal cycling test profiles the payload must withstand during launch aboard the Griffin Lander and surface operations, directly informing the structural design margins of all mechanical assemblies.
ASTM E11	Regolith Processing	Governs the woven wire sieve specifications used in the three-stage vibratory sifting subsystem, which classifies collected regolith to the correct particle size range before it is dispensed into the sintering mold.
ASTM C39	Structural Testing	Provides the compressive strength test procedure used to verify that sintered regolith tiles achieve the required 423.99 kPa structural threshold set by the HLS plume impact requirement.
IEEE Std 1625	Power Systems	Sets the safety, charge management, and performance requirements for the multi-cell lithium-ion battery pack powering the rover drive and sensor systems throughout each collection cycle.
ASTM C862	Thermal / Materials	Guides the preparation of refractory mold test specimens used during ground characterization of the sintering chamber, ensuring material samples are fabricated consistently for repeatable thermal test results.
NASA-STD-8739.10	Avionics / Wiring	Establishes the workmanship and inspection requirements for all electrical harnesses, crimp terminations, and connector installations throughout the avionics core and payload power distribution system.
ISO 6015:2006	Rover Operations	Provides the standardized vocabulary for excavation and material handling operations used in all rover collection system documentation and interface control documents.

Standard	Discipline	Impact on Project Hestia
MSFC-SPEC-494	Avionics / Wiring	Defines solder joint quality, inspection criteria, and process controls for all printed circuit board assemblies in the avionics core, ensuring flight-heritage reliability of the electronics.
KSC-STD-Z-0017	Electrical Grounding	Governs the electrical bonding and grounding architecture of the payload structure to prevent electrostatic discharge events that could damage sensitive electronics or interfere with the RF sintering system.
NASA-HDBK-7009B	Modeling & Simulation	Defines the verification and validation framework applied to all thermal simulation models and structural finite element analyses, ensuring model fidelity meets the confidence level required for flight design decisions.
NASA-STD-7001B	Vibroacoustics	Specifies the vibroacoustic environment the payload must survive during launch, driving the acoustic qualification test levels and informing the design of mechanical damping features in the hinge and panel assemblies.
ASTM C31	Materials Testing	Applied to the preparation and curing of sintered regolith tile specimens during ground demonstration, ensuring samples are fabricated under controlled conditions for valid mechanical property measurements.
NASA-STD-7002B	Systems Testing	Establishes the system-level test and verification requirements for the complete STiM payload prior to flight delivery, including qualification test sequencing and pass/fail acceptance criteria.

A.3 Code

Below is the GitHub link to the code used for this project. This includes the code for the prototype rover and its connection to the motion capture system, the code for optimizing the regolith layer thickness of the sintering process, and the code for determining the specific heat of the regolith based on its composition: `https://github.com/Cosmonick04/VT_Luna/tree/main`